

KLE SOCIETY'S COLLEGE OF SCIENCE & COMMERCE

NAAC Accredited Grade B+

(Affiliated to University of Mumbai)

Plot No. 29, Sector No. 01, Kalamboli, Navi Mumbai

(2025-2026)



DEPARTMENT OF INFORMATION TECHNOLOGY

PRACTICAL BOOK

ON

ROBOTIC PROCESS AUTOMATION

Submitted To

UNIVERSITY OF MUMBAI

BY

MAHESH TANAJI CHAVAN

Seat No: -1314518



MASTER OF SCIENCE INFORMATION TECHNOLOGY

PART – II (SEMESTER - IV)

(2025-2026)

KLE SOCIETY'S COLLEGE OF SCIENCE & COMMERCE

NAAC Accredited Grade B+

(Affiliated to University of Mumbai)

Plot No. 29, Sector No. 01, Kalamboli, Navi Mumbai

(2025-2026)



CERTIFICATE

This is to certify that

Mr/Ms. _____

of Class _____, Semester _____

has successfully completed the practical for the subject _____

in M.Sc. (Information Technology) during the academic year 2025- 2026 as per the syllabus prescribed by the University of Mumbai.

Prof. In charge

Date: _____

Head of the Department

Date: _____

External Examiner

Date: _____

Principal Sign

Date: _____

College Stamp

INDEX

Sr.No	Practicals	Date	Sign
1.	RPA Basics: Sequences and Flowcharts: - Create a simple sequence-based project. - Create a flowchart-based project. - Automate UiPath Number Calculation (Subtraction, Multiplication, Division of numbers). - Create an automation UiPath project using different types of variables (number, datetime, Boolean, generic, array, data table)		
2.	Decision making and looping: - Consider an array of names. We have to find out how many of them start with the letter "a". Create an automation where the number of names starting with "a" is counted and the result is displayed. - Demonstrate switch statement with an example. - Create an automation using DoWhile . - Create an automation to demonstrate use of decision statements (if, for each)		
3.	Types of Recording: - Basic Recording using ToolBar. - Basic Recording using Notepad. - Desktop Recording using Tool bar. - Web Recording manually		
4.	Excel Automation: - Automate the process to extract data from an excel file into a data table and vice versa - Create an automation to Write data to specific cell of an excel sheet. - Create an automation to Read data to specific cell of an excel sheet.. - Create an automation to append data to specific cell of an excel sheet		
5.	Different controls in UiPath: - Implement the attach window activity. - Automate using Anchor Base. - Automate using Element Exists. - Automate using Find Children control.		

6.	Keyboard and Mouse Events: a) Demonstrate the following activities in UiPath: i. Mouse (click, double click and hover) ii. Type into iii. Type Secure text b) Demonstrate the following events in UiPath: i. Element triggering event ii. Image triggering event iii. System Triggering Event c) Automate the process of launching an assistant bot on a keyboard event.		
7.	Screen Scraping and Web Scraping methods: a) Automate the following screen scraping methods using UiPath: i. Full Text ii. Native iii. OCR.		
8.	Exception Handling and Plugins a) Demonstrate exception Handling using UiPath. b) Install and automate any process using UiPath with the following plug-ins: i. Java Plugin ii. Mail Plugin iii. PDF Plugin iv. Excel Plugin v. Word Plugin vi. Credential Management		
9.	a) Automate the process of send Email. b) Automate the process of logging and taking screenshots in UiPath. c) Automate any process using State Machine in UiPath. d) Demonstrate the use of publish utility. .		
10.	Orchestrator management and mini project a) Deploy bots to Orchestrator b) Run jobs from Orchestrator c) Queue Introduction i. Add items to Queue. ii. Get Queue item from Orchestrator Build UiPath Chatbot using Google dialogflow		

Practical No.1

1. a. Create a simple sequence-based project.

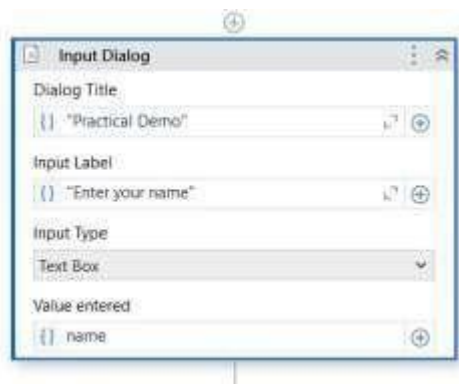
A Sequence is a group of logical steps. Each step represents an action or a piece of work. A Sequence is used for processes that happen in linear succession, that is, one after the other. Among the three types of projects in UiPath, Sequences are the smallest. An illustration of how to build a Sequence in UiPath Studio is presented in the following section.

In the following example, we will make a simple project that asks for the name of the user and then displays his or her response:

Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel.



Step 2: Search for Input dialog in the Search panel of the Activities panel. Drag and drop the Input dialog activity inside the Sequence (the Input dialog activity is a dialog box that appears with a message or a question, in response to which the user is required to put in his or her reply) Write the appropriate message on the Label of this Input dialog to ask for the user's name. for e.g. Enter your name.



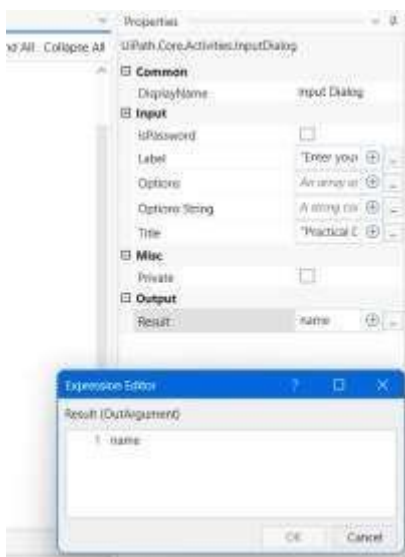
Step 3: Drag and drop a Message box activity into the Sequence. (A Message box, as the name suggests, displays a given text. In this case, we will use it to display the text/reply that the user has given in the Input dialog box on being asked his/her name.) Connect Input Dialog Activity to the Message Box Activity.



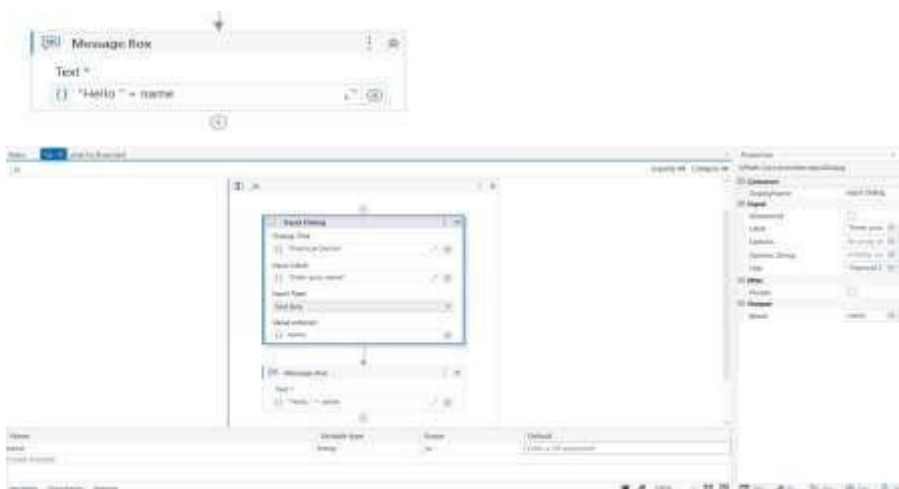
Step 4: Next, create a variable and give it the desired name. This variable will receive the text that the user has entered in the Input dialog box in response to our question, that is, the user's name.



Step 5: We now have to specify the Result property (in the Properties panel) of the Input dialog box. On specifying the variable name there, it will receive the text that the user entered. Click on the dotted icon that appears on the right side of the Result property. Now, specify the variable name:

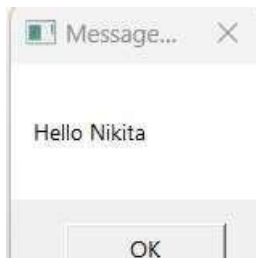


Step 6: Specify the variable name that we have created in the Text area of the Message box (the Text area of the Message box is used to input text that will be displayed in the Message box).



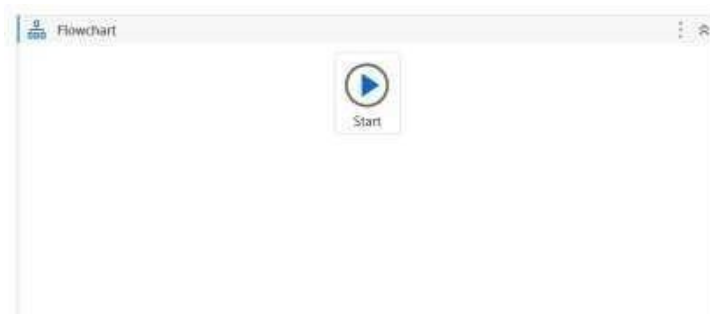
Step 7: Hit the Run button and see the result.

OUTPUT



1. b. Create a flowchart-based project

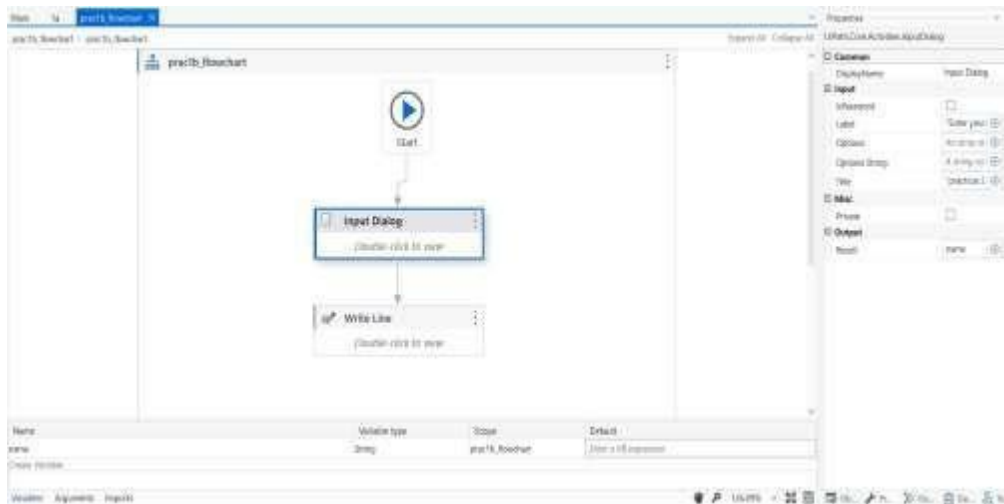
Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Flowchart activity from the Activities panel.



Step 2: Search for Input dialog in the Search panel of the Activities panel. Drag and drop the Input dialog activity inside the Sequence. Write the appropriate message on the Label of this Input dialog to ask for the user's name. for e.g. Enter your name.

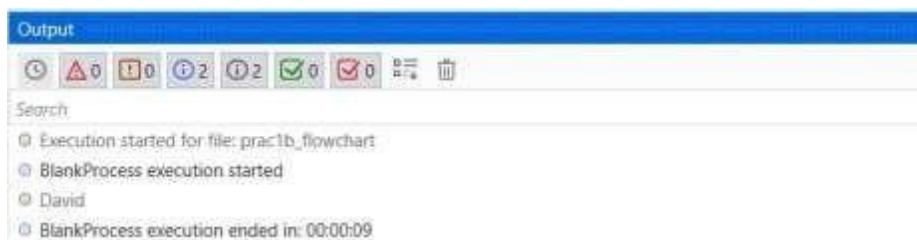
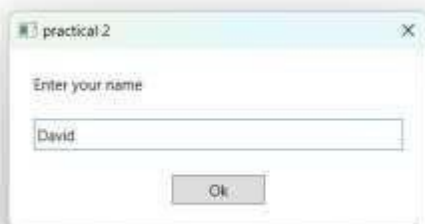


Step 3: Drag and drop a Write Line activity into the designer panel. (A Write Line, as the name suggests, displays a given text into a output panel. In this case, we will use it to display the text/reply that the user has given in the Input dialog box on being asked his/her name.) Connect Input Dialog Activity to the Write Line Activity.



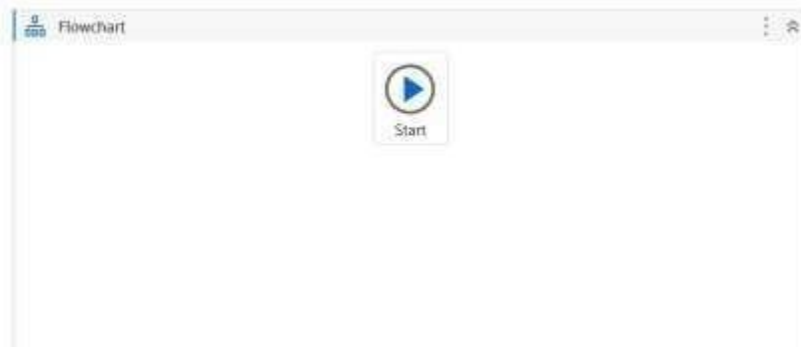
Step 3: Hit the Run button and see the result.

OUTPUT:

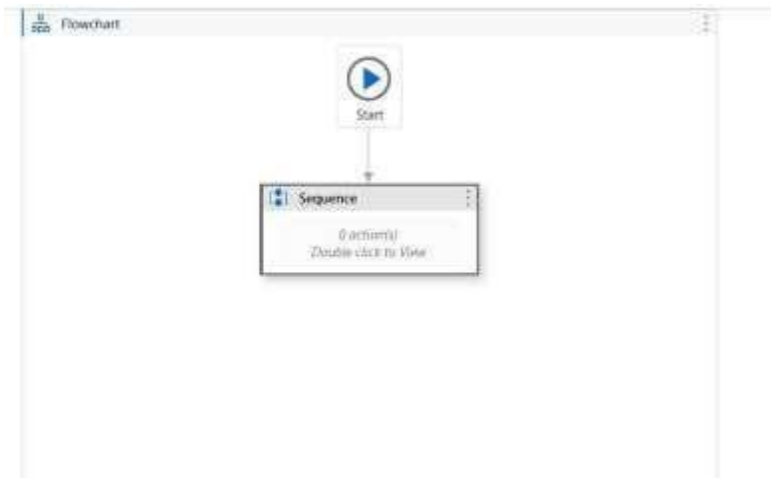


Example 2, To enter two numbers and check whether their sum is less than 20. Perform the following steps:

Step 1: First, add a Flowchart from the Activities panel into the Designer panel.



Step 2: Add a Sequence activity within the Flowchart.



Step 3: Take two Input dialog activities (for entering the numbers to be added) inside the Sequence activity



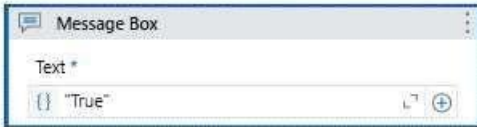
Step 4: Create the variables x and y to save the values.

Name	Variable type	Scope	Default
x	int32	mathematical_flowchart	Enter a VB expression
y	int32	mathematical_flowchart	Enter a VB expression

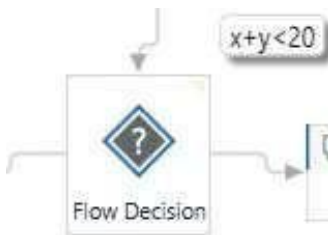
Create Variable

Variables Arguments Imports

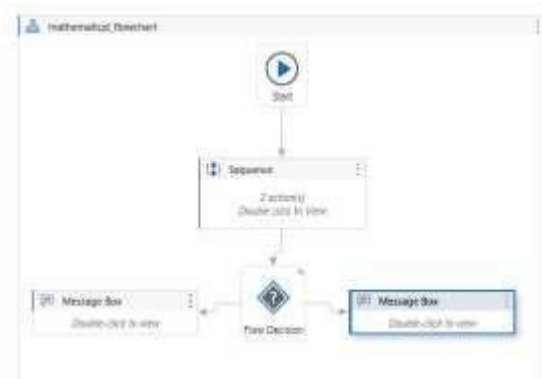
Step 5: Next, add a Message box activity to perform a mathematical operation. In our case, the sum of the two numbers is less than 20: $x + y < 20$



Step 6: Now, add a Flow Decision activity to check the mathematical operation.



Step 7: If true, the Flow Decision will flow toward the true branch. Otherwise, it will flow towards the false branch.





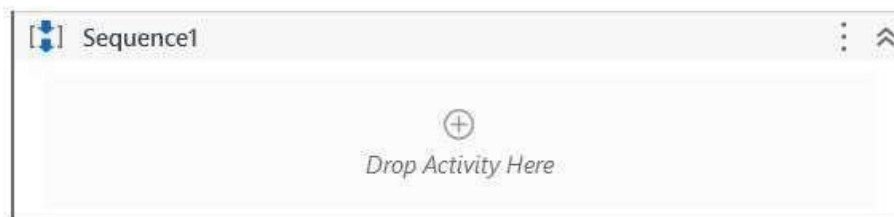
Step 7: Hit the Run button and see the result.

Output:



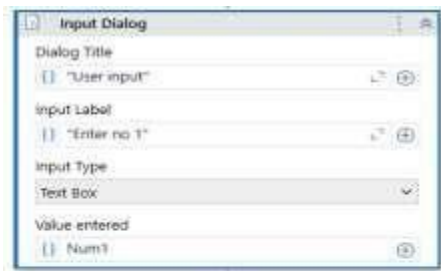
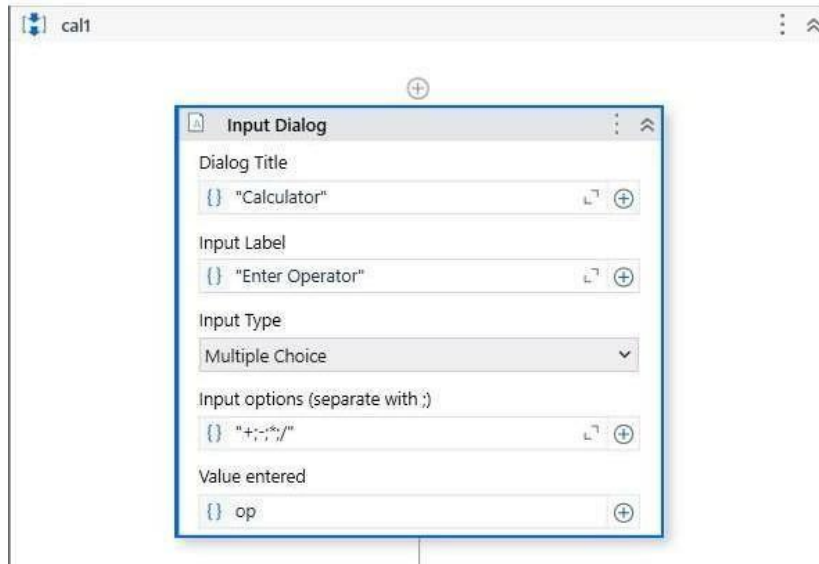
1. c. Automate UiPath Number Calculation (Subtraction, Multiplication, Division of numbers).

Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel



Step 2: Search for Input dialog in the Search panel of the Activities panel.

Drag and drop three Input dialog activity inside the Designer panel of the Sequence. Write the appropriate message on the Label of the Input dialog to ask for the user's input. For first Input dialog to select one operator from multiple option in input dialog box, select input type as Multiple choice and Write Input Options as "+;-;*;/". Next, create a variable and give it the desired name. This variable will receive the text that the user has entered in the Input dialog box in response to our question, that is, the operator, num1 & num2. Specify these variables in value entered field of Input dialog.

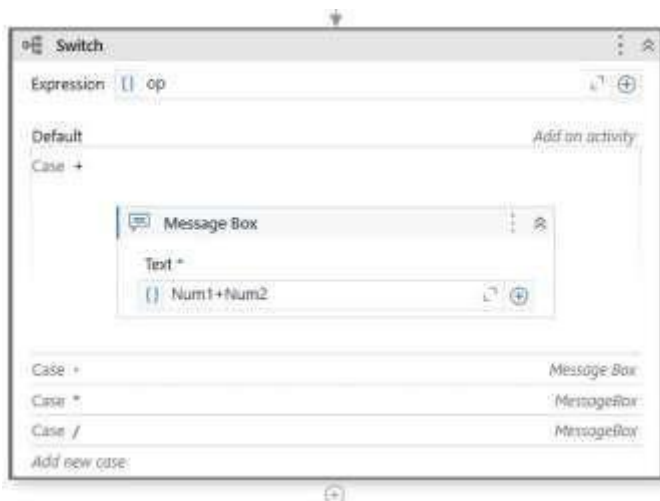


Name	Variable type	Scope	Default
op	String	cal1	Enter a VB expression
Num1	Int32	cal1	Enter a VB expression
Num2	Int32	cal1	Enter a VB expression

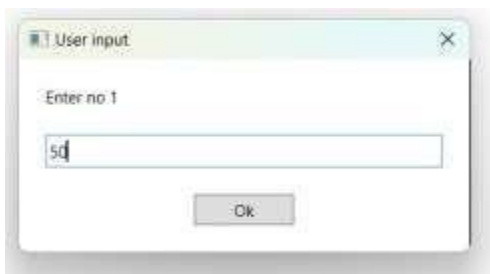
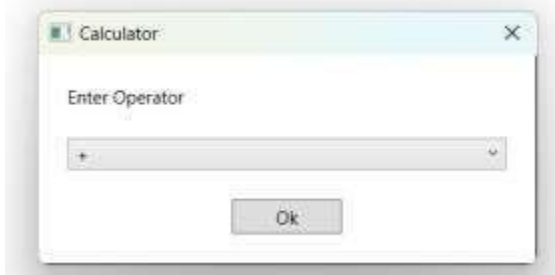
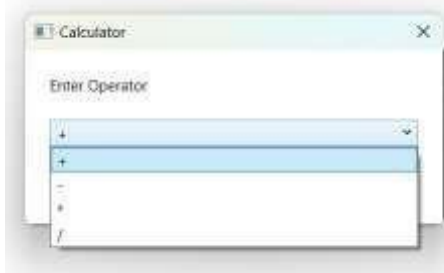
Step 4: Drag and drop Switch activity from Activity panel into the designer panel.

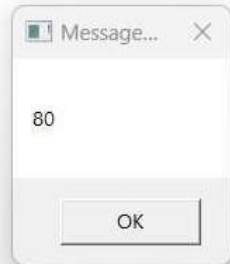
- Specify variable name in Expression field of the switch activity

- ii. Add four case for four different operators i.e, Num1 +Num2 ,
Num1 - Num2, Num1 * Num2 and Num1 / Num2



Step 5: Hit the Run button and see the result





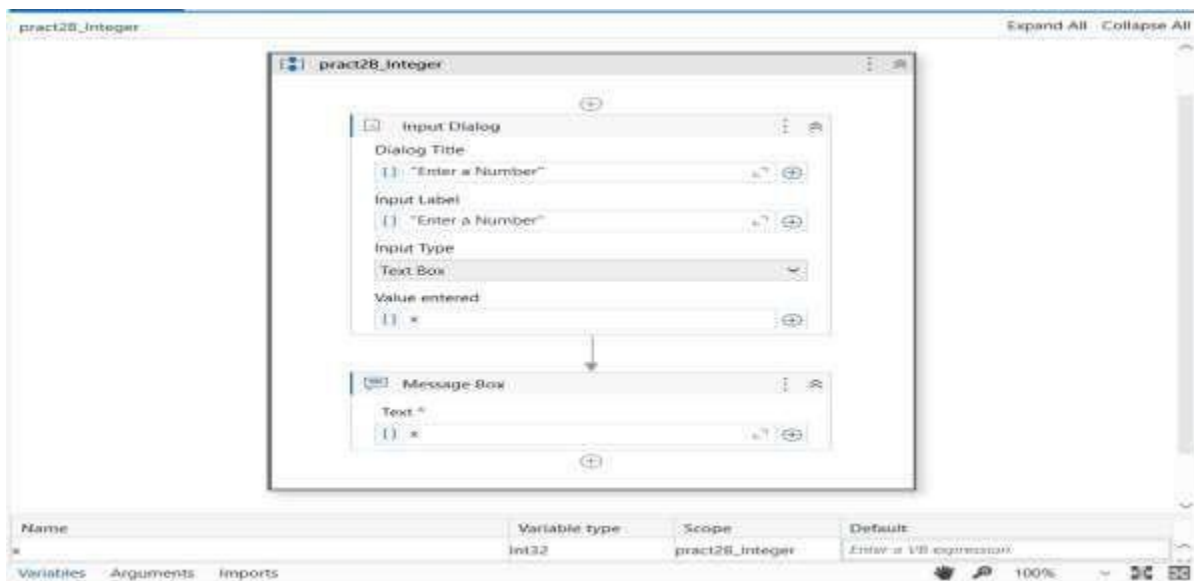
1.d. Create an automation UiPath project using different types of variables (number, datetime, Boolean, generic, array, data table)

1] Number

a) Integer

Step 1: Drag and drop input Dialog box and create variable x with integer datatype.

Step 2: Drop a message box and enter x display entered value



Output: -



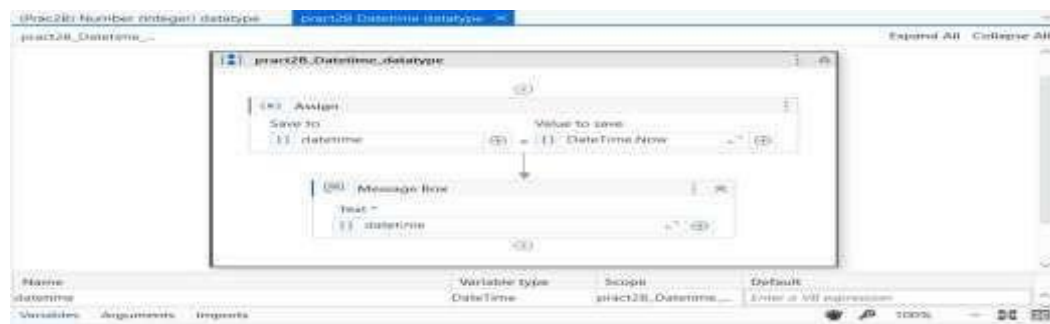
2] Datetime

Step 1: Drag and drop Assign activity.

Step 2: Create a variable datetime with DateTime datatype.

Step 3: Save DateTime.Now in datetime variable

Step 4: Display with message box.



OUTPUT:



3] Generic

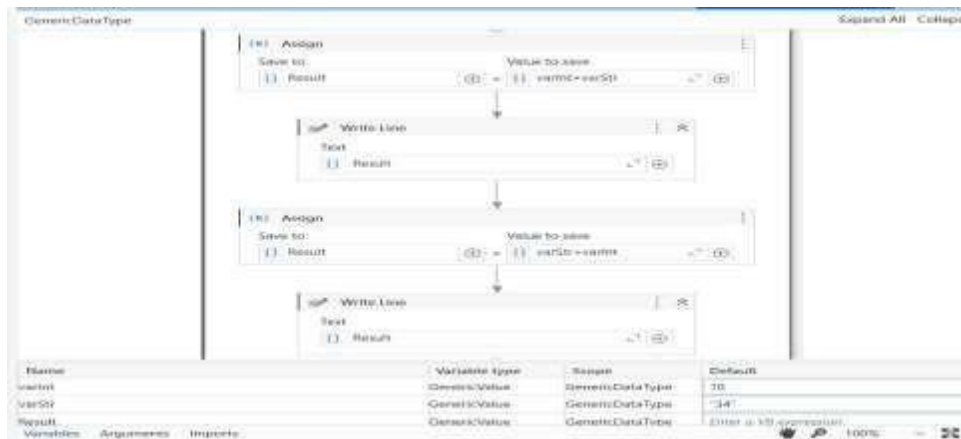
Step 1: Create a new Sequence.

Step 2:- Create 3 variables with GenericValue datatype i.e., varInt as 10 integer as default value, varStr as “34” String as default value, and Result.

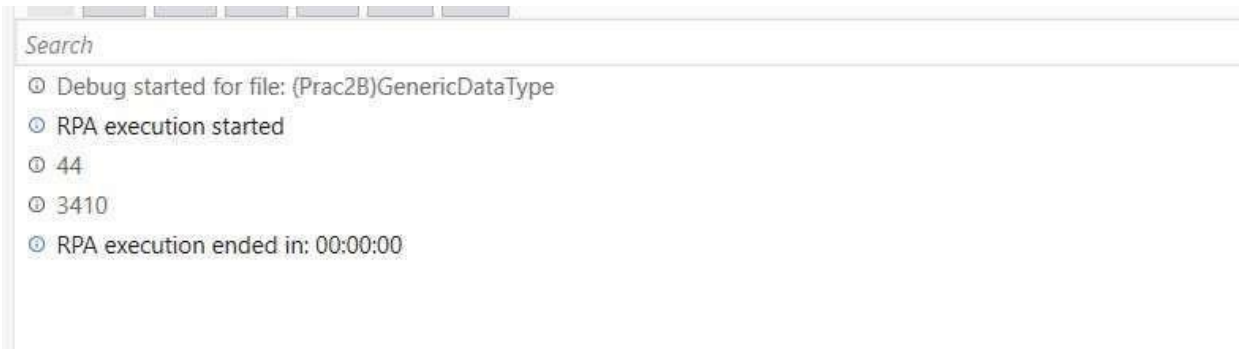
Step 3:- Assign value to Result in 2 cases

Case 1: varInt+varStr

Case 2: varStr+varInt



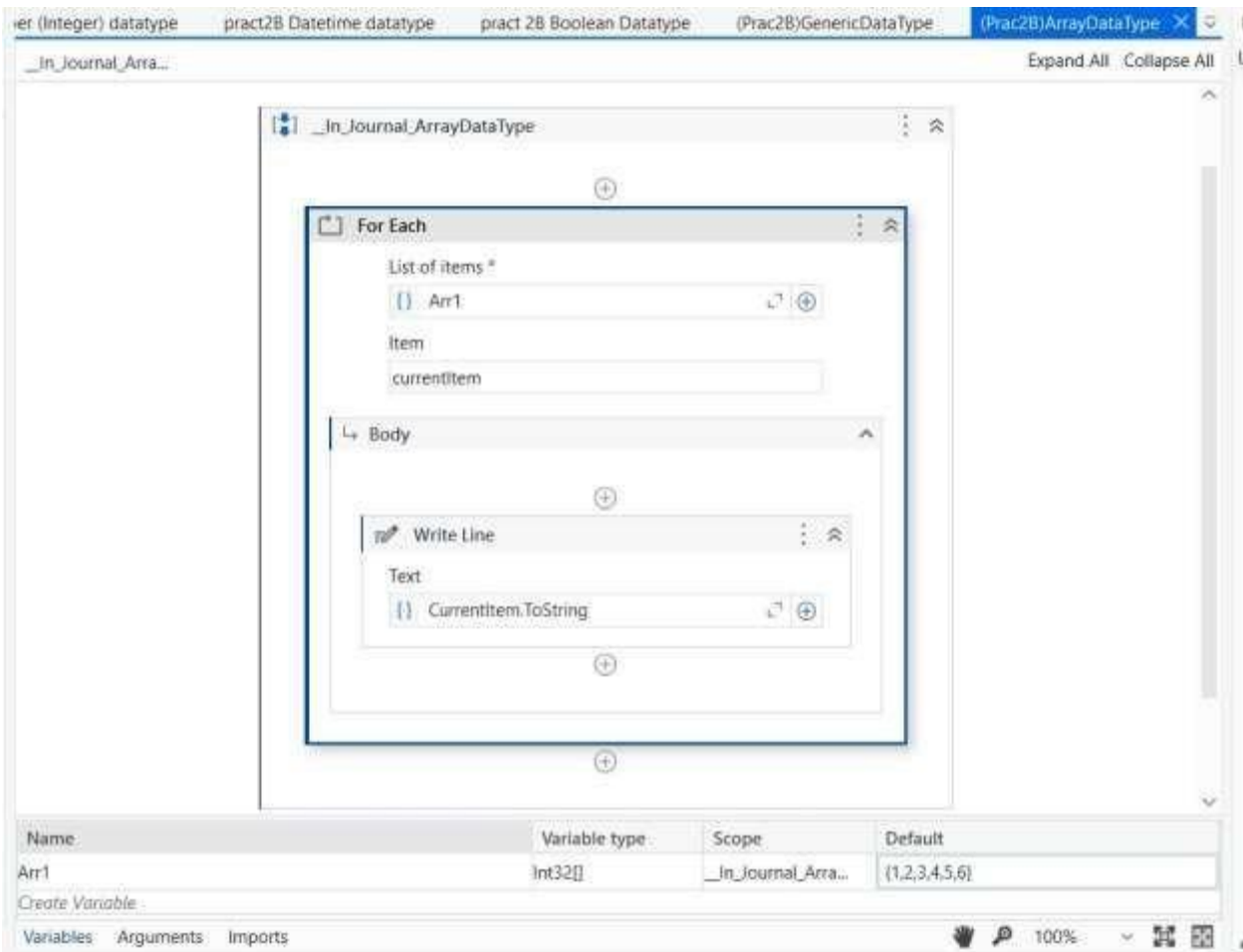
OUTPUT:



4] Array

Step 1: Create an integer Array (Arr1) and store {1,2,3,4,5,6}

Step 2: Drag and drop For Each activity. Give created Arr1 as input. In Body, drop Write Line activity and write CurrentItem.ToString to convert array into String for displaying purpose.



OUTPUT:

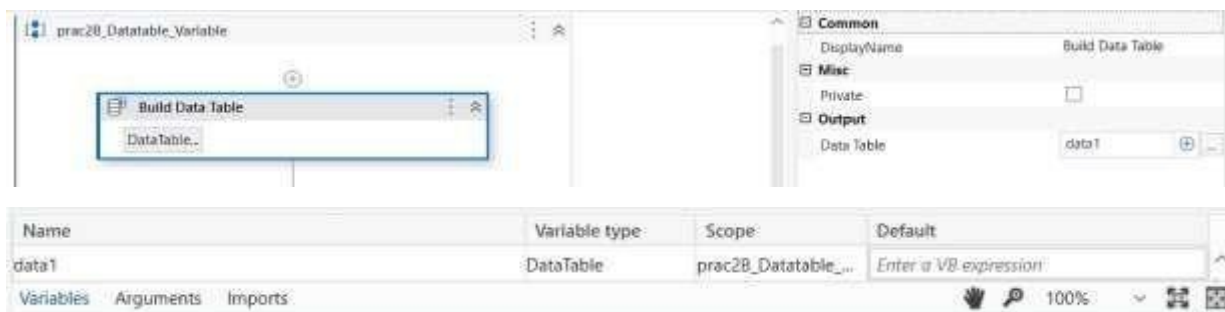
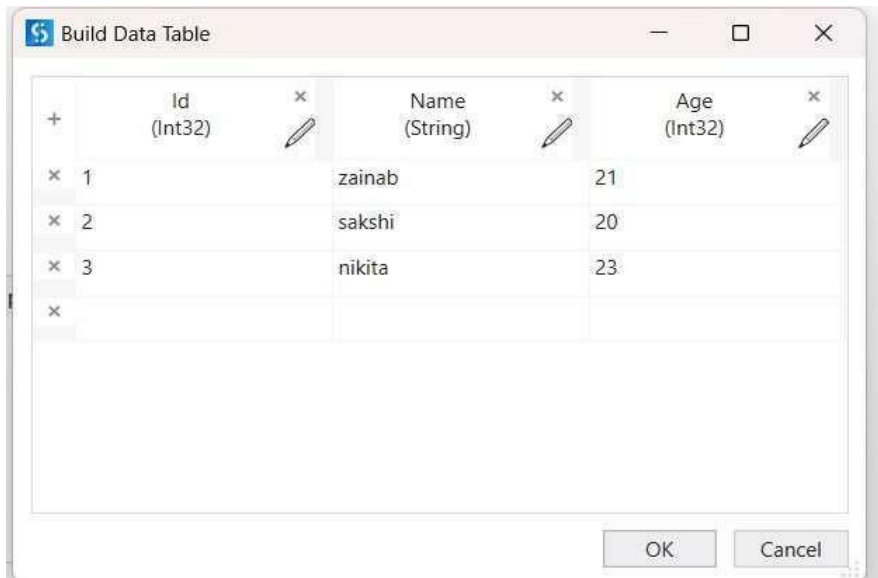
```

① Debug started for file: (Prac2B)ArrayDataType
① RPA execution started
① 1
① 2
① 3
① 4
① 5
① 6
① RPA execution ended in: 00:00:00

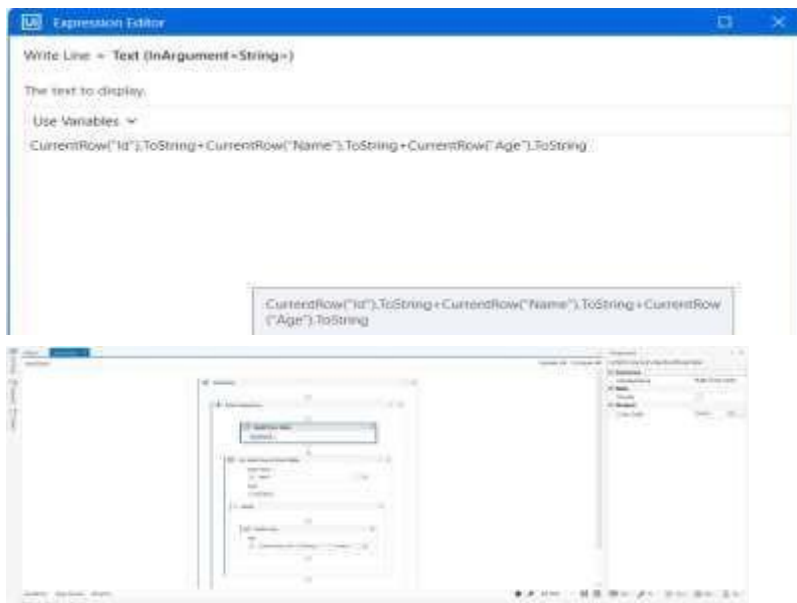
```

5] DataTable

Step 1: Drag and drop Build Data Table activity, create a table as and store it in variable (data1) with DataTable datatype.

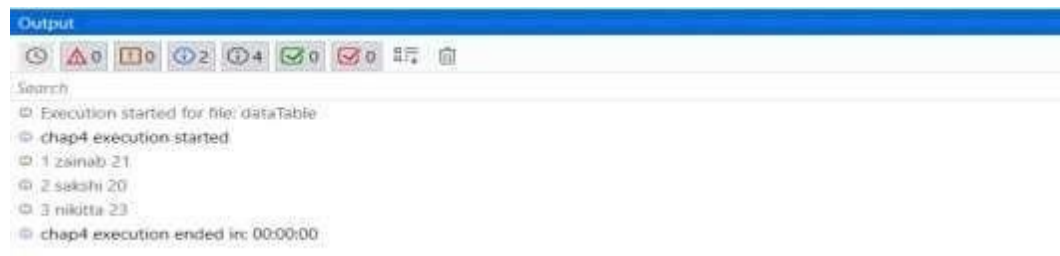


Step 2: Drop For Each Row in Data Table activity. Give data1 as input In Body, drop Write line to display output



Step 3: Run.

OUTPUT:



Practical No.2

2. a. Consider an array of names. We have to find out how many of them start with the letter "a". Create an automation where the number of names starting with "a" is counted and the result is displayed.

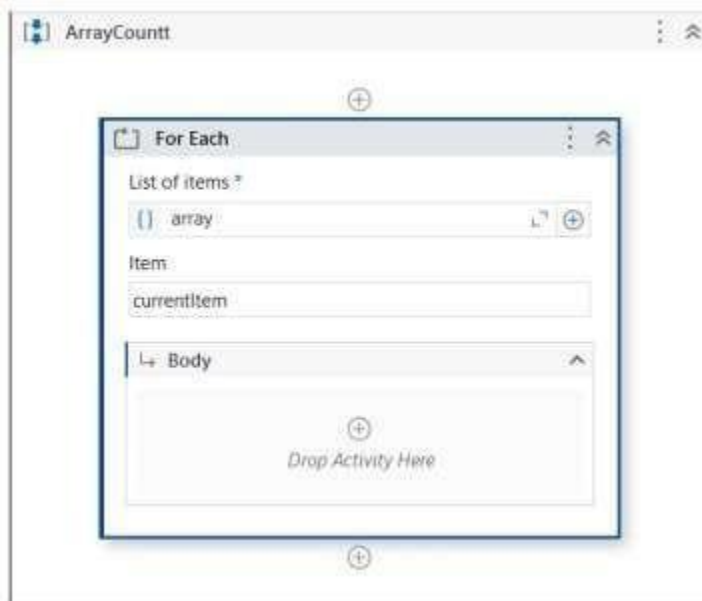
Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel.



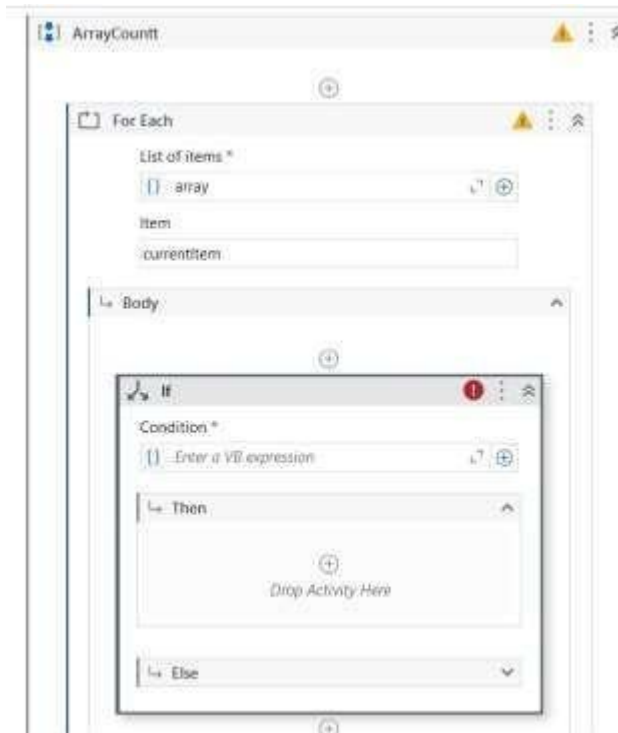
Step 2: Create a variable. Give it a name (in our case, we will create an array of type string and name the variable as array). Set the variable type to Array of [T]. When asked for the type of array, select String. Also, initialize the array in the Default section of the variable by giving it a default values. For example, {"zainab","vignesh","shruti","divyanshu"} .Create a variable of type integer Count for storing the result. Set the variable type to Int32.

Name	Variable type	Scope	Default
array	String()	Array	{"zainab","vignesh","shruti","divyanshu"}
count	Int32	Array	0
Create Variable			
Variables Arguments Imports			

Step 3: Drag and drop a For each activity inside the Sequence. Also, specify the array name in the expression box of the For each activity. The For each activity is used to iterate over the array. It will pick up one name from the array each time until it reaches the end:



Step 4: Drag and drop the If activity from the Activities panel and place it inside the For each activity at the location where Drop activity here is mentioned. Specify the condition in the expression box of the If activity. The If activity is used to check for a particular condition/expression. If that expression is satisfied, the Then block will be executed. Otherwise, the Else block will be executed.

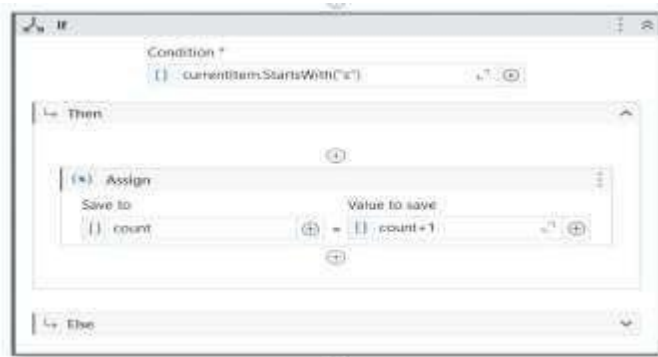


Step 5: Specify the expression as `currentItem.StartsWith("s")`. This expression specifies the name present in the item variable starts with the letter s. The For each activity iterates over the array, picks up one name at a time, and stores it as a variable, `currentItem`:



Step 6: Now, we are going to use the count variable and increment it each time a name from an array starts with the letter s. For this, we have to use the Assign activity. Drag and drop

the Assign activity inside the If activity. Set the To property to count (variable name) and the Value property to count+1 (to increment its value) of the Assign activity:



Step 7: Just drag and drop a Message box activity inside the Sequence activity. Specify the count variable in the expression box of the Message box activity. But remember, the variable that we have created is of type Int32, so, it cannot be used with the Message box activity without converting it to a string. To convert it to a string, we have the '.toString' method available in UiPath Studio. Just apply it to the variable and select '.toString':



Step 8: Hit the Run button or press F5 and see the result

Output: -

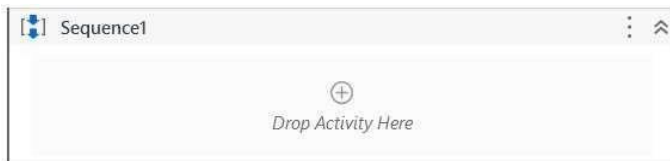


2. b. Demonstrate switch statement with an example.

The Switch activity can be used to make a choice. When we have various options available and want to execute one option, we frequently use the Switch activity. The Switch activity is very useful in the categorization of data according to one's own choice.

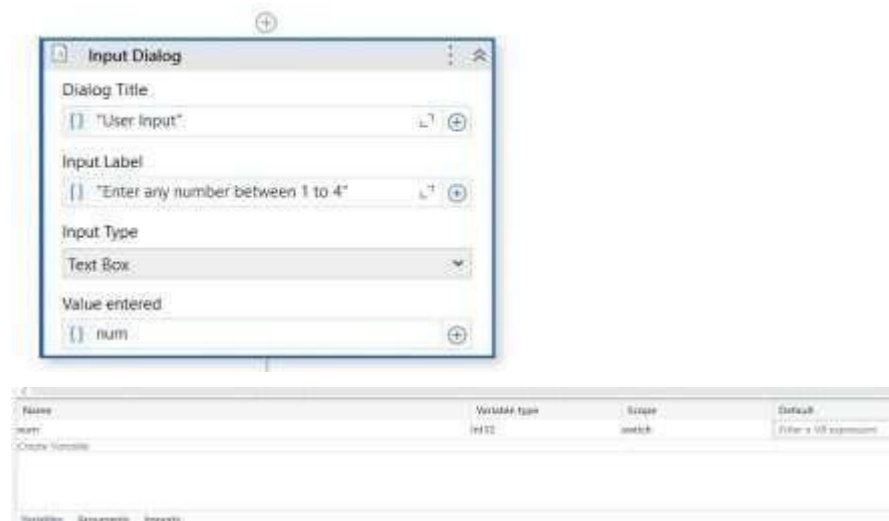
Example:

Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel.

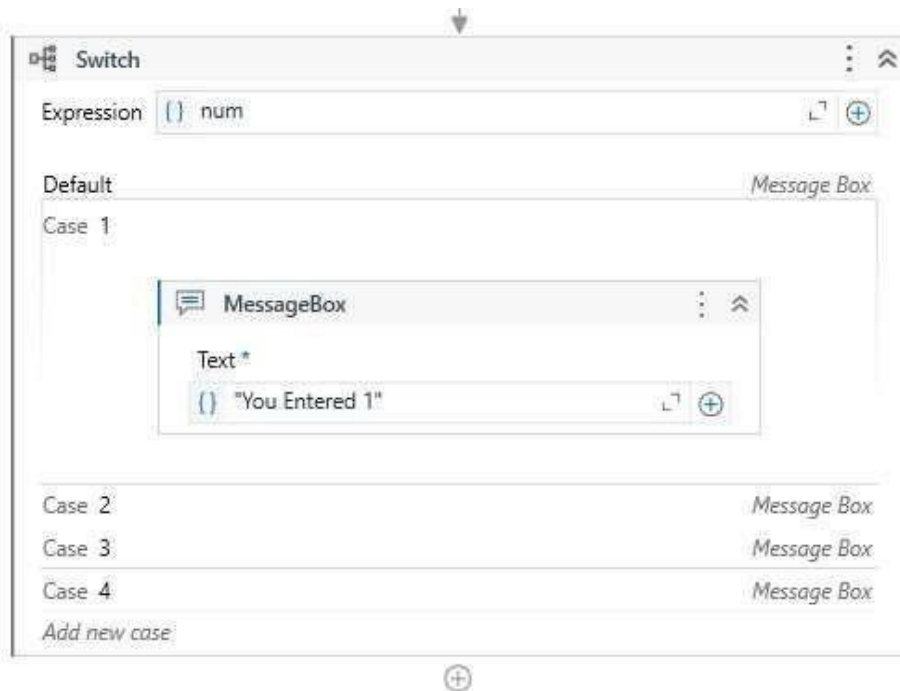


Step 2: Search for Input dialog in the Search panel of the Activities panel.

- i. Drag and drop Input dialog activity inside the Designer panel of the Sequence.
- ii. Write the appropriate message on the Label of the Input dialog to ask for the user's input. Select input type as Text Box
- iii. Next, create a variable and give it the desired name. This variable will receive the text that the user has entered in the Input dialog box in response to our question, that is, the operator, num1 & num2.
- iv. Specify these variables in value entered field of Input dialog.



Step 3: Drag and drop Switch activity from Activity panel into the designer panel. Specify variable name in Expression field of the switch activity and Add four case.



Step 4: Click the Run button to check the output.

Output: -



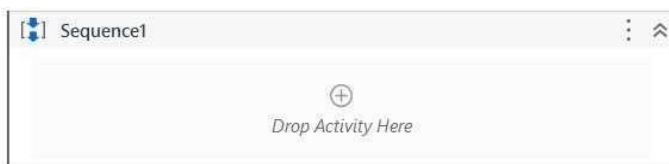
2. c. Create an automation using DoWhile Activity.

The Do while activity is used in automation when it is required to execute a statement based on the fulfillment of a certain condition. How it differs from the While activity is that it executes a statement, then checks whether the condition is fulfilled. If the condition is not fulfilled, it exits the loop.

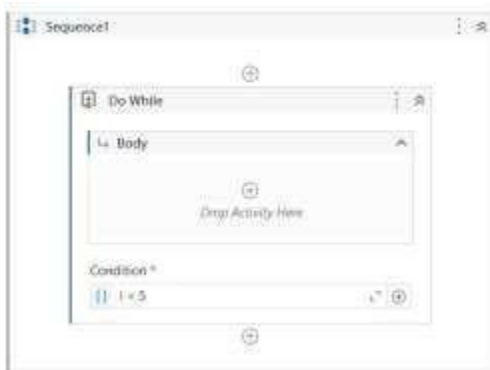
Example:

Perform the following steps to generate all multiples of 2, less than 20 :

Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel.



Step 2: Drag and drop Do While activity from the activity panel to the designer panel. Create new integer variable x and set its default value to 0.

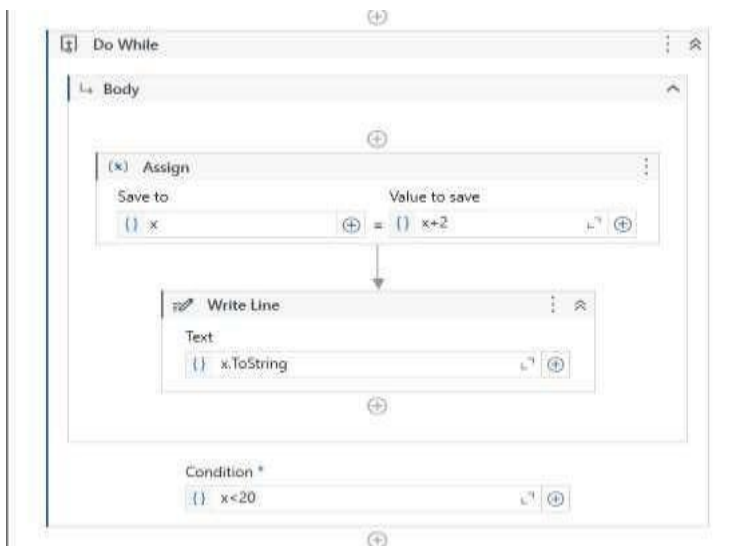


Name	Variable type	Scope	Default
x	int32	DoWhileActivity	Create a V1 expression
Create variable			
Variables Arguments Imports			

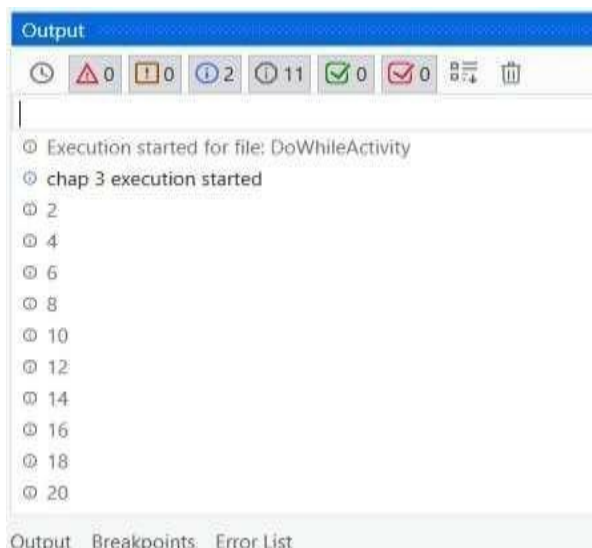
Step 3: In the body section of the Do while activity, add an Assign activity. Now, select the Assign activity. Set x in the value section of the Assign activity to increment the result each time by 2 until the loop is executed.

Step 4: Add a Write line activity inside the Assign activity. In the text field of the Write line activity, type x.ToString.

Step 5: In the condition section, set the condition x<20. The loop will continue until the condition holds true:



Step 6: On clicking the Run button, the output displayed will be as follows **Output:-**



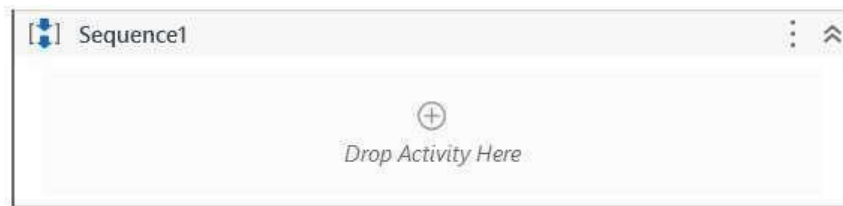
2. d. Create an automation to demonstrate use of decision statements.

1]The If activity

The If activity consists of a statement with two conditions: true or false. If the statement is true, then the first condition is executed; if not, the second condition is executed. This is useful when we have to take decisions on the basis of statements.

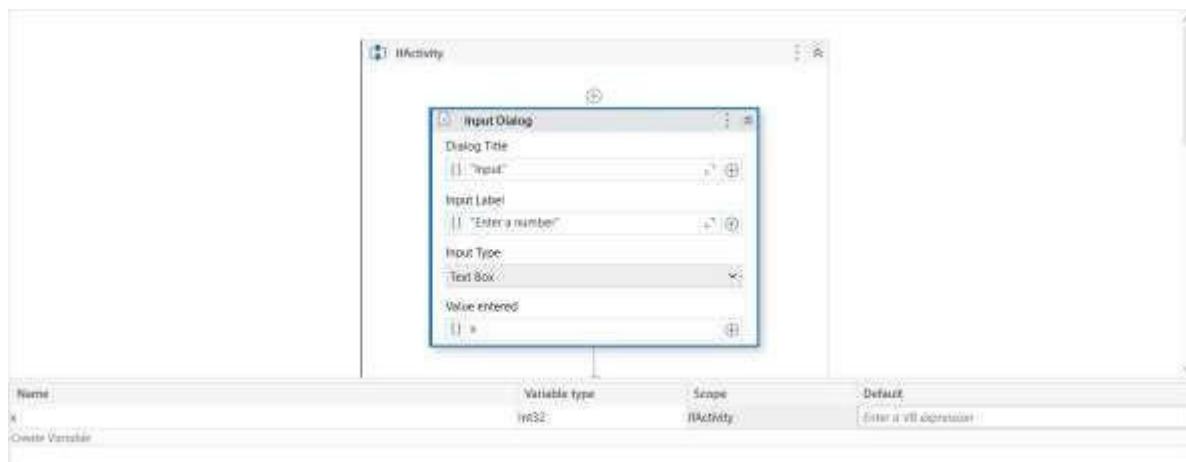
Perform the following steps to check whether the number is less than 5:

Step 1: Open UiPath Studio and click on Blank to start a fresh project. Give it a meaningful name. On the Designer panel, drag and drop a Sequence activity from the Activities panel.



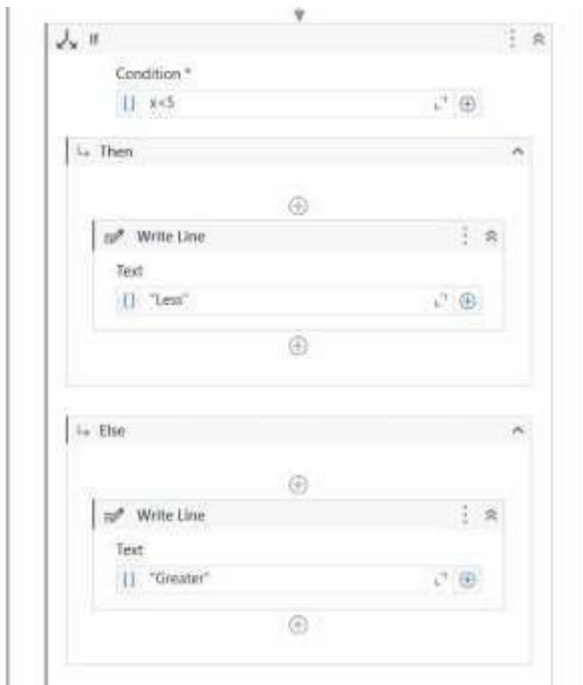
Step 2: Search for Input dialog in the Search panel of the Activities panel.

- i. Drag and drop Input dialog activity inside the Designer panel of the Sequence.
- ii. Write the appropriate message on the Label of the Input dialog to ask for the user's input.
- iii. Select input type as Text Box.
- iv. Next, create a variable and give it the desired name. This variable will receive the text that the user has entered in the Input dialog box in response to our question, that is, the operator, num1 & num2. Specify these variables in value entered field of Input dialog



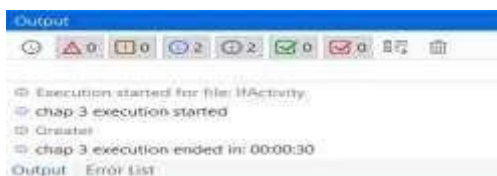
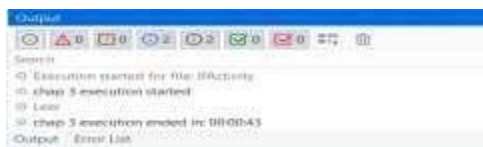
Step 3: Now add the If activity to the Designer panel:

Step 4: In the condition part, $x < 5$, check whether it is true or false. Add two Write line activities and type “Less” in one and “false” in the other



Step 7: Click the Run button to check the output. If the condition holds true then it will show the true value; otherwise, it will show the false value

Output: -

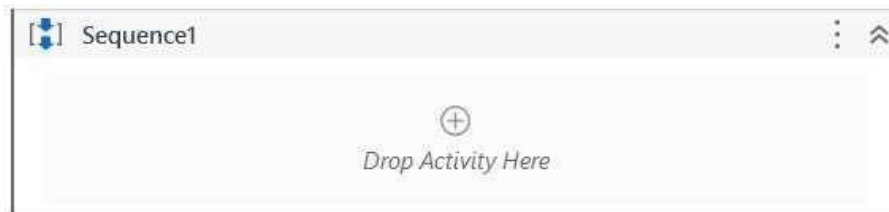


2] The For each activity

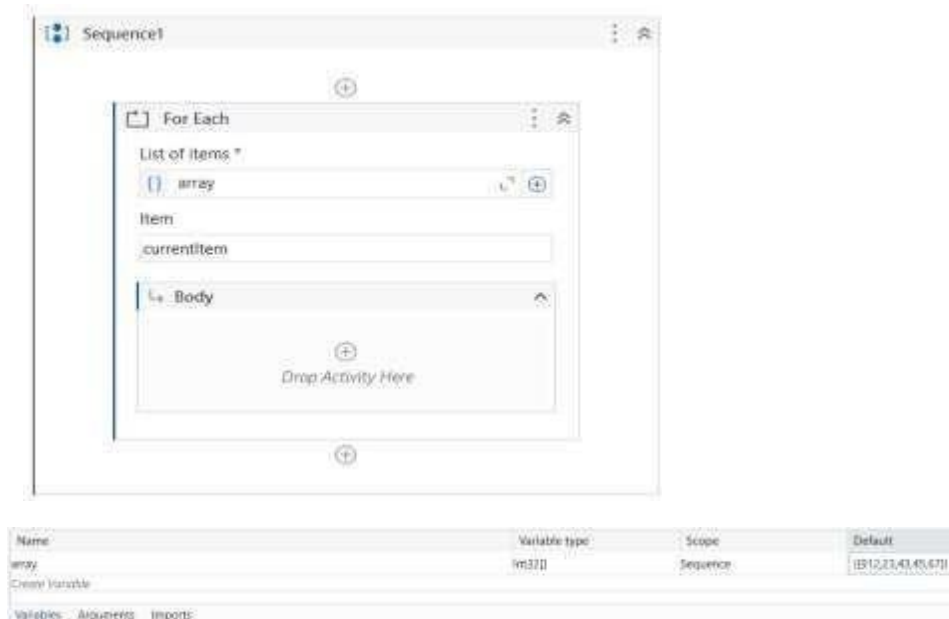
The For each activity works by iterating each element from the collection of items or list of elements, one at a time. In the process, it will execute all the actions that are available inside the body. Thus, it iterates through the data and processes each piece of information separately.

Example :To use the For each activity to go through a collection of even numbers and display each element one at a time. Perform the following steps:

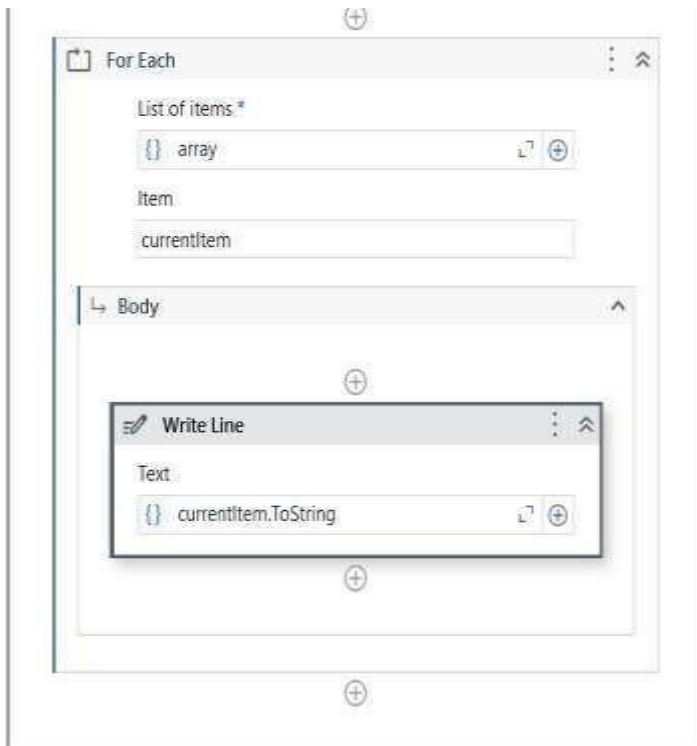
Step 1: Start with a Blank project in UiPath. Add a Sequence activity to the Designer panel.



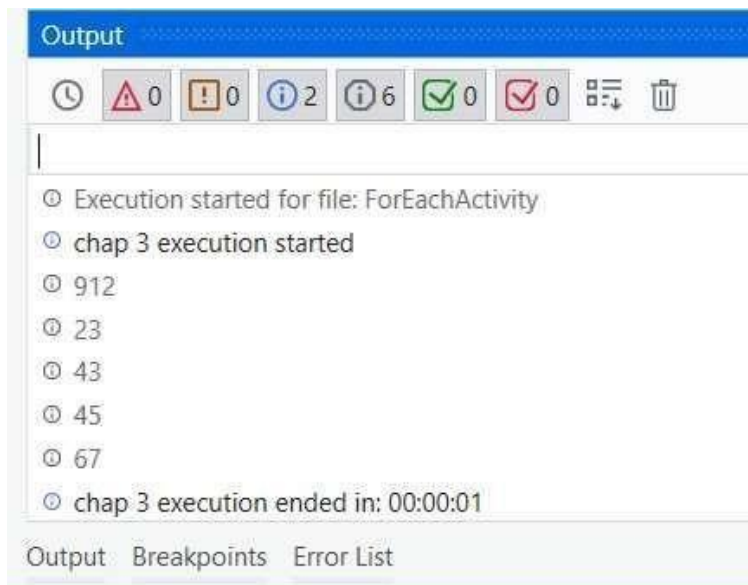
Step 2: Next, add a For each activity within the Sequence and create an integer type array variable, array. In the default value of the variable, put in ({912,23,43,45,67})



Step 3: Add a Write line activity to the Designer Panel (this activity is used to display the results). 6. In the Text field of the Write line activity, type `currentItem.ToString` to display the output.



Step 4: Now, run the program. You will see that each number of the array is displayed one by one because of the use of the For each activity:



Practical No.3

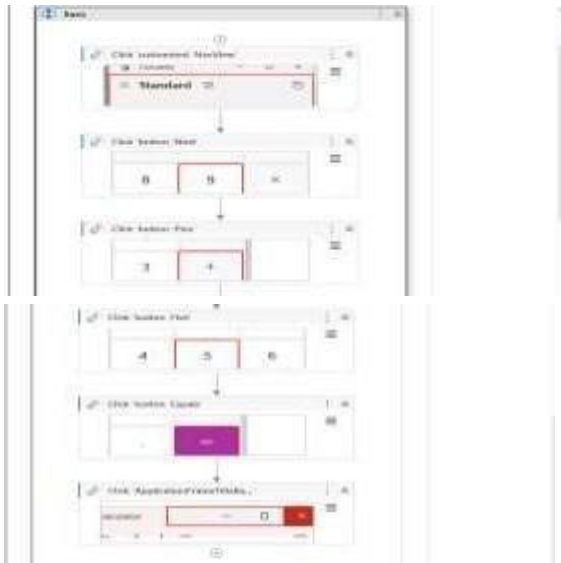
Types of Recording:

3. a. Basic Recording using Toolbar

Step 1: Open a Calculator.

Step 2: Create a new Sequence.

Step 3: Start Basic recording by clicking on recording from design ribbon. Indicate on Calculator and start performing actions. Press Esc. Save & Exit.



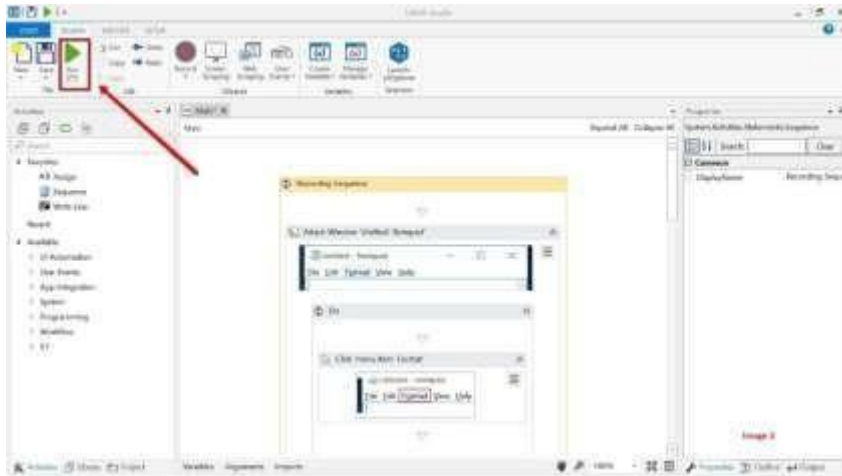
Step 4: Run



3. b. Basic Recording using Notepad

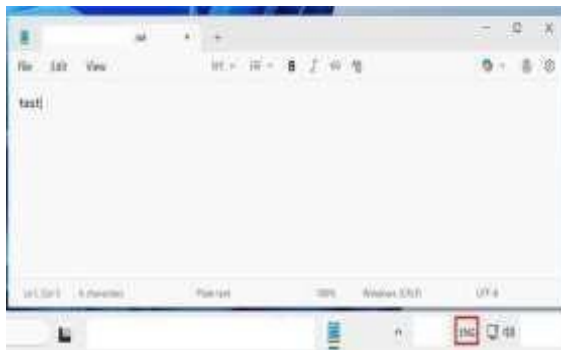
Step 1: Create a New Process

Open UiPath Studio and create a new Process project named **BasicRecording_Notepad**. From the Design tab, select **Recording** → **Basic**.



Step 2: Open Notepad and Start Recording

Launch Notepad and indicate the Notepad window in UiPath. Record the actions such as clicking inside the text area.

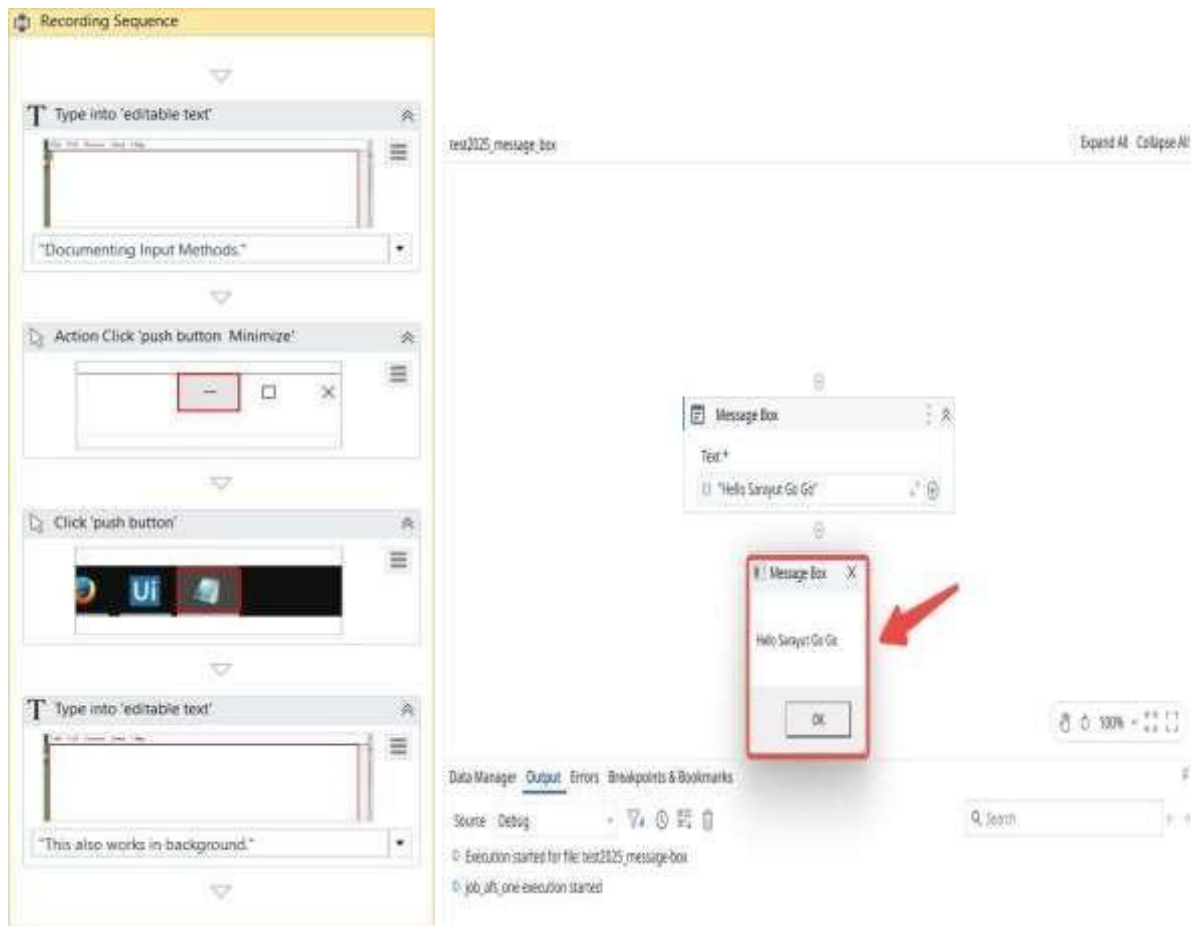


Step 3: Type Text in Notepad

Using the recording session, enter the required text, for example: Hello, this text is entered using UiPath Basic Recording.

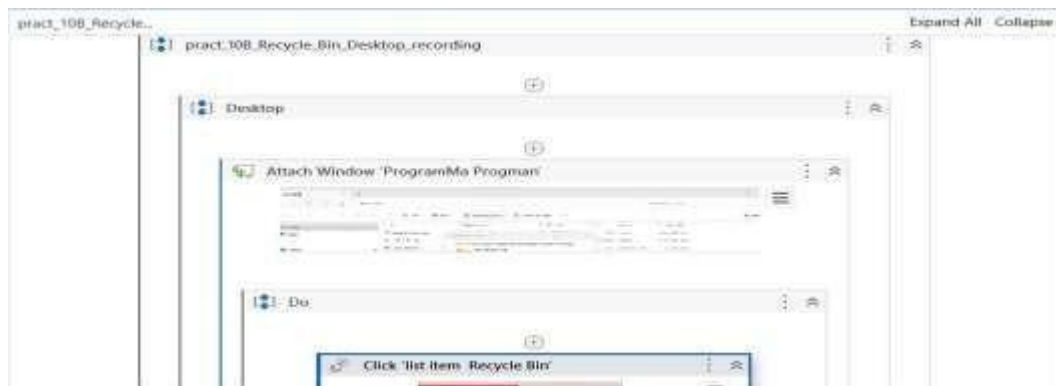
Step 4: Save and Run the Workflow

Stop the recording, save the generated workflow, and click **Run** to execute the automation.



3. c. Desktop Recording using Tool bar

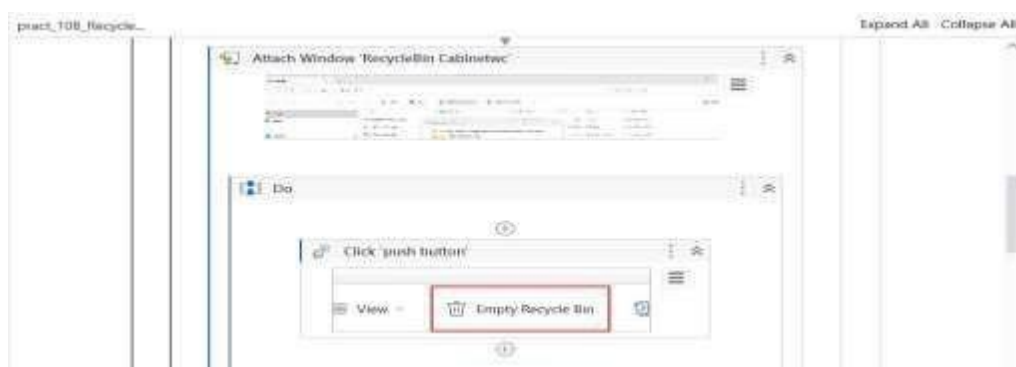
Step 1: Click on Desktop Recording



Step 2: Indicate on recycle bin and press Enter.

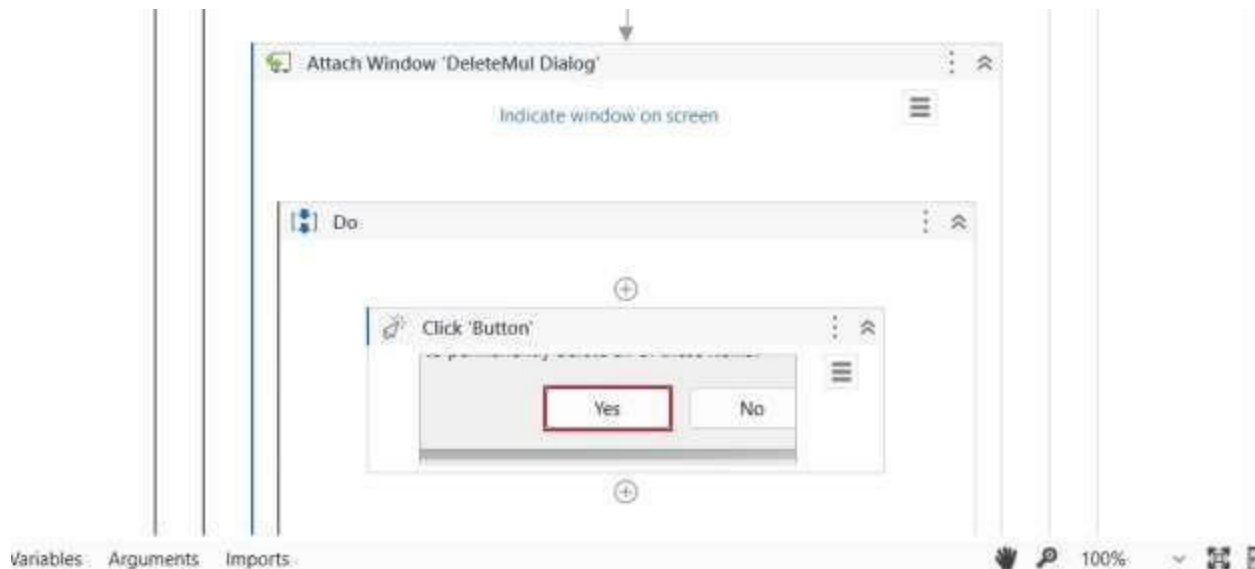


Step 3: Click on Empty Recycle Bin



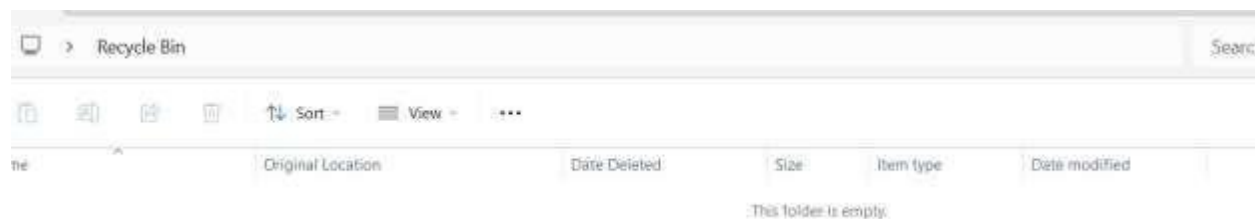
Step 4: It will ask for confirmation. Click on yes. Press Esc.

Save & Exit.



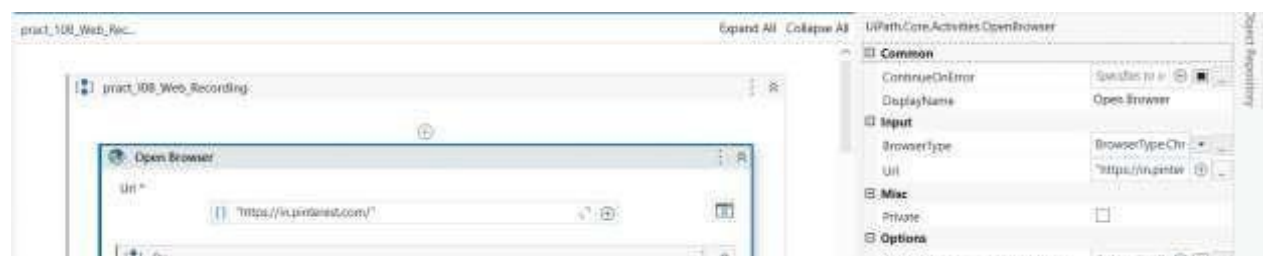
Step 5: Run.

Output: -

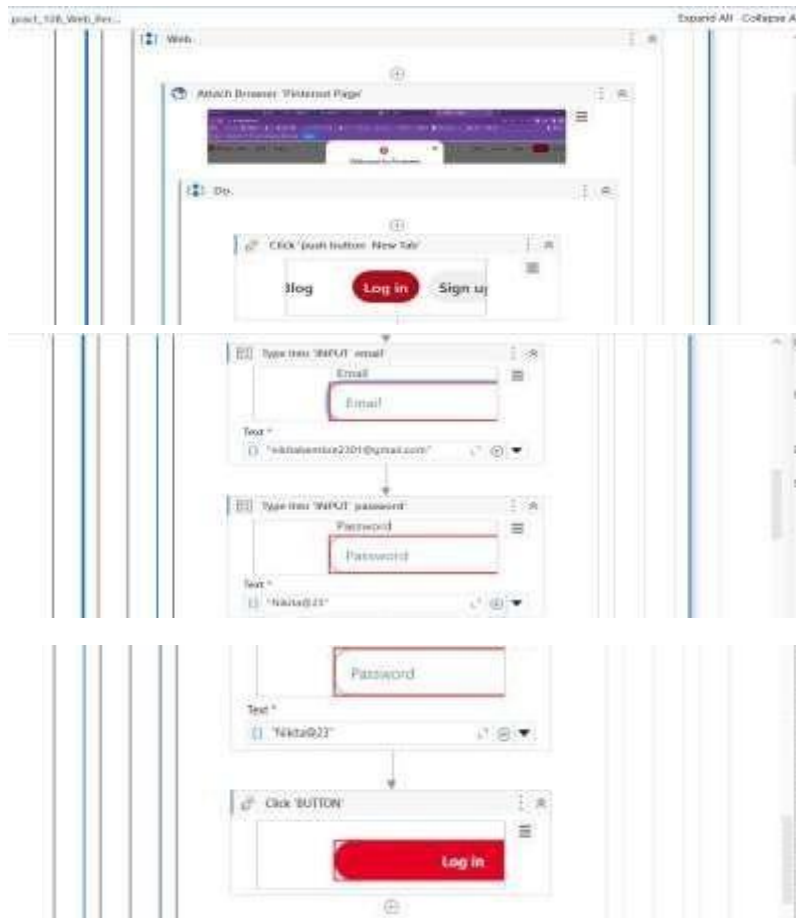


3. d. Web Recording manually

Step 1: Drag and drop Open Browser activity, Enter <https://in.pinterest.com/> and select Browser type as Chrome.

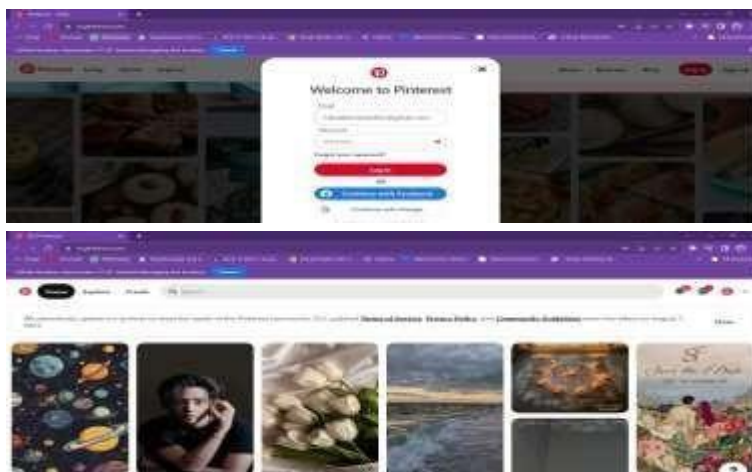


Step 2: Select Web Recording from Design Ribbon. Start recording. Click on Login. select Email field enter your email id then select on password enter your password and then click on log in. Press esc. Save & Exit.



Step 3: Run.

Output:-



Practical No.4

Excel Automation:

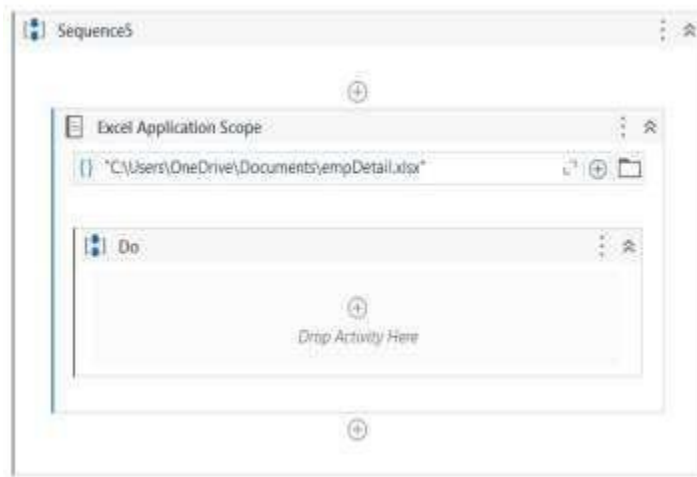
4. a. Automate the process to extract data from an excel file into a data table and vice versa.

To extract data from an Excel file into a data table and vice versa. We will achieve this by:

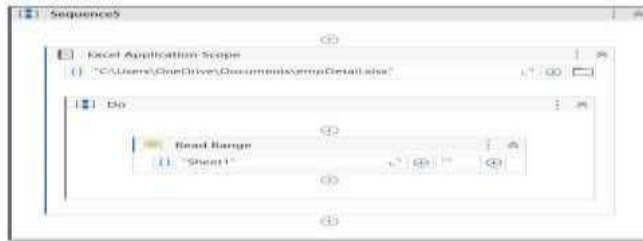
- Reading an Excel file and creating a data table using data from the Excel file
- Creating a data table and then writing all its data to an Excel file

1. Reading an Excel file and creating a data table using data from the Excel file We have an existing Excel file and we are going to use it in our project:

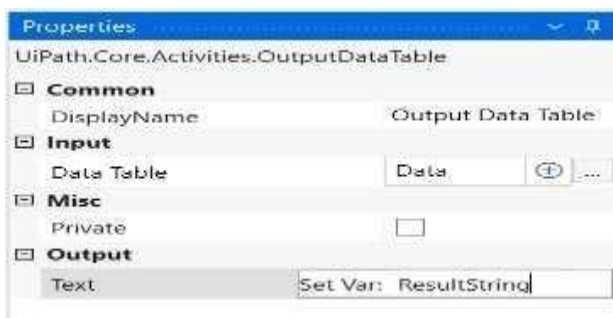
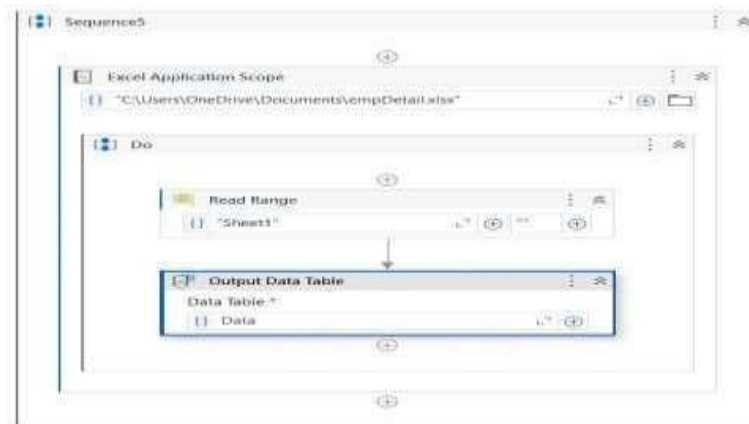
Step 1: Drag and drop the Sequence activity on the main Designer window. Also, drag and drop the Excel application scope inside the sequence.



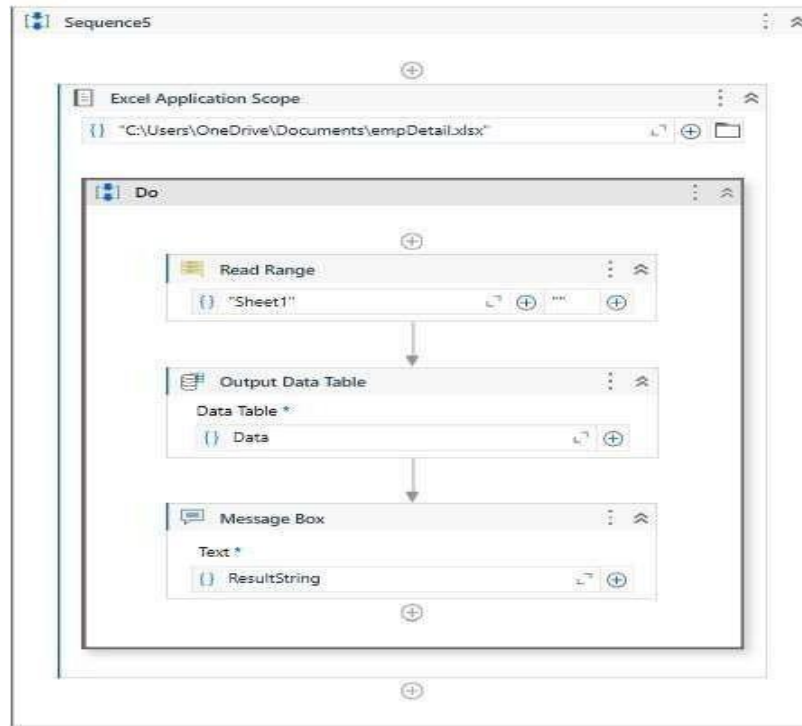
Step 2: Double-click on the Excel application scope. You have to specify the path of your workbook/Excel file. Drag and drop the Read Range activity from the Activities panel inside the Excel application scope. The Read Range activity will read the entire Excel sheet. We also have the option of specifying our range. Create a variable of type data table and specify it in the Output property of the Read Range activity. This variable will receive the data table produced by the Read Range activity:



Step 3: Drag and drop the Output Data Table activity inside the Excel application scope activity. Now, we have to specify two properties of the Output Data Table activity: the Data Table property and the text property. The Data Table property of the Output Data Table activity is used to convert the Data Table into string format. The text property is used to supply its value in a string format. We have to receive this value in order to consume it. For this, let us create a variable of type string. Give it a meaningful name:

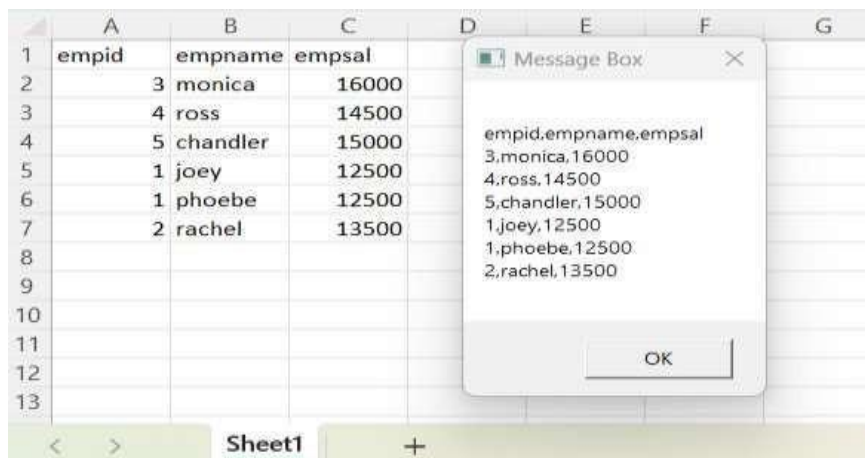


Step 4: Drag and drop a Message box activity inside the Excel application scope activity. Also, specify the string variable's name that we created earlier inside the Message box activity.



Step 5: That's it. Press F5 to see the result. A window displaying the Excel file data will pop up.

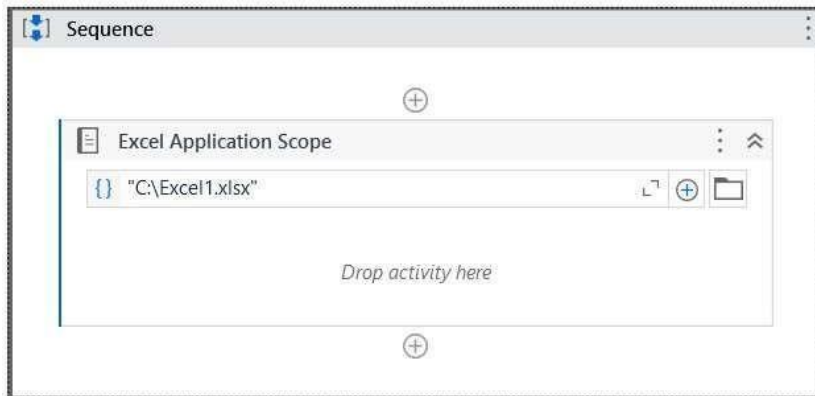
Output:



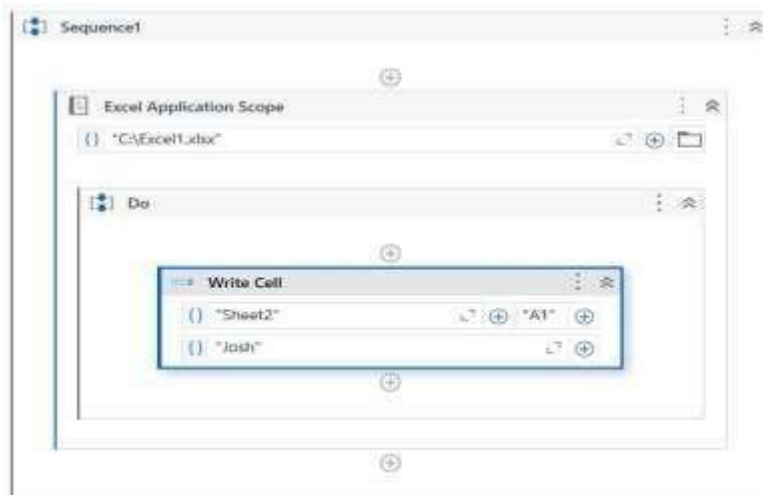
4. b. Create an automation to Write data to specific cell of an excel sheet.

This activity is used to write a value in a cell of an Excel file:

1. Drag and drop a Sequence activity on the main Designer panel. Also, drag and drop an Excel application scope inside the sequence activity.



Step 2: Drag and drop a Write Cell activity inside the Excel application scope. Specify the cell value in which we want to write in the Range property of the Write Cell activity. Also, specify the value of the Value property:



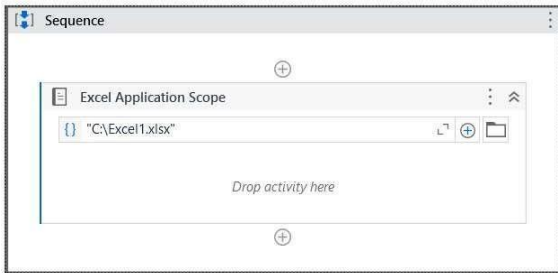
Step 3: Press F5 and see the result. Open the Excel file to see the changes:

4. c. Create an automation to Read data to specific cell of an excel sheet.

This is used to read the value of a cell from an Excel file.

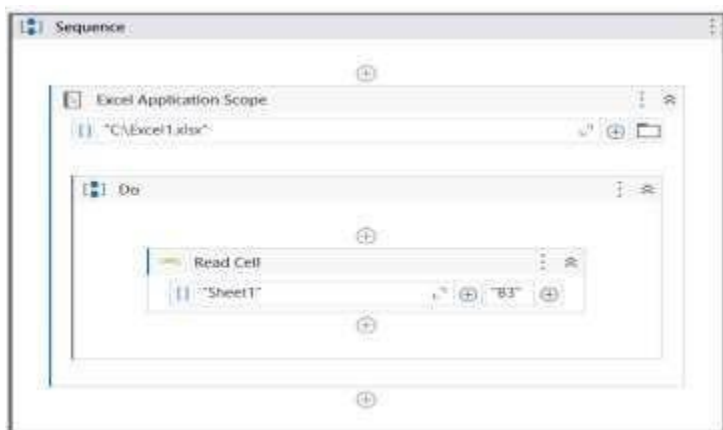
Example: To read the value of B3 cell from sheet1 to b3 cell of sheet2, perform the following steps:

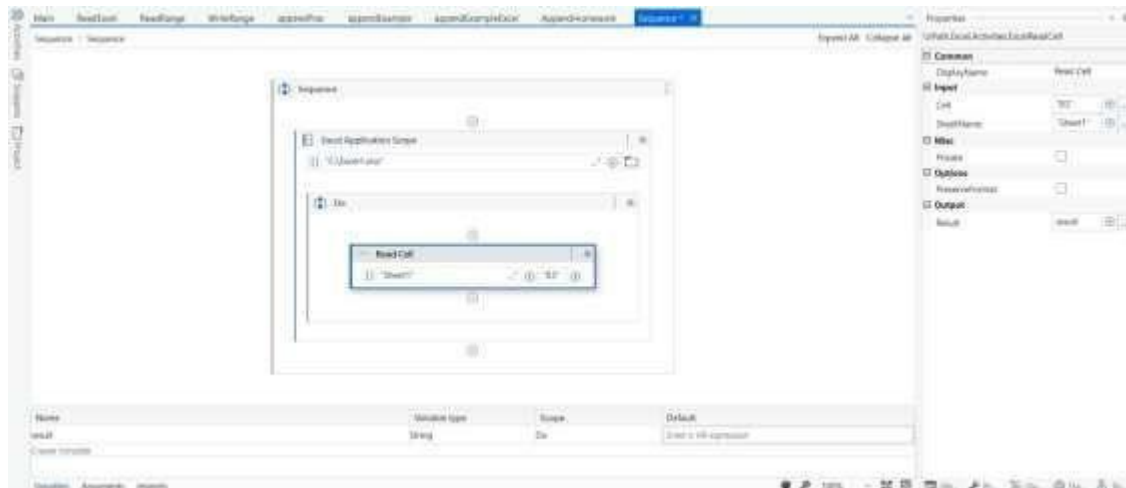
Step 1: Drag and drop a Sequence activity on the main Designer panel. Also, drag and drop an Excel application scope inside the Sequence.



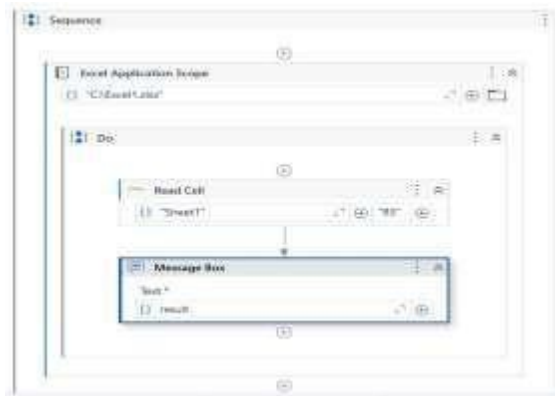
Double click on Excel application scope. It is a good practice to use the Excel application scope when using Excel activities inside our project.

Step 2: Drag and drop the Read Cell activity inside the Excel application scope activity. Specify the range value in the cell text box of the Read Cell activity. Create a variable of type string to hold the result produced by the Read Cell activity. In our case, we have created a 'result' variable. Specify the Output property of the Read Cell activity by providing the variable's name that we have created:

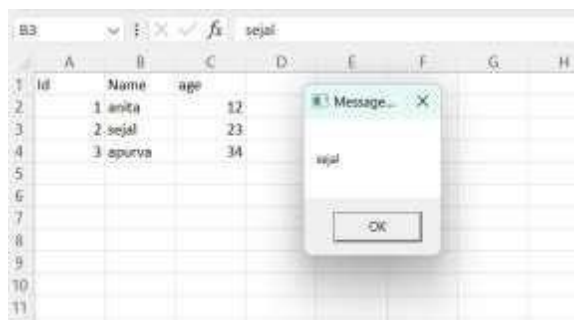




Step 3: Drag and drop a Message box activity inside the Excel application scope activity and specify the string variable's name (which we created earlier) in the expression box of the Message box activity.



Step 4: That's it. Press F5 to see the result.



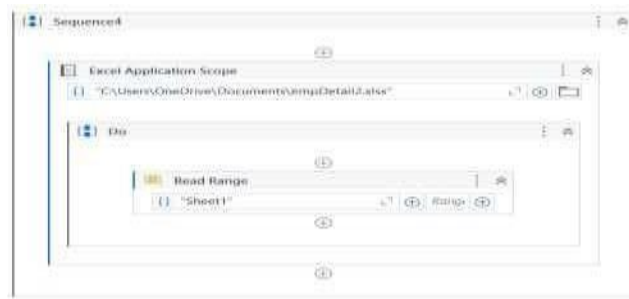
4. d. Create an automation to append data to specific cell of an excel sheet.

This is used to add more data into an existing Excel file. The data will be appended to the end.

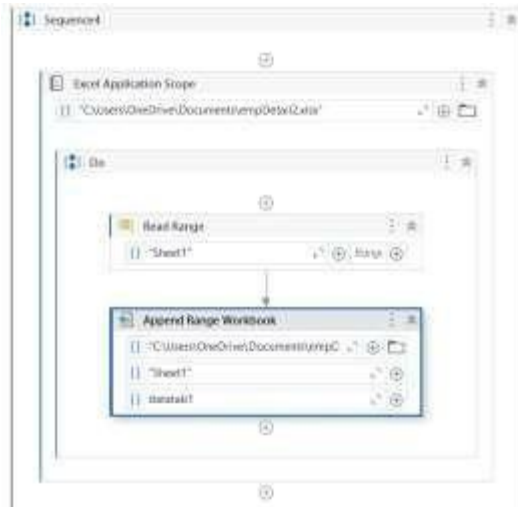
Step 1: Drag and drop the Sequence activity on the main Designer window. Also, drag and drop the Excel application scope inside the sequence activity. The Append Range activity requires a data table. In this program, we are going to use another sample Excel file, which has some raw data. Then, we will read this Excel file and append the data to another Excel file. First, we have to read its contents:

	A	B	C	D
1	empid	empname	empsal	
2		1 phoebe	12500	
3		2 rachel	13500	
4				
5				
6				
7				
8				
9				
10				
11				

Step 2: Drag and drop the Read Range activity inside the Excel application scope activity. The Read Range activity produces a data table. We have to receive this data table in order to consume it. Create a data table variable and specify it in the Output property of the Read Range activity:



Step 3: Drag and drop the Append Range activity inside the Excel application scope activity. Specify the Excel file path in the Append Range activity (in which we want to append the data). Also, specify the data table (which is generated by the Read Range activity):



Step 4: That's it. Press F5 to see the result:

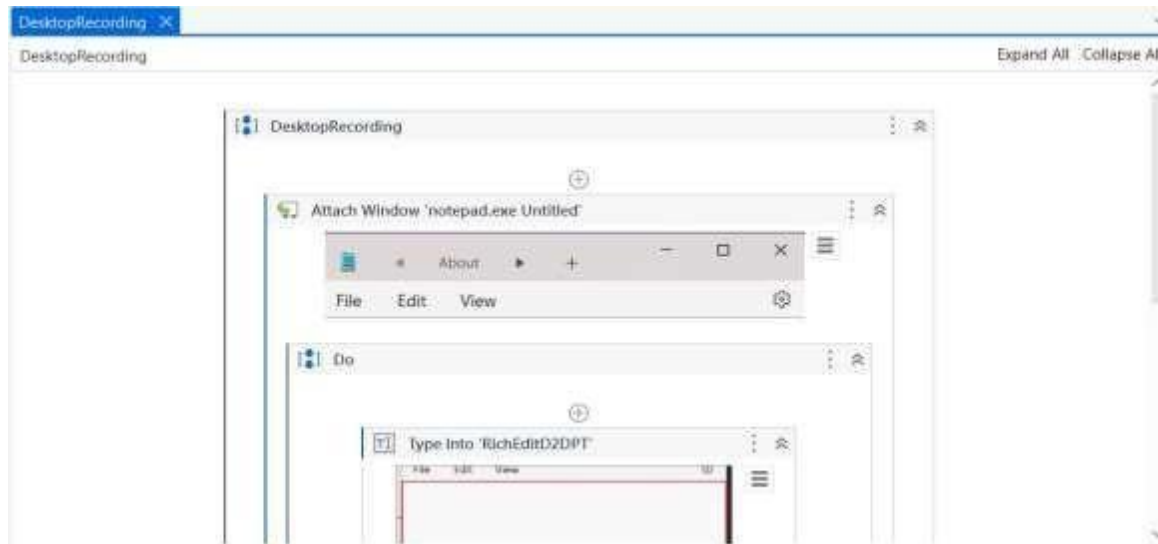
Output: We can clearly see that the data has been appended successfully to the Excel sheet.

	A	B	C	D
1	empid	empname	empsal	
2	3	monica	16000	
3	4	ross	14500	
4	5	chandler	15000	
5	1	joey	12500	
6	1	phoebe	12500	
7	2	rachel	13500	
8				
9				

Practical No. 5

Different controls in UiPath:

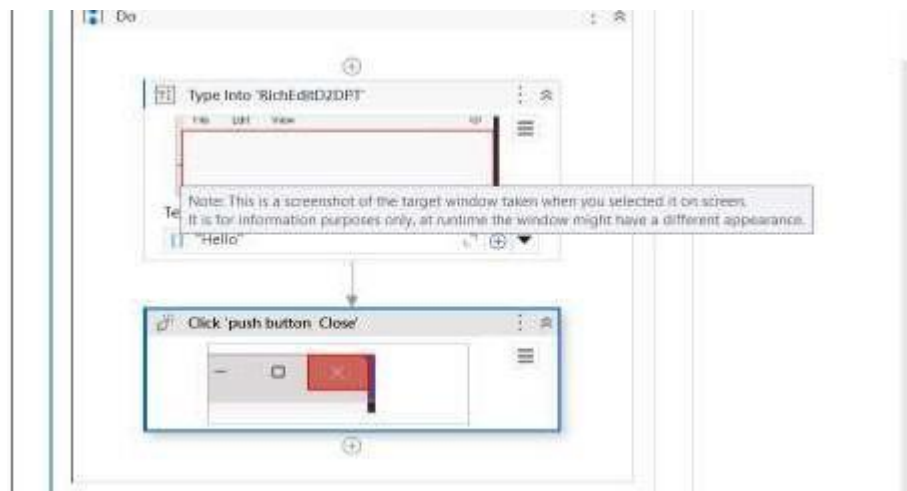
5.a. Implement the attach window activity.



Step 1: Drag and drop Attach Window activity.

Indicate it on blank notepad window.

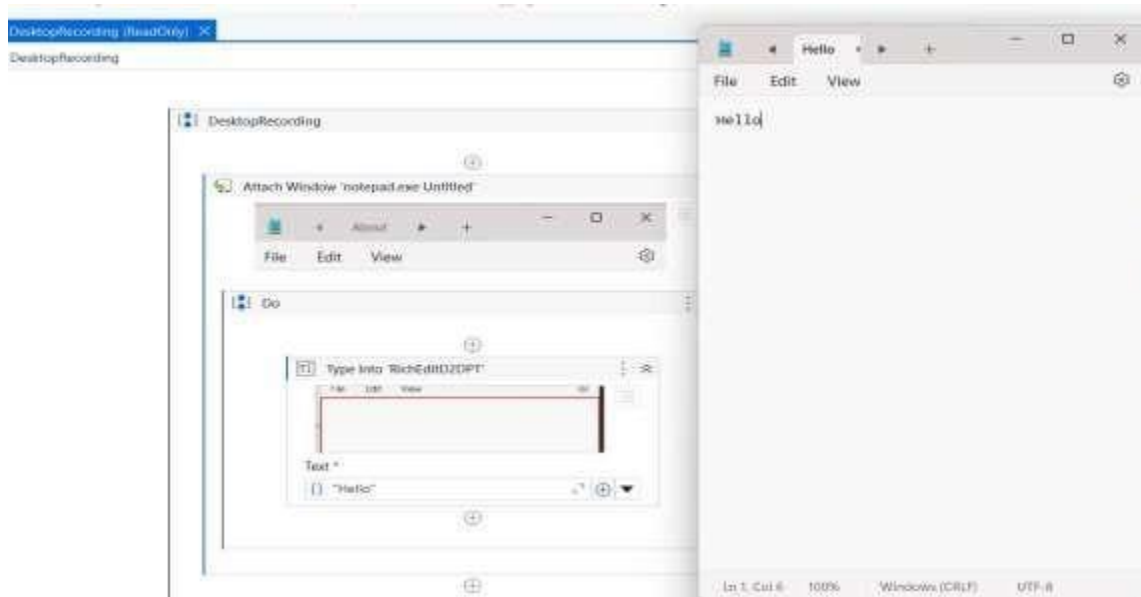
Step 2: In Do, drop an Type Into activity



In Type into activity write “Hello”

Step 3: Then drop an click activity, indicate it on close. Run

Output:-



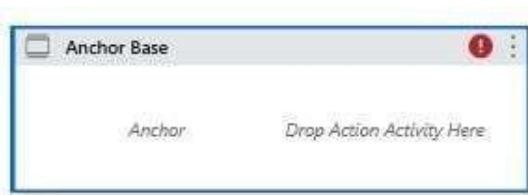
Then notepad window will get closed automatically because of click activity.

5.b. Automate using Anchor Base.

Step 1: Drag and drop a Flowchart activity on the Designer panel of a blank project. Also, drag and drop an Anchor base control from the Activities panel. Connect the Anchor base control with Start.

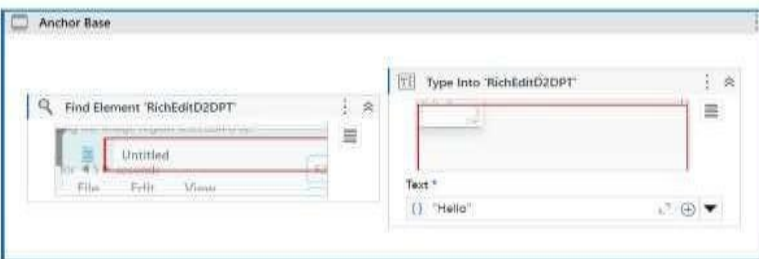
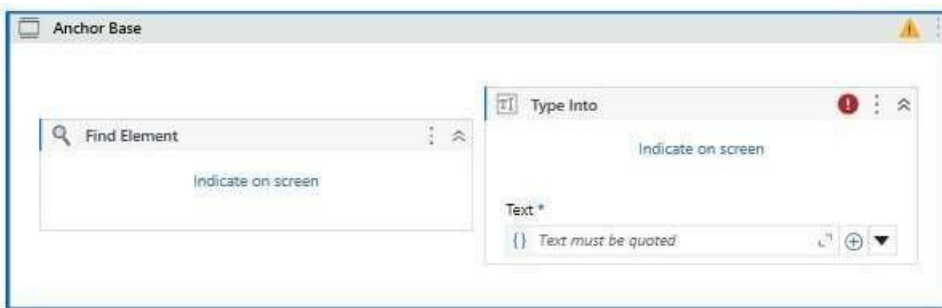


Step 2: Double-click on the Anchor base control:



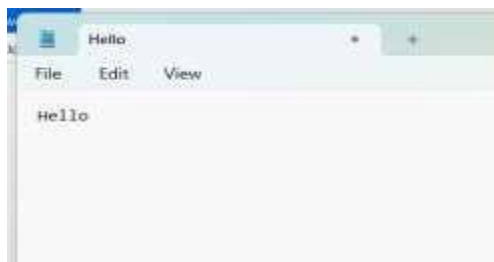
Step 3: There are two activities that we have to supply to the Anchor base control: Anchor and action activities.

Step 4: Drag and drop the Anchor base activity (for example; Find Element activity) in the Anchor field and Action activity (for example; Type into) in the Drop Action Activity Here field of the Anchor base control.

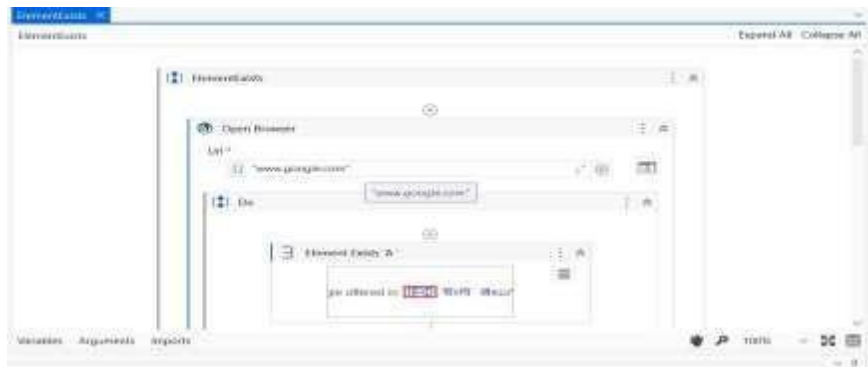


Step 5: Press F5 to see the results.

Output:



5.c. Automate using Element Exists.

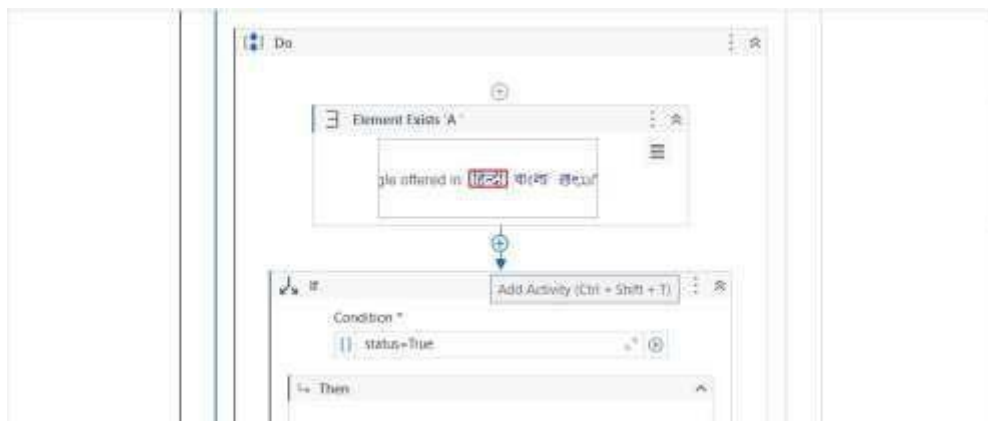


Step 1: Drag and drop Open Browser activity.

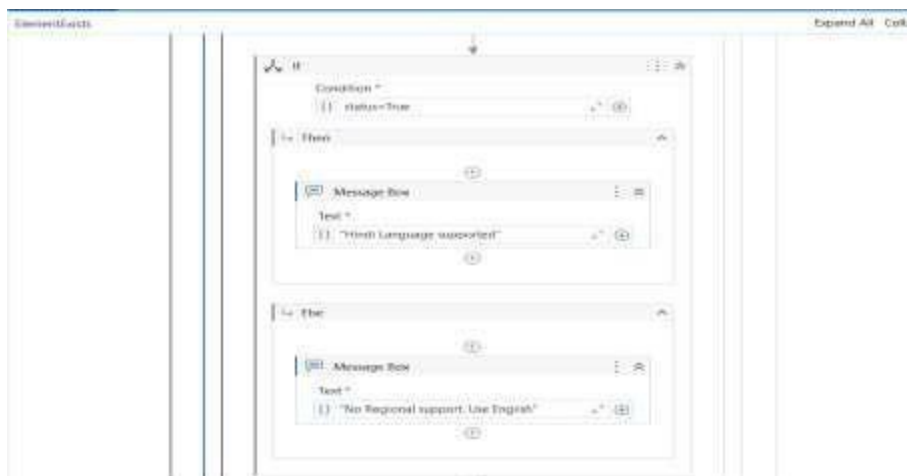
Give url of Google.

Step 2: In Do, Drop an activity called Element Exist.

Indicate on Hindi element.



Step 3: Drop an If activity. Condition: status=True



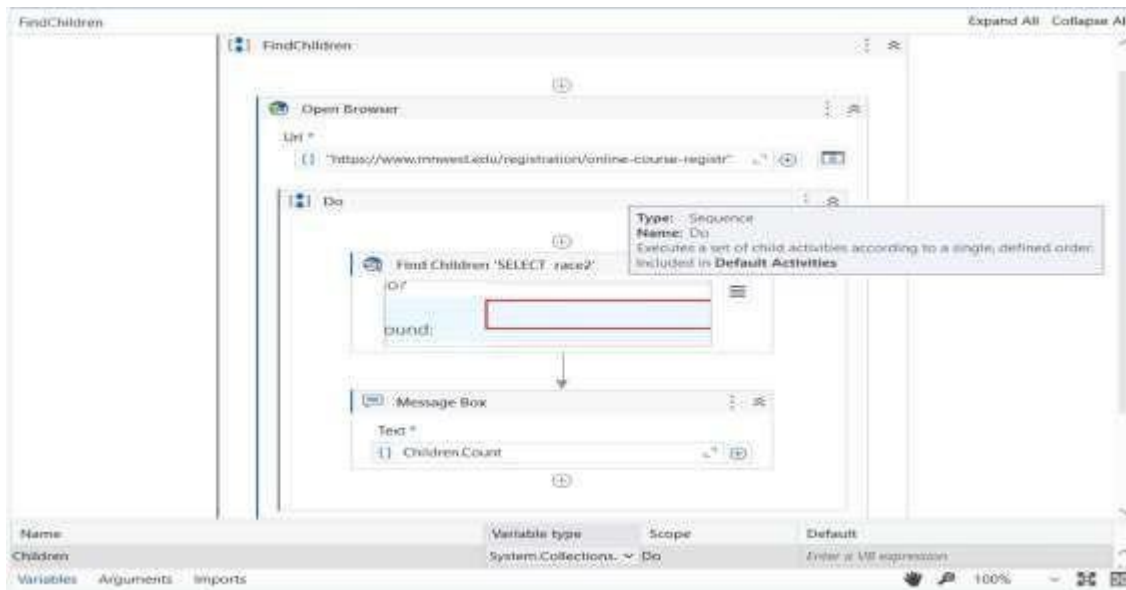
Then Element triggered Else Element not exist.

Step 4: Run.

Output:-



5.d. Automate using Find Children control.



Step 1: Drag and drop Open Browser activity.

Enter url <https://www.mnwest.edu/registration/online-course-registr>.

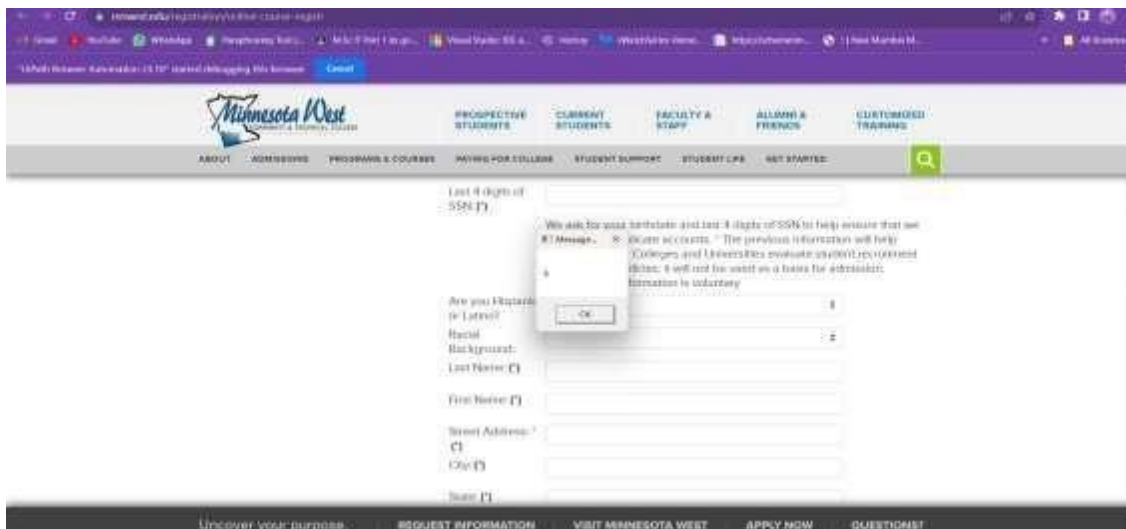
Step 2: In Do, drop Find Children activity. Indicate on the drop down box.

Create a variable Children to store the elements. Give datatype as generic.

Step 3: Drop a message box activity. { Children.Count } Count is inbuilt function to count the elements.

Step 4: Run

Output:-

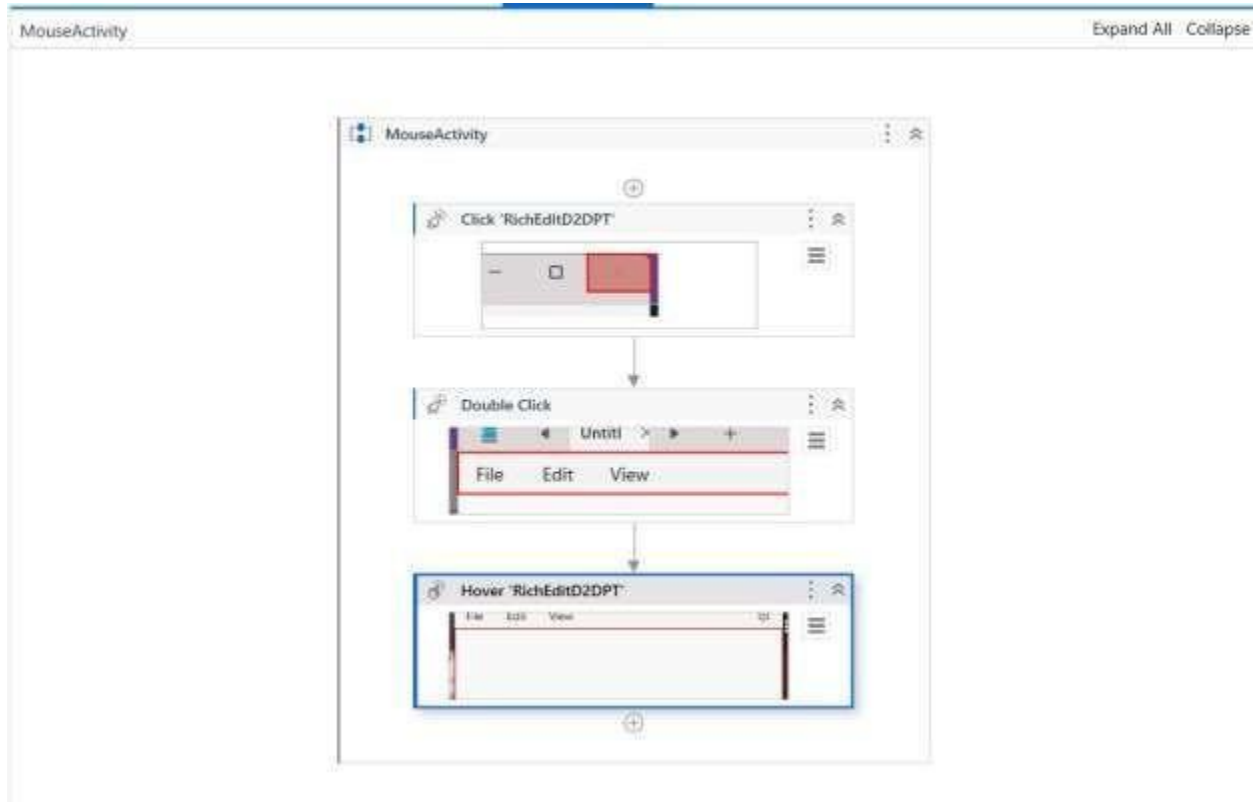


Practical No.6

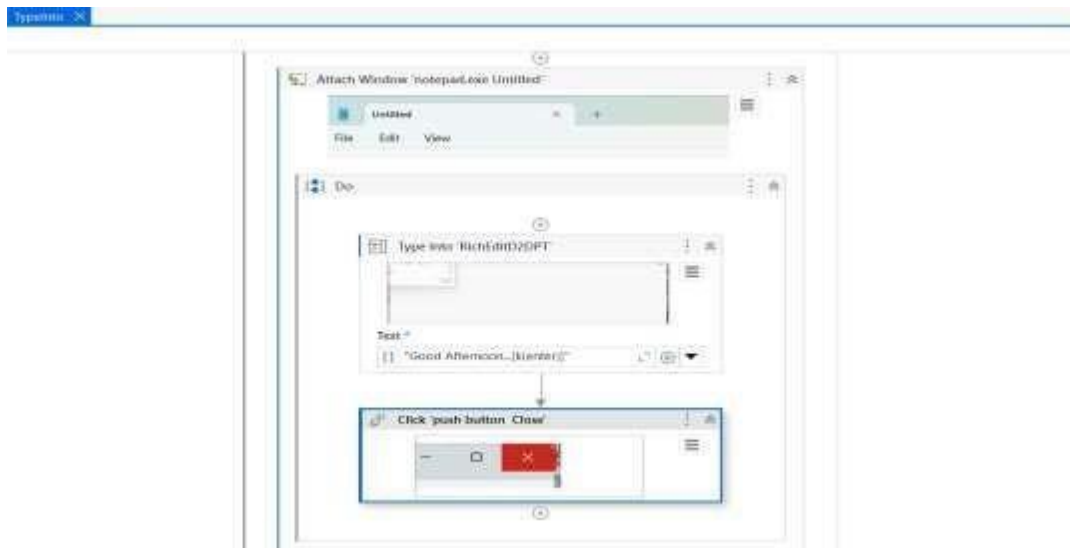
Keyboard and Mouse Events:

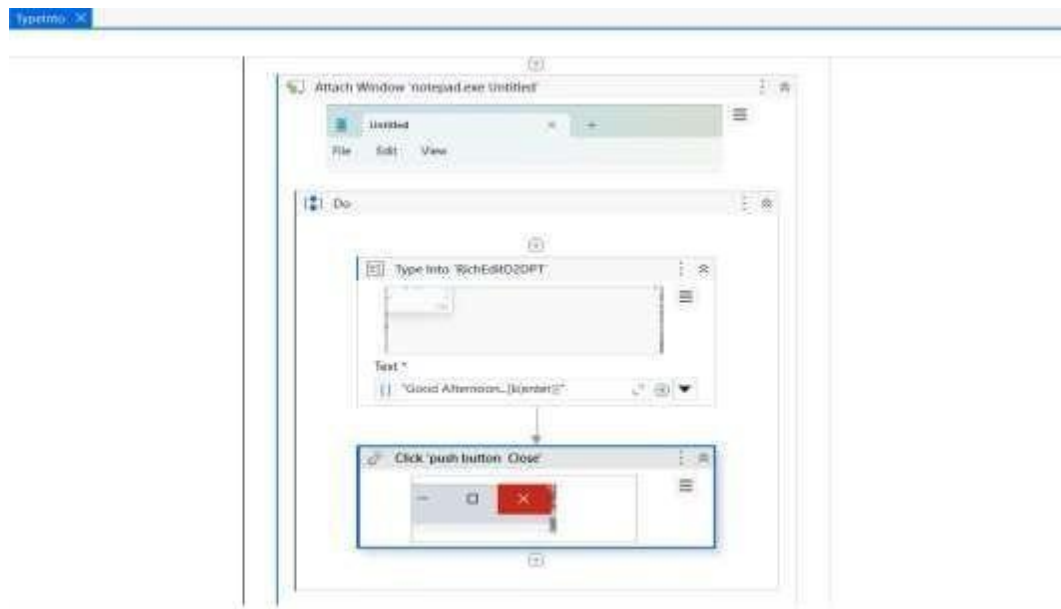
6. a. Demonstrate the following activities in UiPath:

i. Mouse (click, double click and hover)



ii. Type into





Step 1: Drag and drop Attach Window activity.

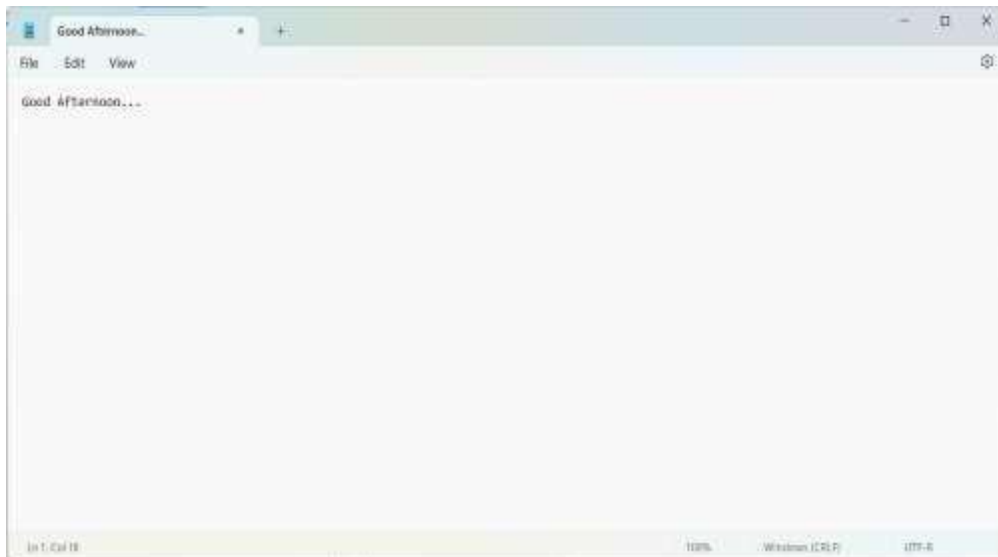
Indicate blank notepad window.

Step 2: Drop Type into activity, indicate notepad text area.

Step 3: Drop Click activity indicate on close.

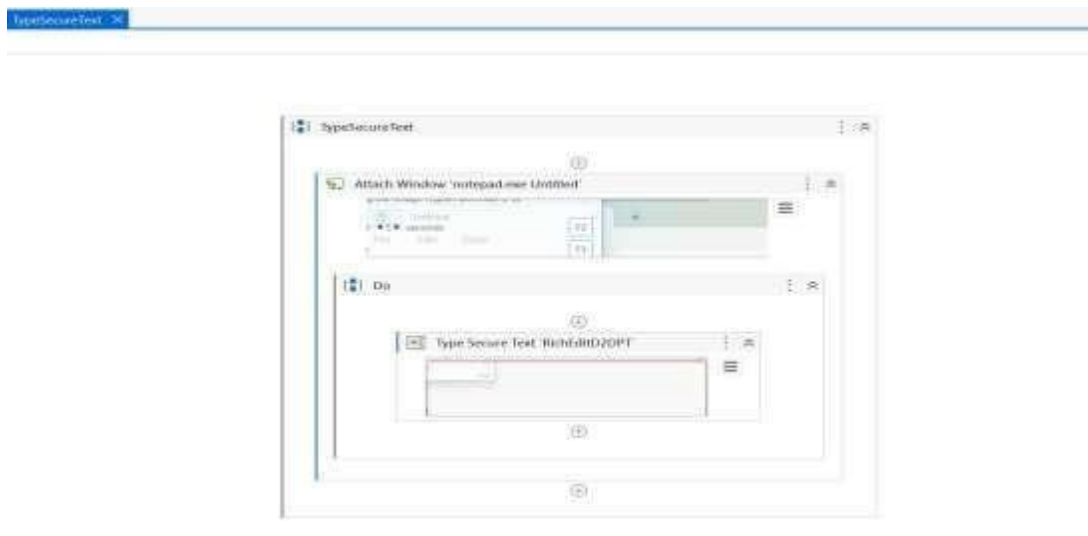
Step 4: Run

Output:-



Then notepad window will get closed automatically because of click activity.

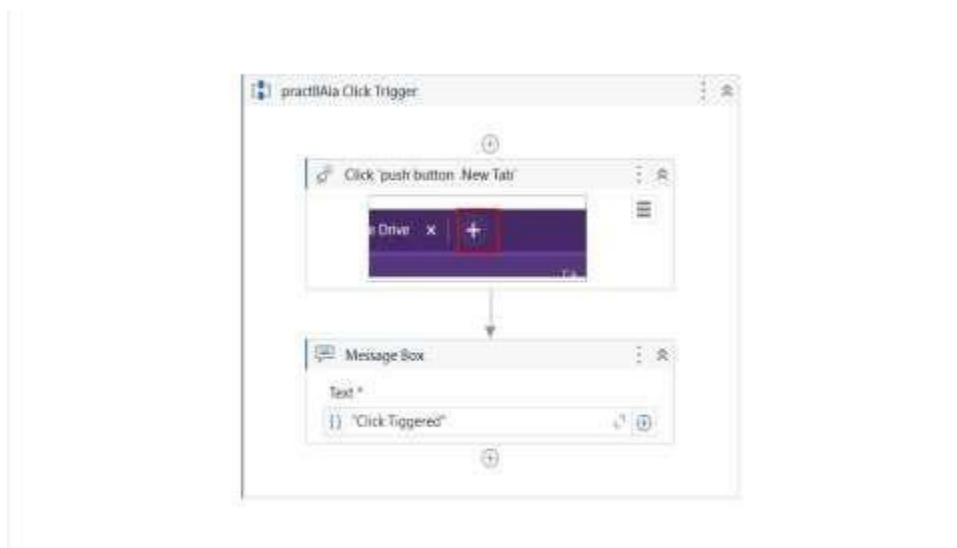
iii. Type Secure text



6. b. Demonstrate the following events in UiPath:

i. Element triggering event

a) Click trigger



Step 1: Drag and drop Click activity indicate on anything.

Step 2: Drop message box enter “Click Triggerd”.

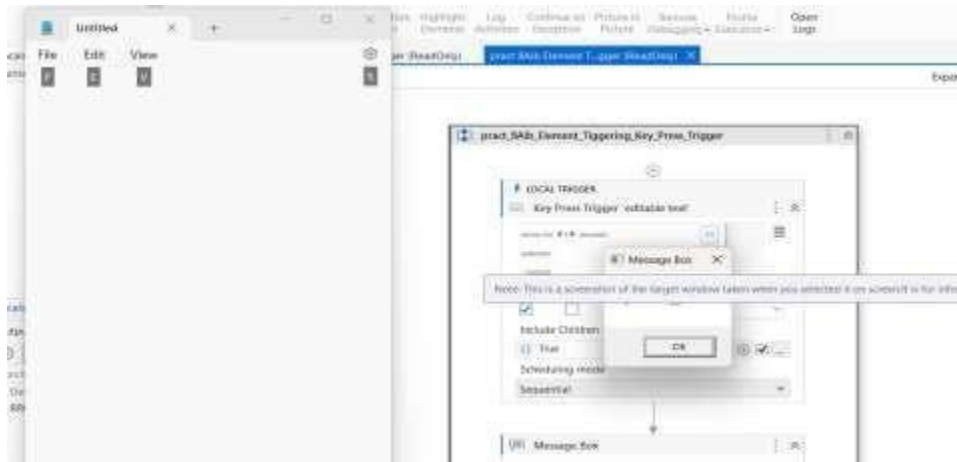
Step 3: Run.

The screenshot shows a software configuration window titled "LOCAL TRIGGER". Inside, there is a section for "Key Press Trigger 'editable test'". The configuration includes a "delay for" of 10 seconds, a "selection" of the letter 'a', and checkboxes for "Alt", "Ctrl", "Shift", and "Win" (all are unchecked). Below these are "Include Children" (set to "True") and "Scheduling mode" (set to "Sequential"). An arrow points from this configuration window to a "Message Box" window below it, which displays the text "Key Press Triggered".

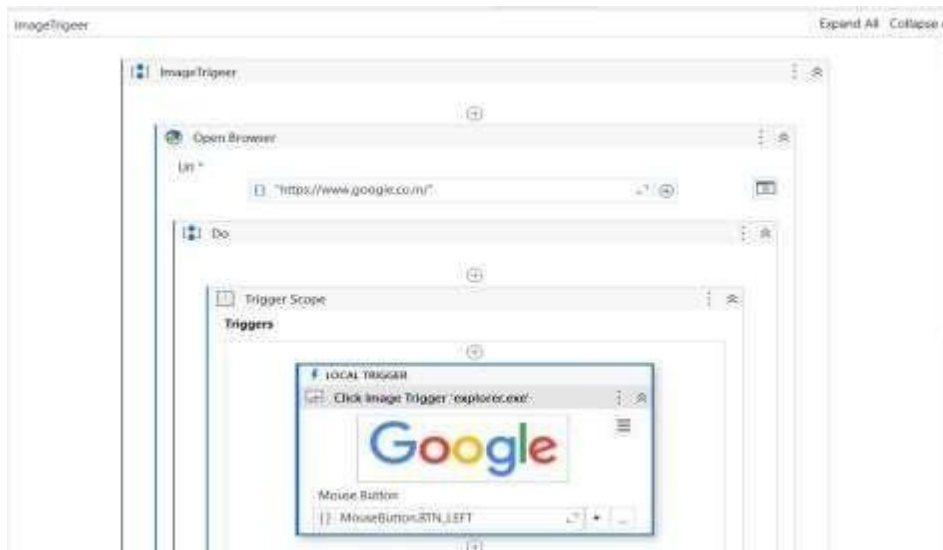
Step 2: Drop message box activity, enter Text as “Key Press Triggered”.

MAHESH TANAJI CHAVAN

Output:-



ii. Image triggering event

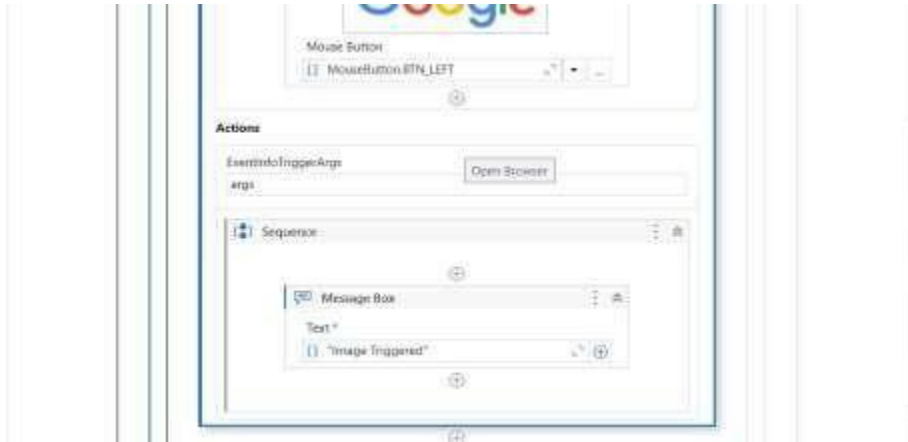


Step 1: Drag and drop Open Browser activity, give Google's url.

Step 2: Drop Trigger scope activity,

In triggers indicate Google's Logo image.

In Action drop message box in text "Image Triggered".



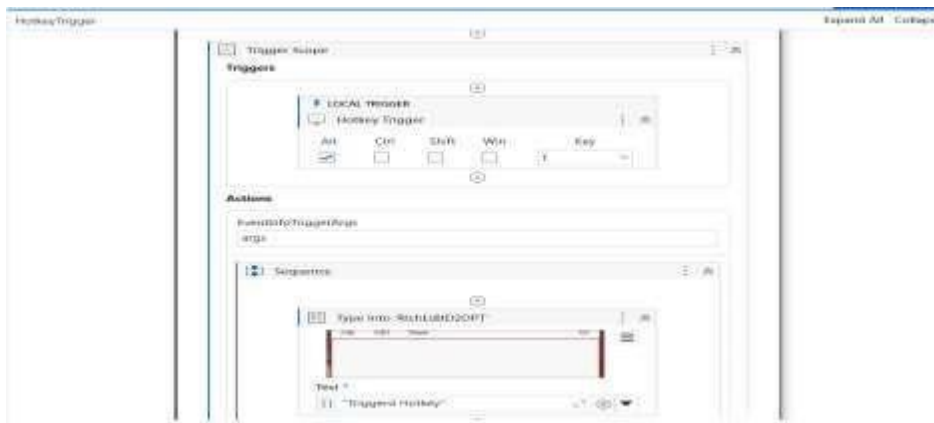
Step 3: Run.

Output:-



iii. System Triggering Event

a) Hotkey Trigger



Step 1: Drag and drop Trigger scope.

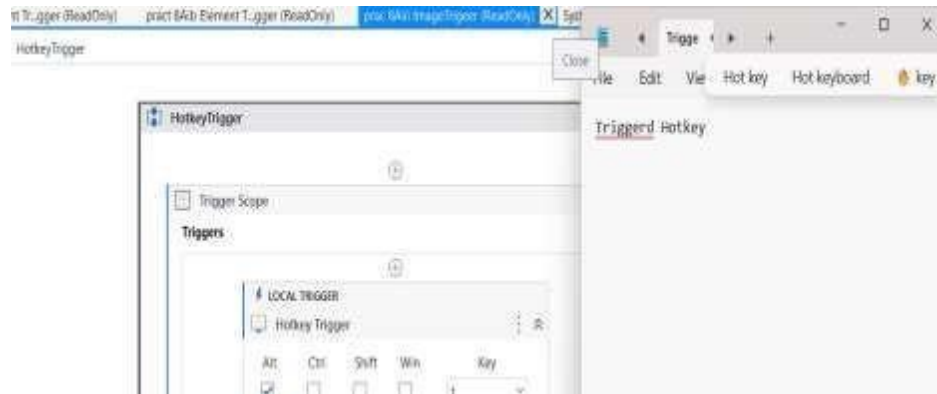
In Triggers, drop Hotkey Trigger and select Alt and enter key as (t).

In Actions, drop Type into activity indicate on notepad and write text as “Triggered Hotkey”.

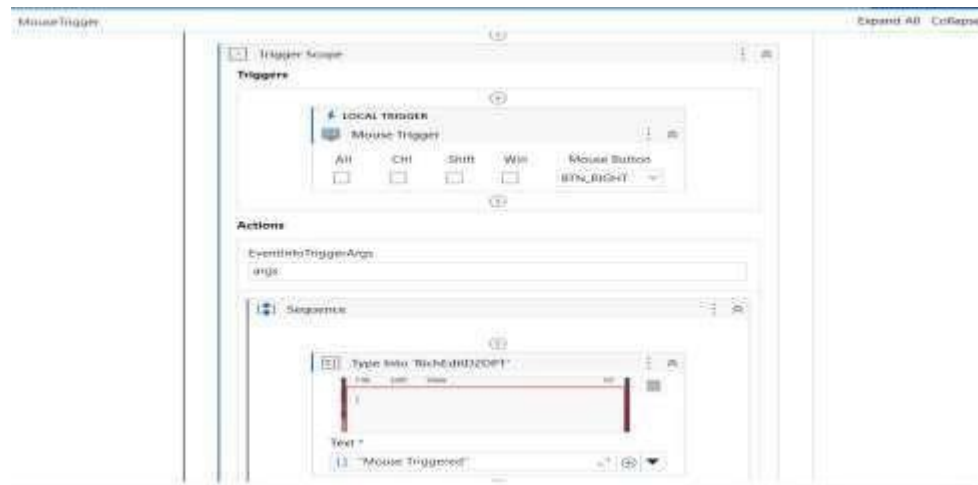
Step 2: Run.

Output:-

Press Alt+t on your keyboard.



b) Mouse Trigger



Step 1: Drag and drop trigger scope.

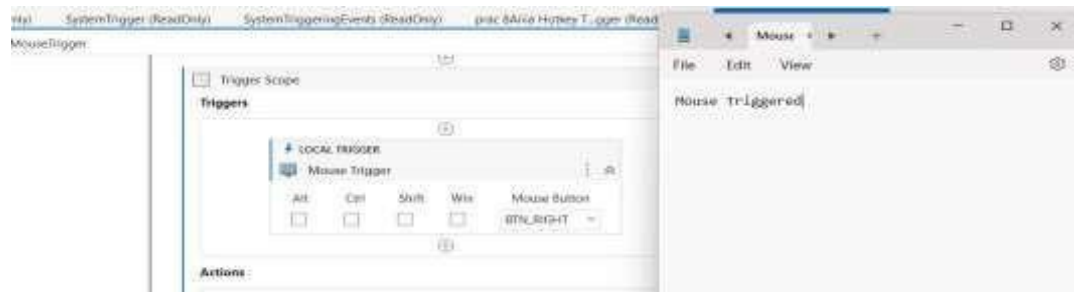
In Triggers, drop Mouse Trigger activity. Select mouse button BTN_RIGHT

In Actions, drop Type into activity indicate it on notepad, write text as “Mouse Triggered”.

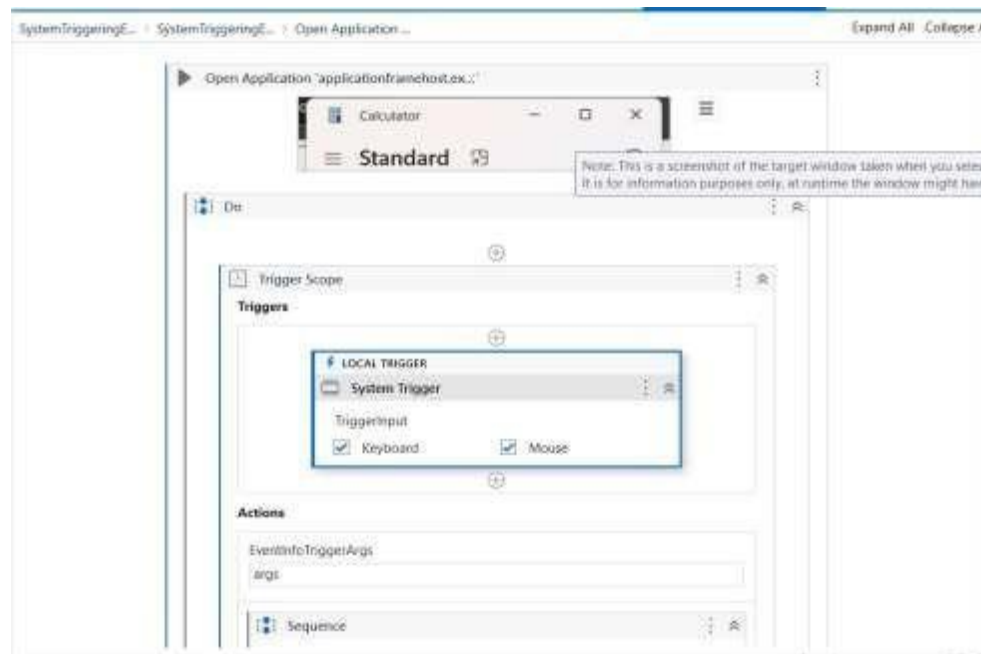
Step 2: Run.

Output:-

Right Click on notepad.



c) System Trigger



Step 1: Drag and drop Open Application activity, indicate calculator on it.

In Do, drop an Trigger scope activity.

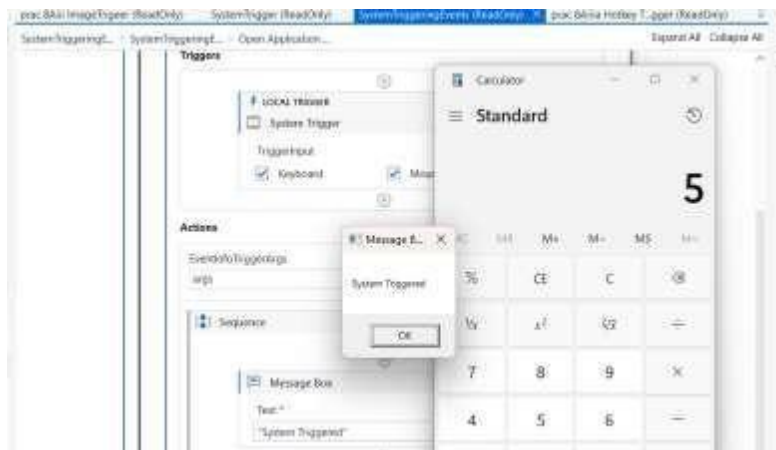
In Triggers, drop System Trigger activity select both keyboard and mouse, so it will trigger both when we click any keyboard key or mouse button.



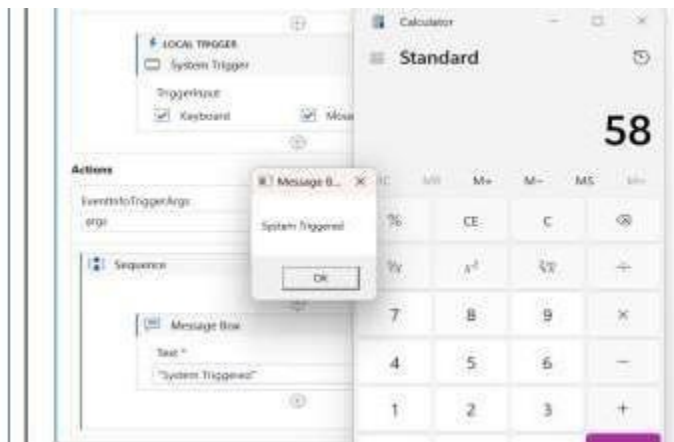
In Actions, drop message box in text write “System Triggered”.

Step 2: Run

Output:-

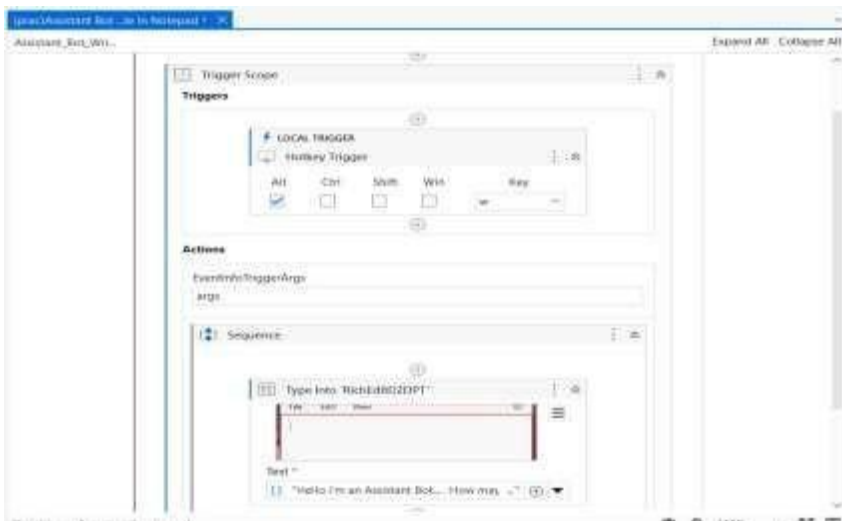


Clicked with keyboard.



Clicked mouse.

6. c. Automate the process of launching an assistant bot on a keyboard event.



Step 1: Create a new Sequence.

Step 2: Drag and drop Trigger scope.

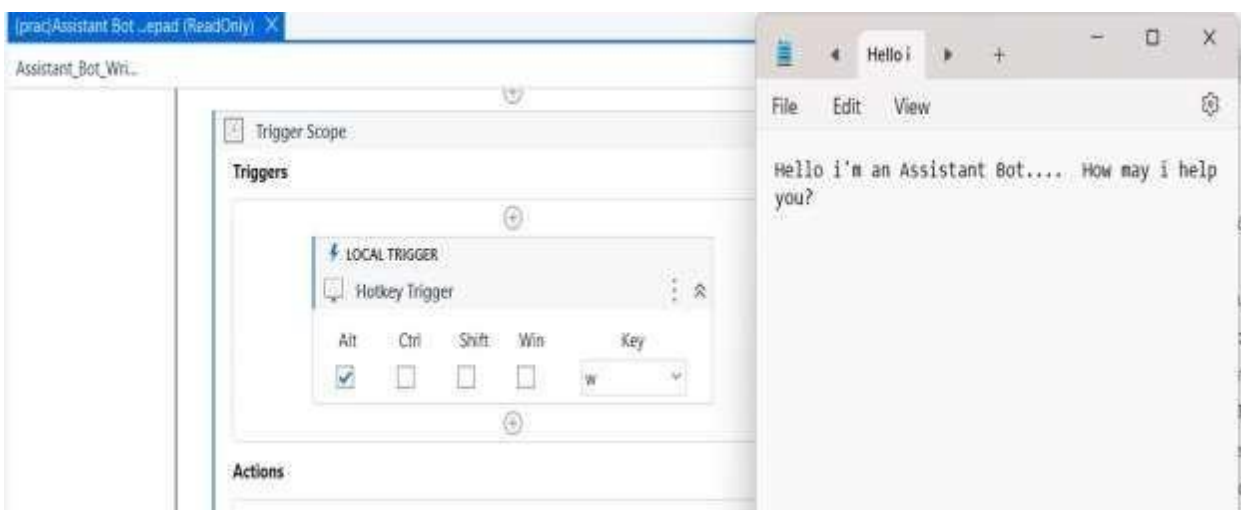
In Triggers, drop Hotkey Trigger. Select Alt and enter (w) as key.

In Actions, drop Type Into activity indicate it on blank notepad window write text as “Hello I’m an Assistant Bot... How may I help you?”.

Step 3: Run

Output:-

After clicking Alt+w



Practical No. 7

Screen Scraping and Web Scraping methods:

7. a. Automate the following screen scraping methods using UiPath:

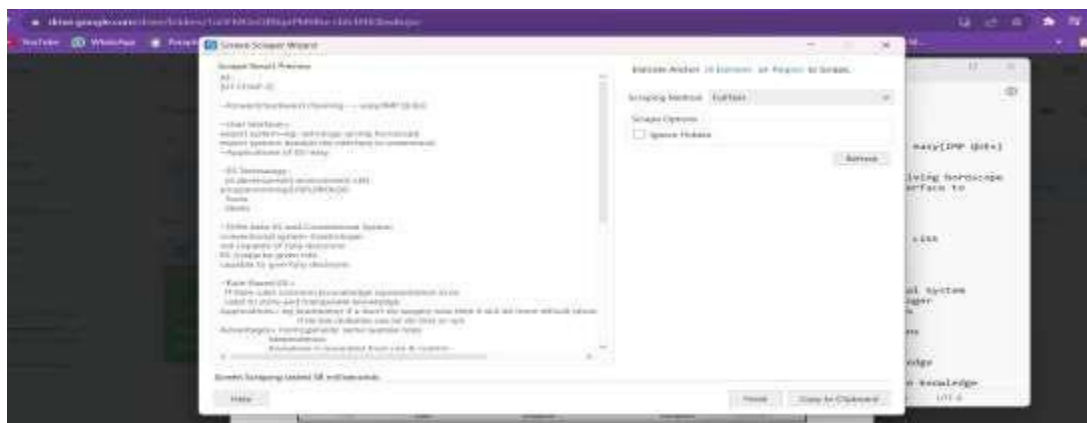
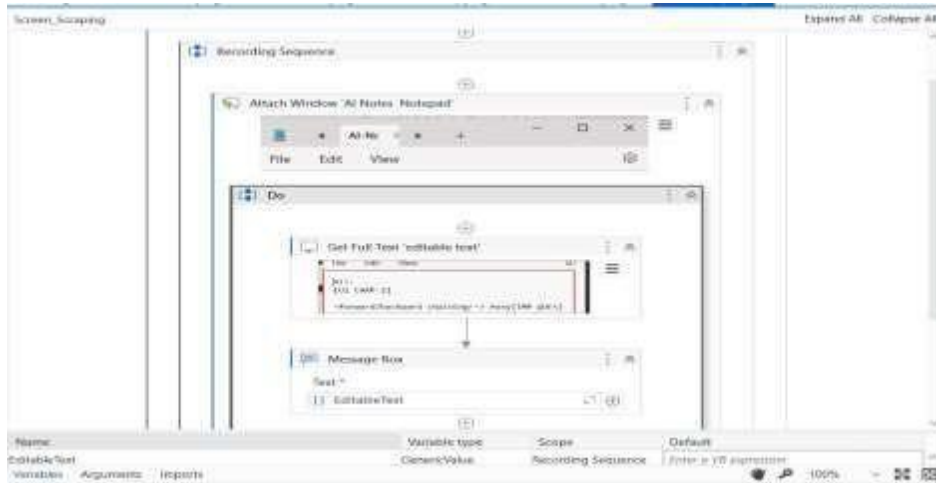
i. Full Text

Step 1: Create a sequence.

Step 2: Select Screen Scraping function from Design ribbon. Indicate it on notepad file that has text in it.

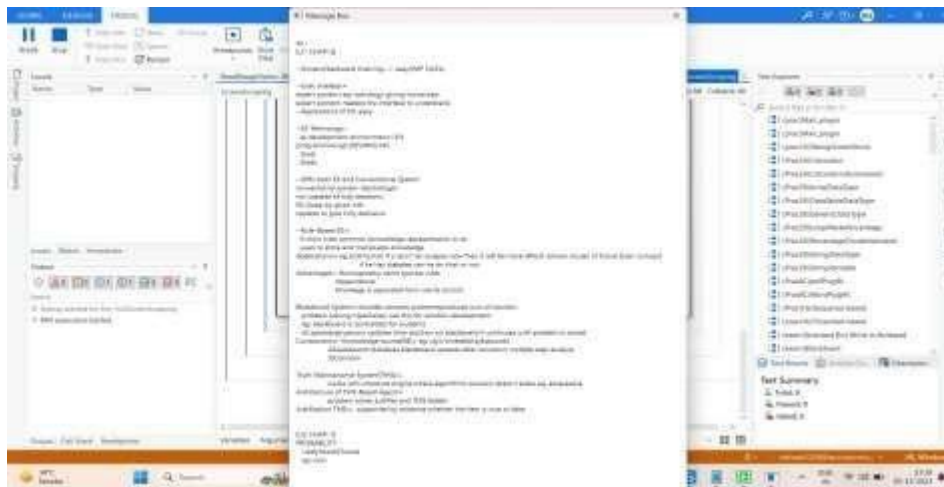
Step 3: Then one window will appear, select Full test scraping method and finish. Then it will automatically create a variable as Editable Text to store the captured value.

Step 4: Then drop an message box activity write Editable Text.

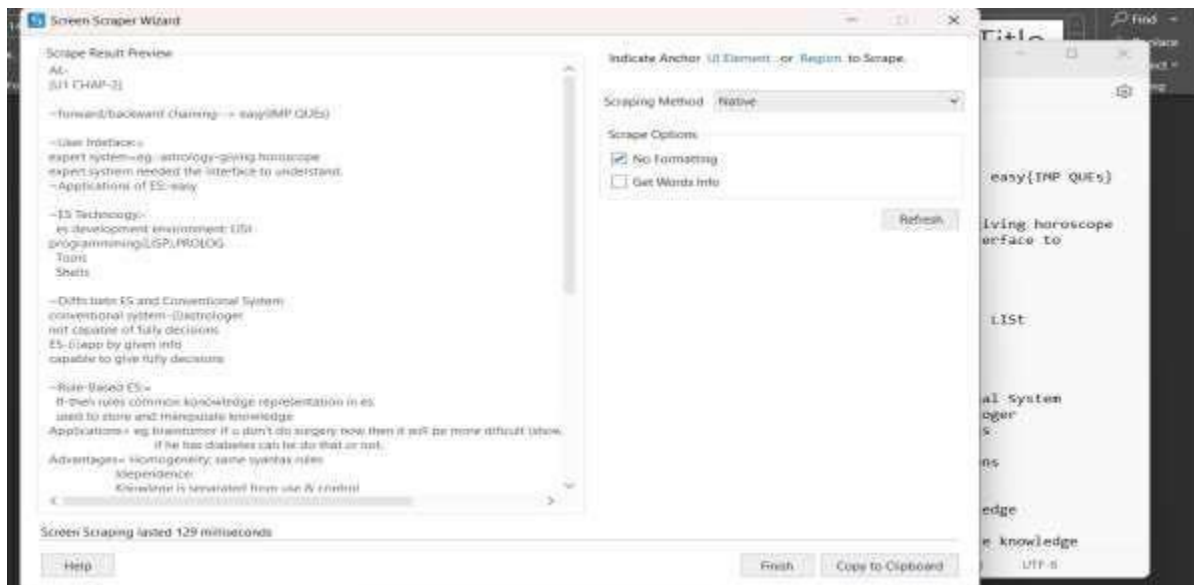


Step 5: Run

OUTPUT:



ii. Native



iii. OCR

Step 1: Create a sequence.

Step 2: Select Screen Scraping function from Design ribbon. Indicate it on notepad file that has text in it.

Step 3: Then one window will appear, select OCR scraping method and finish. Then it will automatically create a variable as Editable Text to store the captured value.

Step 4: Then drop an message box activity write editable Text

Practical No. 8

PDF Automation and Exception Handling:

8. a. Demonstrate Exception Handling using UiPath

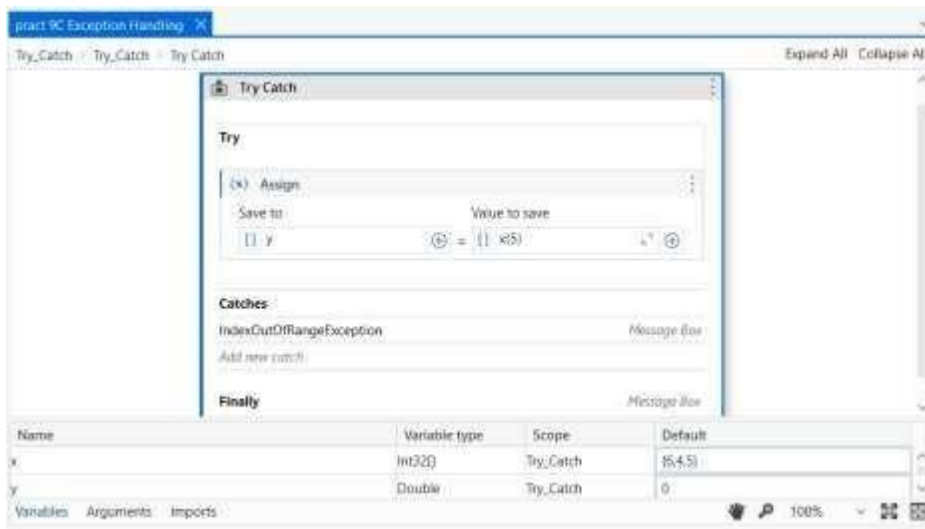
Handling IndexOutOfRangeException.

Step 1: Create a new Sequence.

Step 2: Create an array of x{6,4,5} and create 1 more variable y.

Step 3: Drag and drop Try Catch activity

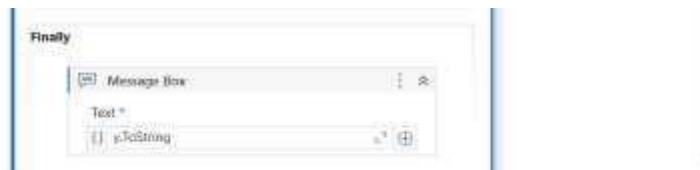
Step 4: In Try, drop assign activity save x(5) in y but x's length is 3 so it will give IndexOutOfRangeException.



Step 5: In Catches, add catch IndexOutOfRangeException. In that drop message box exception.ToString

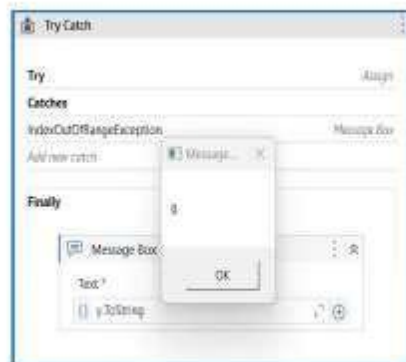
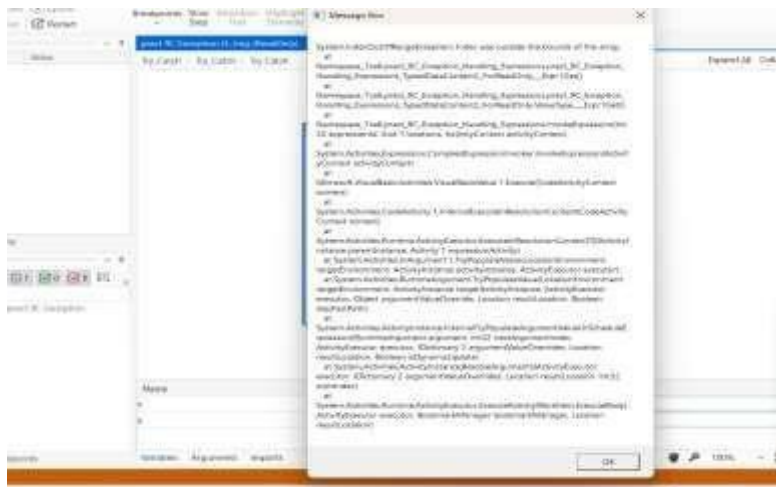


Step 6: In Finally, drop Message Box y.ToString



Step 7:Run

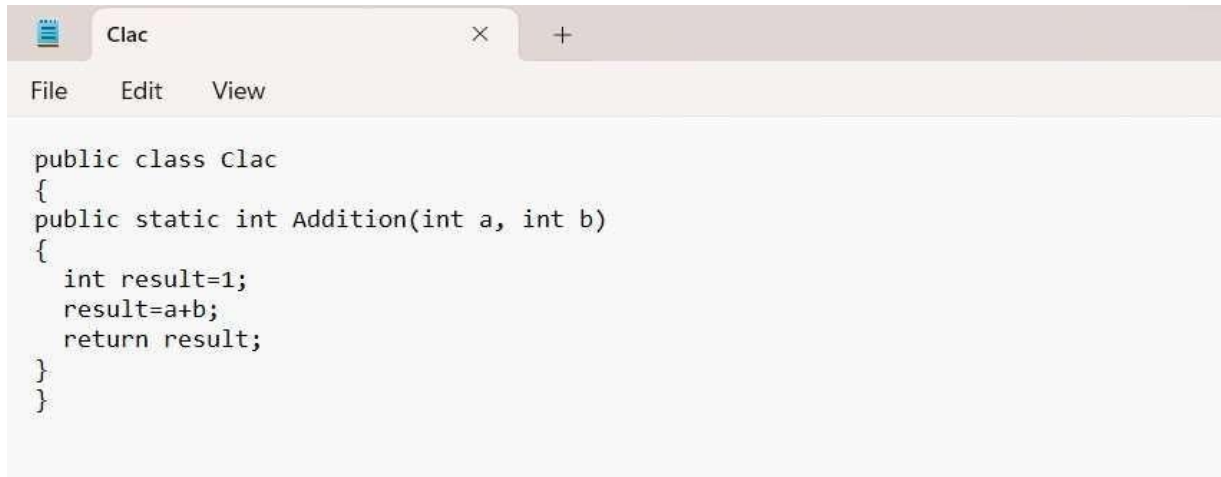
Output:-



8. b. Install and automate any process using UiPath with the following plug-ins

i. Java Plugin

Step 1: Write a java program to add 2 values, create a parameterized method for it.
(Addition)



```
public class Clac
{
    public static int Addition(int a, int b)
    {
        int result=1;
        result=a+b;
        return result;
    }
}
```

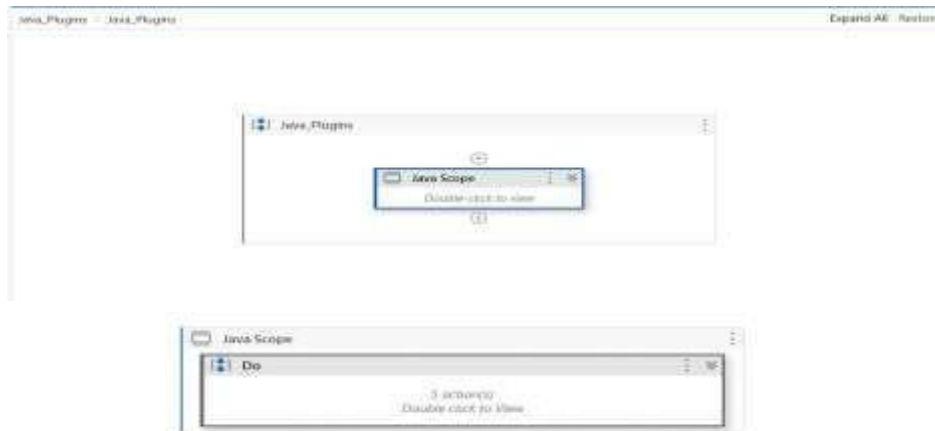
Step 2: Run the program and create a jar file of this program by using **jar -cvf Clac.jar Clac.class** command.

Open UiPath Studio.

Step 3: Install UiPath.Java.Activities package from manage packages.

Step 4: Create a sequence.

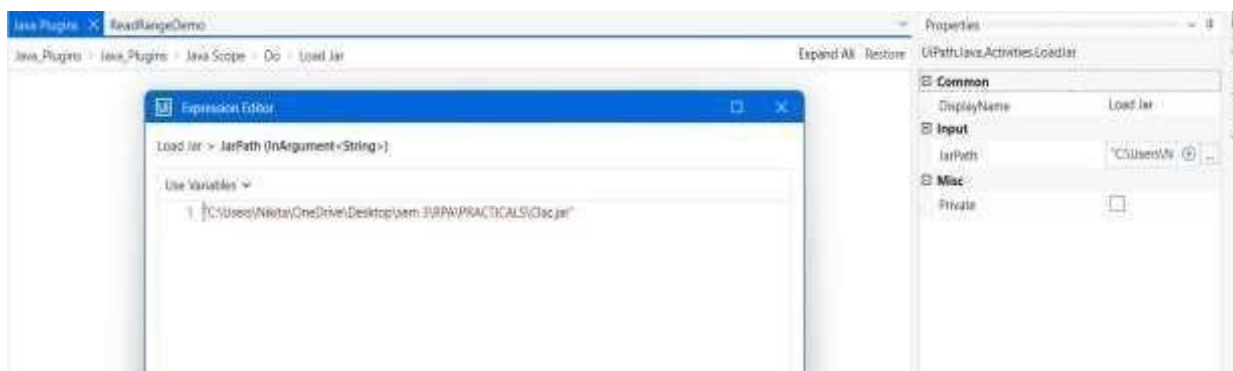
Step 5: Drag and drop Java Scope activity.



Step 6: In Do,



Step 7: In Load Jar, Load the program's jar file as

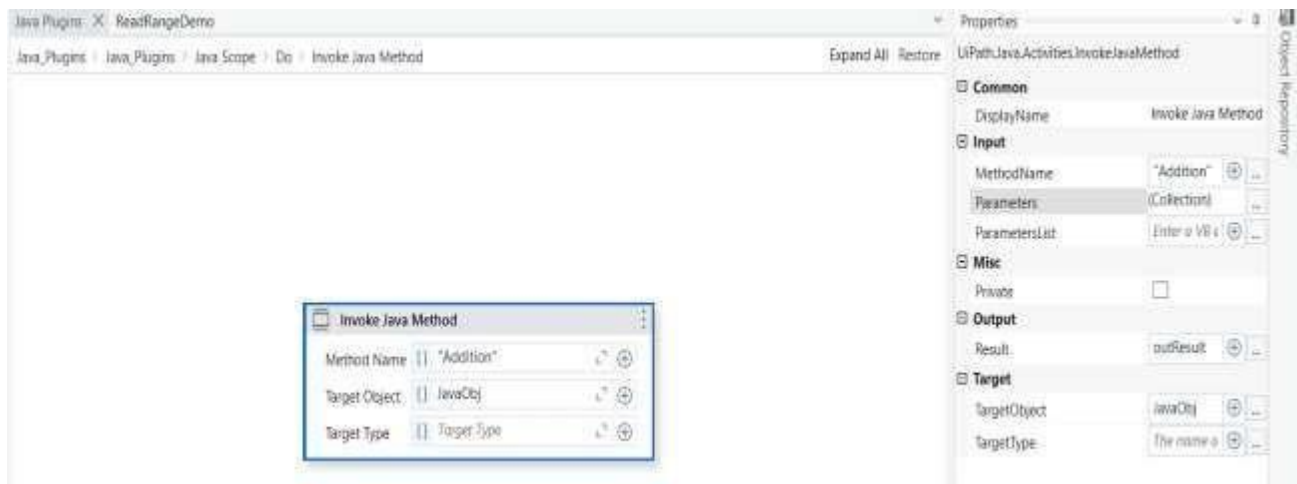


Step 8: In Create Object,

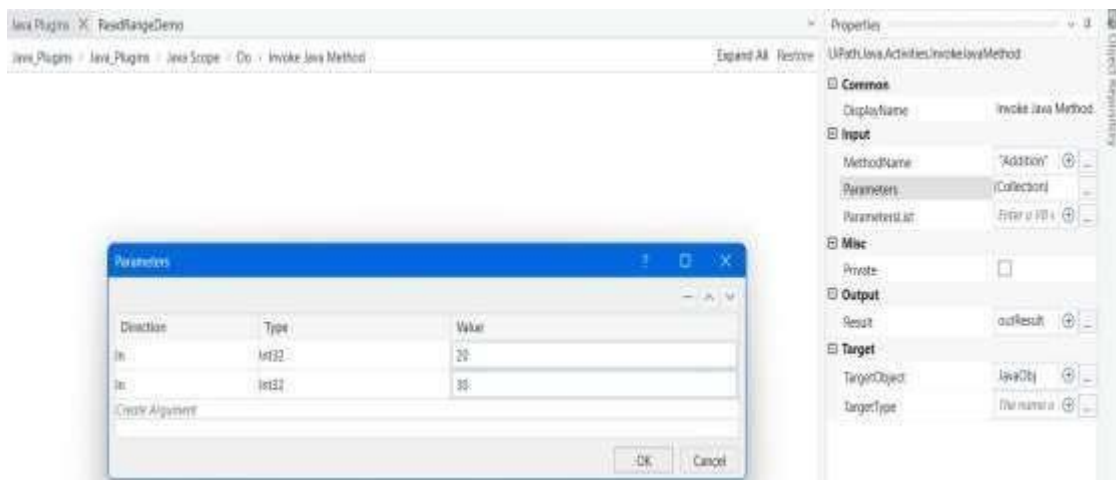


Give TargetType same as class name “**Clac**” Create variable **JavaObj** store the result in Output property and variable type should be java object.

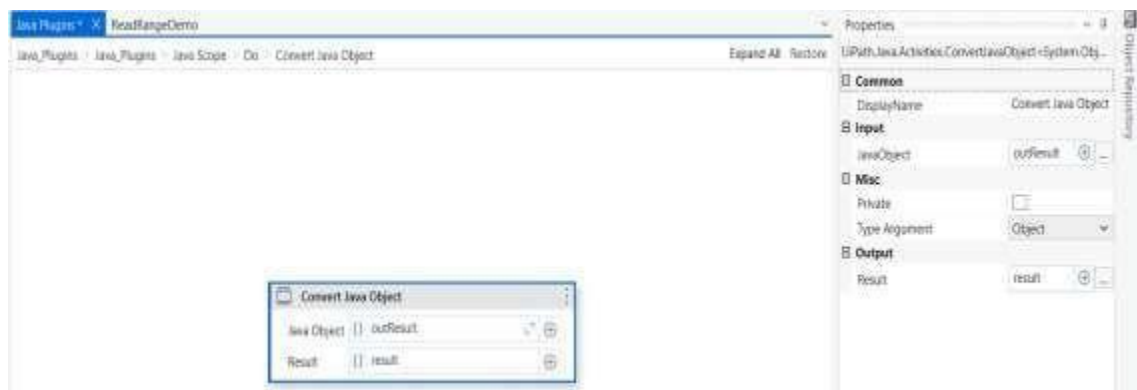
Step 9: In Invoke Java Method, write method name “**Addition**” and Target Object is JavaObj where we stored java object.



Step 10: To give Parameters to Addition method give two parameters as mentioned below in input property. Create a variable to store result as “outResult”.



Step 11: In Convert java Object give outResult in Java Object and create a variable to store the converted java object as result.

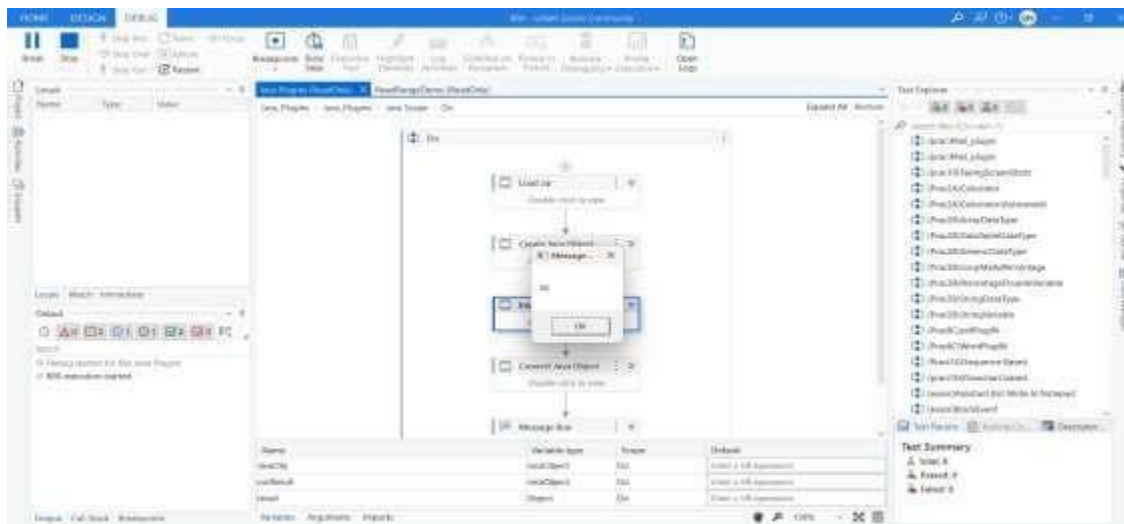


Step 12: Lastly, display the output by using message box activity in text field write **result.ToString** to convert int into String .



Step 13: R\$un.

Output:-

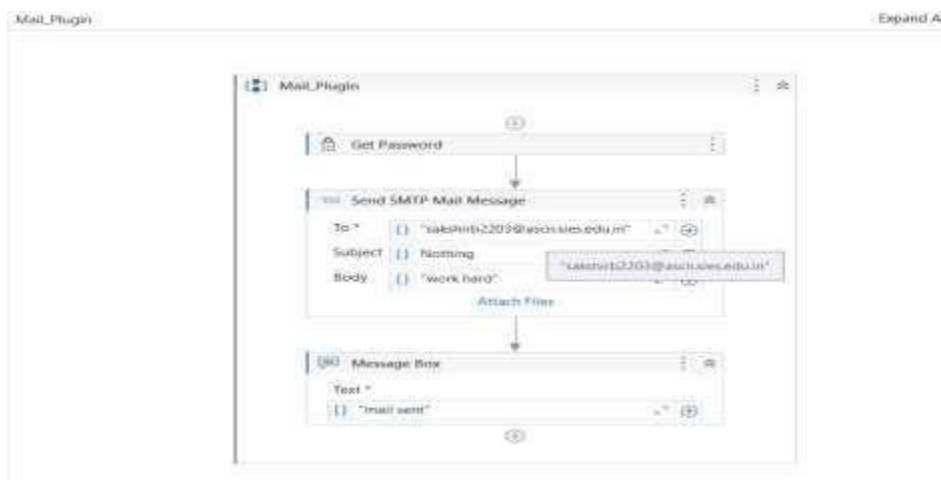


Result is 50 because we gave parameters as 20 and 30.

ii. Mail Plugin

Step 1: Install UiPath.Mail.Activities.

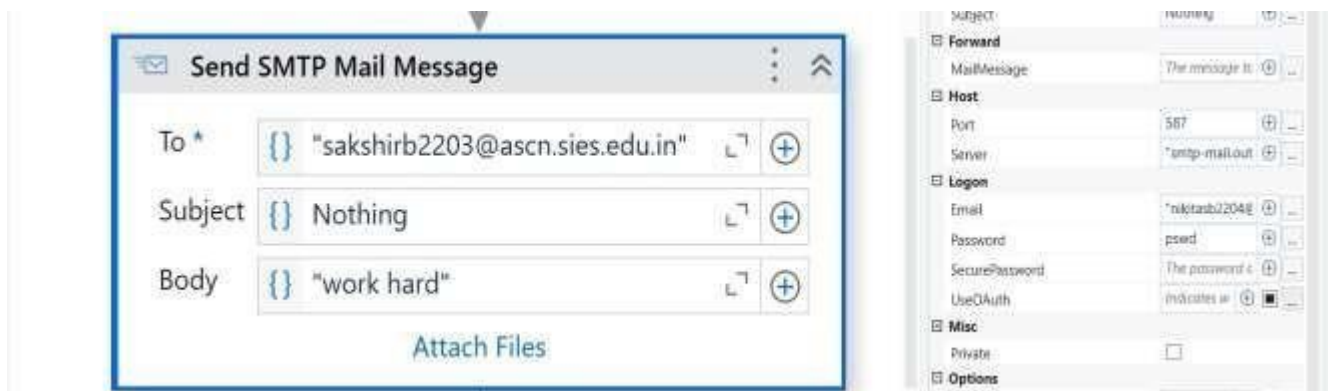
Step 2: Drag and drop Get Password activity.



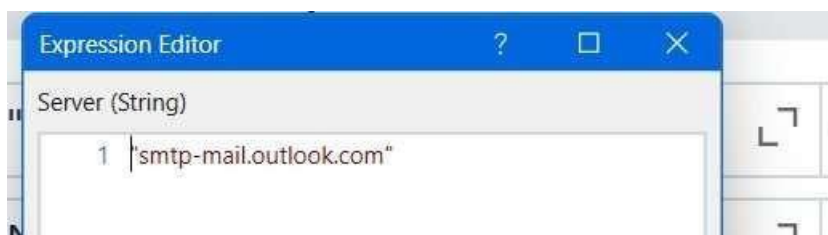
Step 3: In Get Password properties, enter password and create variable to store the password.



Step 4: In Send SMTP Mail Message properties



Step 5: Enter port as 587, Server as

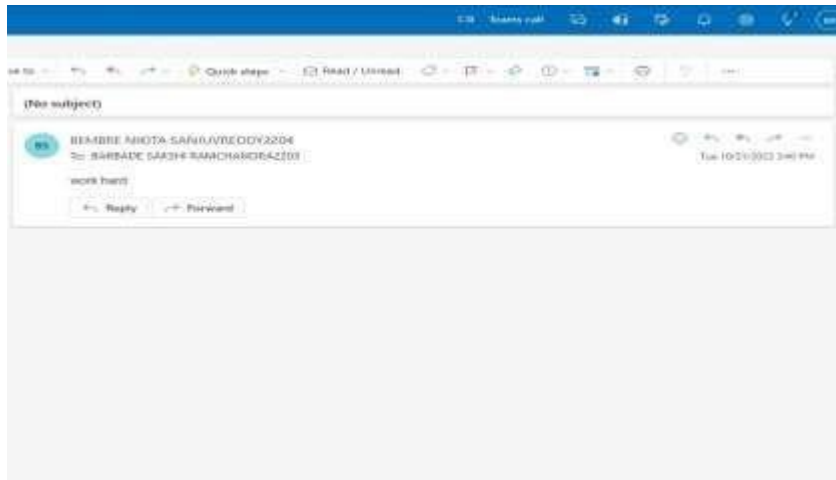


In email, enter your email id and in password give pswd as input where password is stored.

Step 6: Lastly, In Message box “Mail Sent”.

Step 7: Run

Output:-

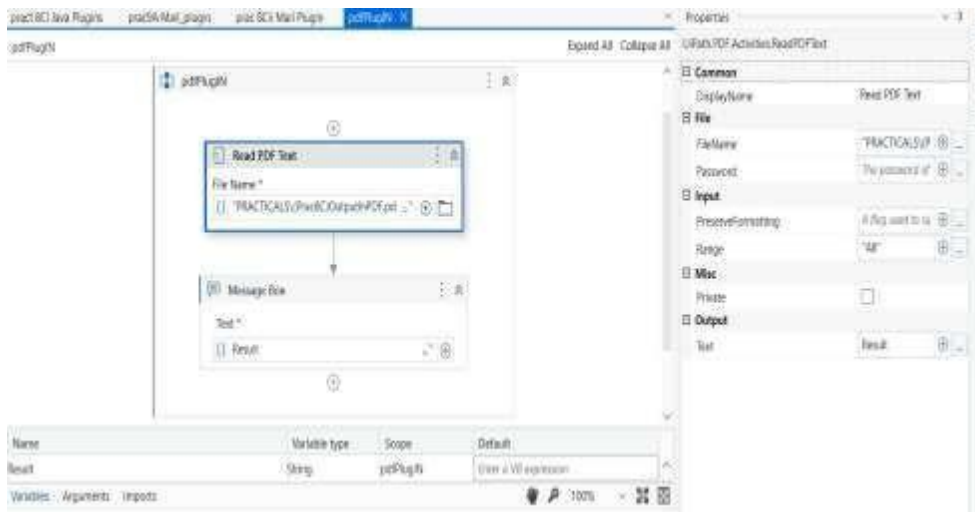


iii. PDF Plugin

Step 1: Install UiPath.PDF.Activities package.

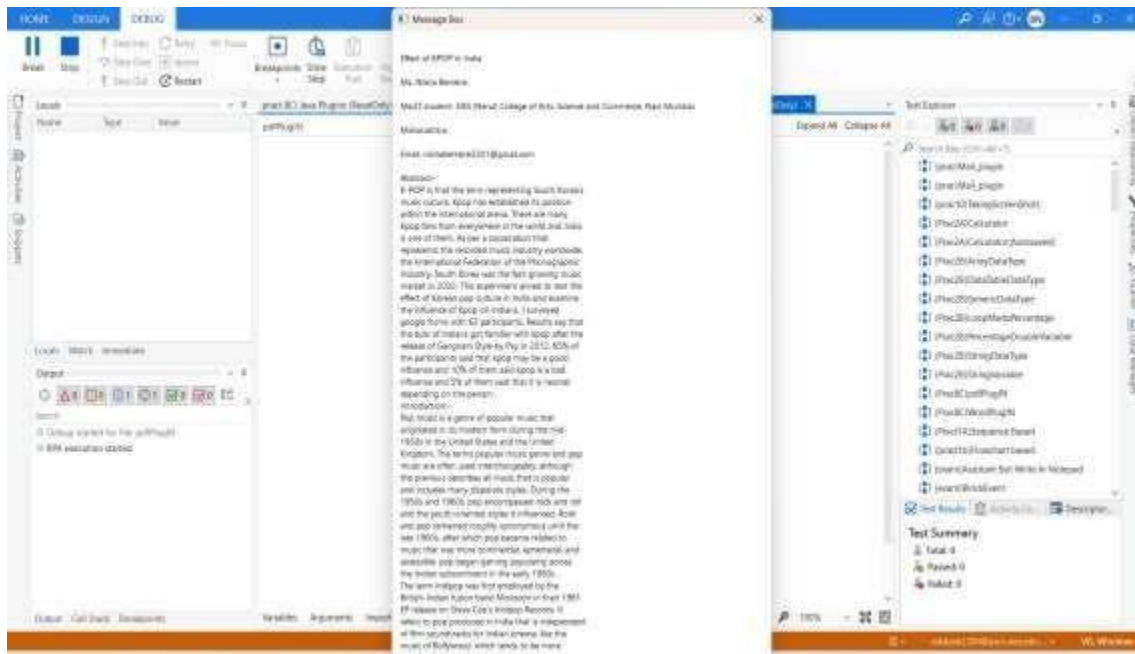
Step 2: Drag and drop Read PDF Text activity give 1 pdf as input and create a variable Result to store it. Give range as “All”.

Step 3: Drop message box and display the Result.



Step 4: Run.

Output:-



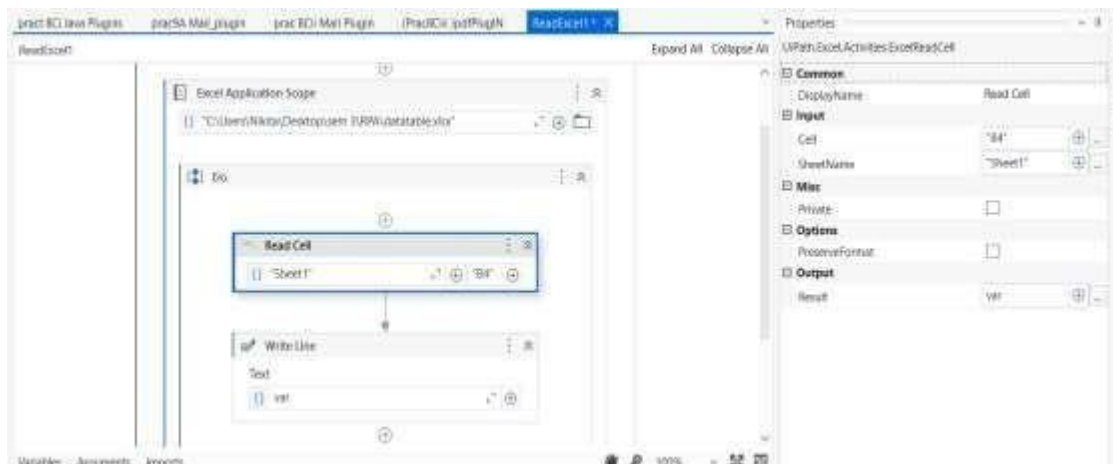
iv. Excel Plugin

Step 1: Create Excel sheet.

Step 2: Drag and drop Excel Application Scope, give created excel sheet as input.

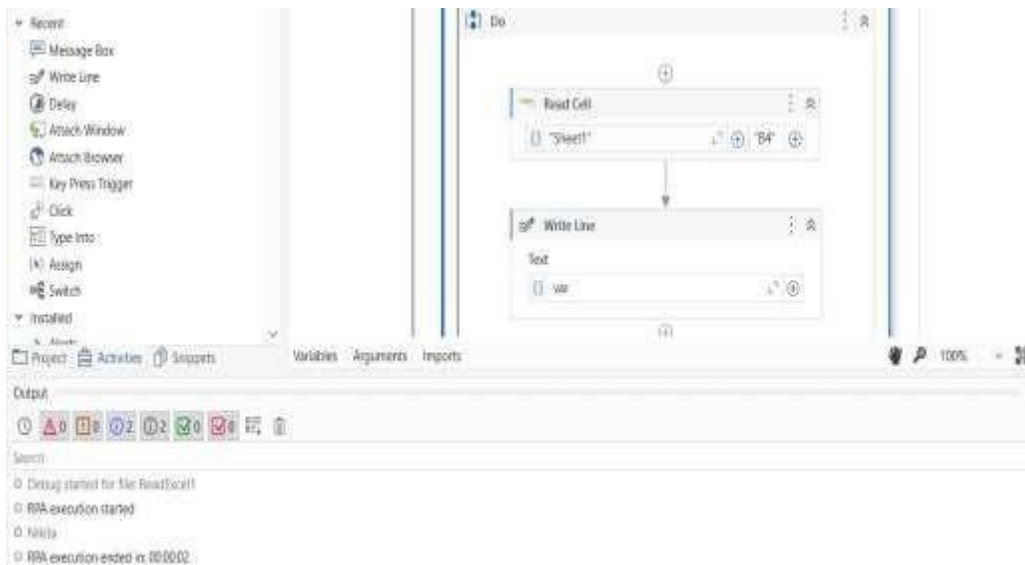
Step 3: In Do, drop Read cell activity enter the cell to read. Create a variable var to store the readed value.

Drop write to display var.



Step 4: Run.

Output:-



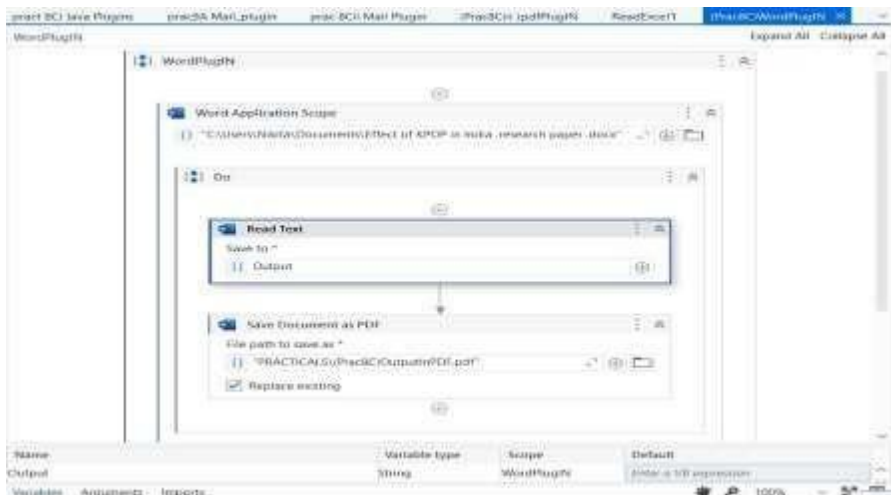
v. Word Plugin

Step 1: Install UiPath.Word.Activities package.

Step 2: Drag and drop Word Application Scope. Give doc as input.

Step 3: In Do, drop Read Text activity, create a variable Output to store the value.

Step 4: Drop an Save Document as PDF activity, it will convert document in pdf. Give Blank pdf file as input. If pdf is not empty then select Replace existing.



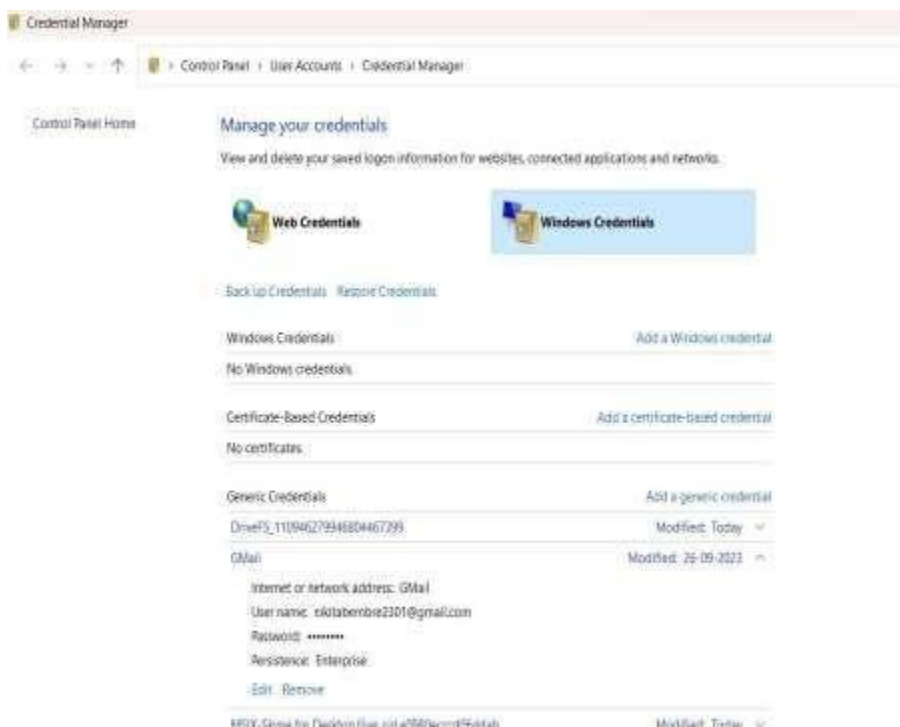
Step 5: Run

Output:-



vi. Credential Management

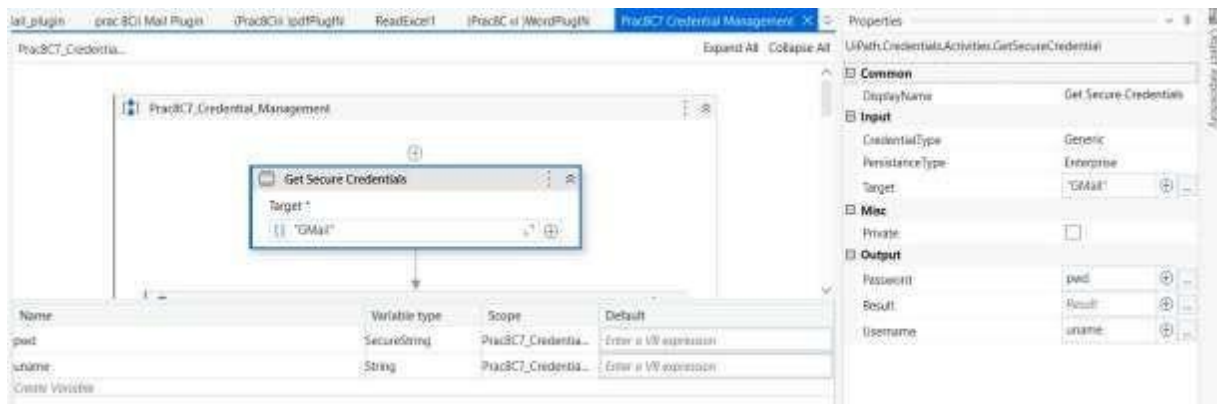
Step 1: Create Windows Credentials as (Add a generic credential) GMail.



Step 2: Open UiPath, Create a new Sequence.

Step 3: Drag and drop Get Secure Credential. Give created credential as input.

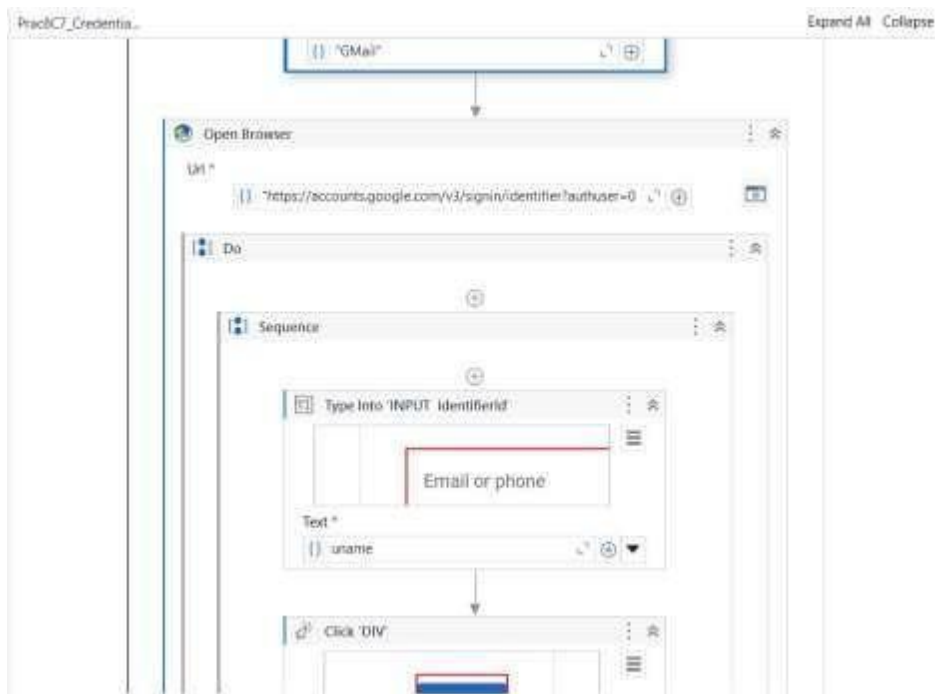
Step 4: In properties, give create 2 variables uname and pwd to store the username and password respectively. give pwd SecureString as variable type.



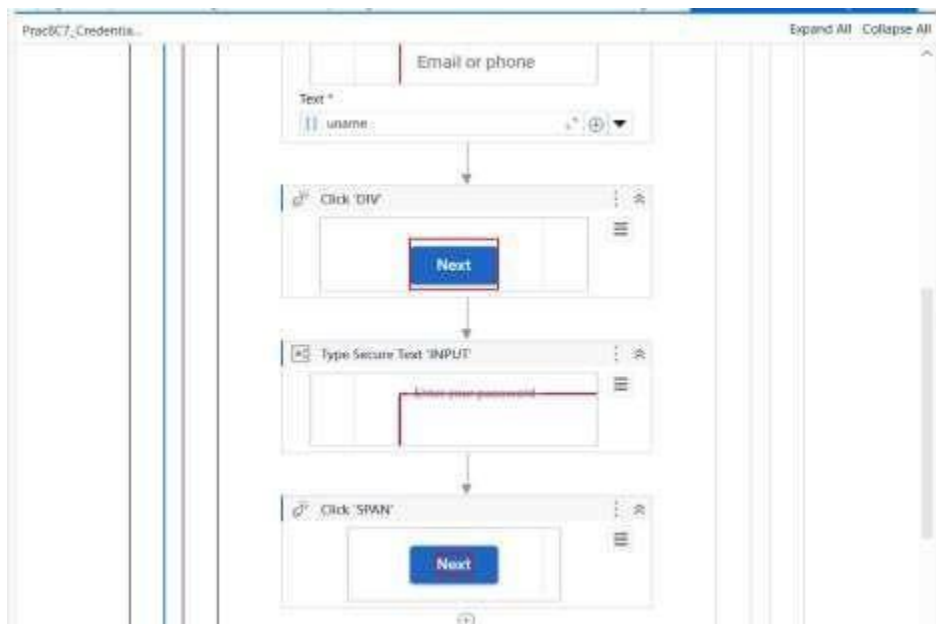
Step 5: Drop an Open browser activity, give gmail login page url.

Step 6: In Do, drop a sequence.

In that, drop Type into activity indicate it on Email field. Drop click activity to click Next button.

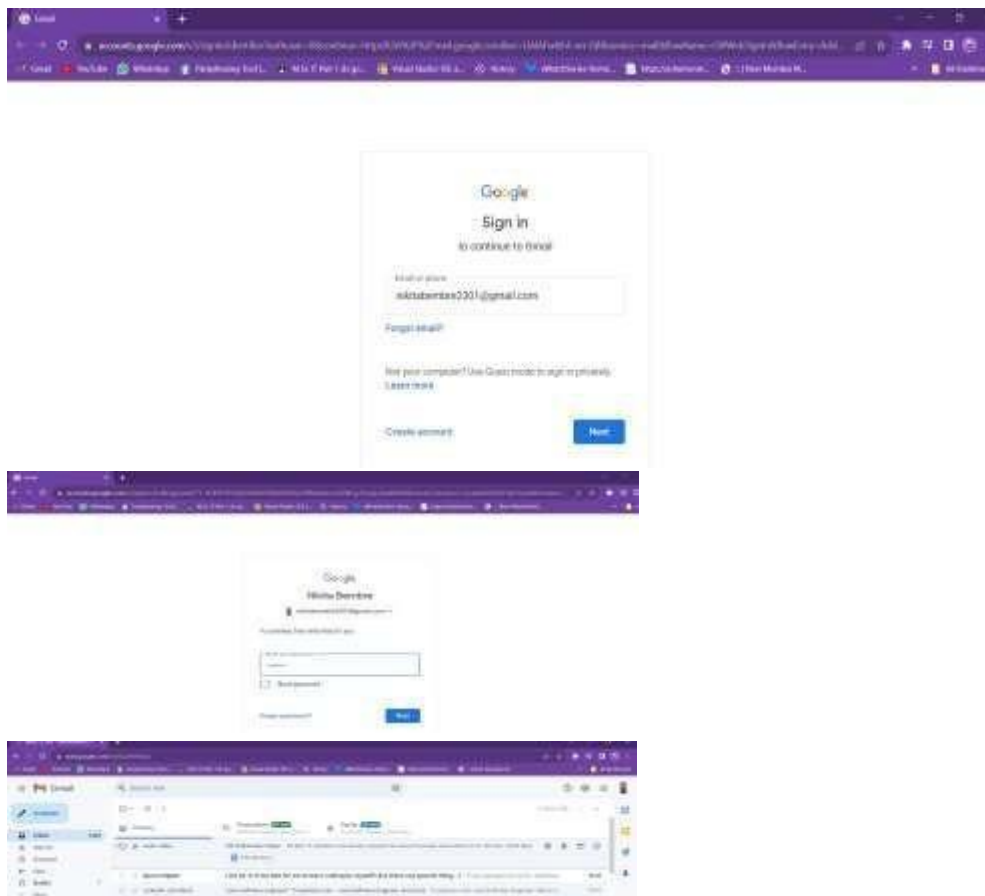


Step 7: Drop Type Secure Text activity and indicate it on Password field, again drop click activity to click Next button.



Step 8: Run.

Output:- Logged In.



Practical No. 9

9. a. Automate the process of send mail event (on any email).

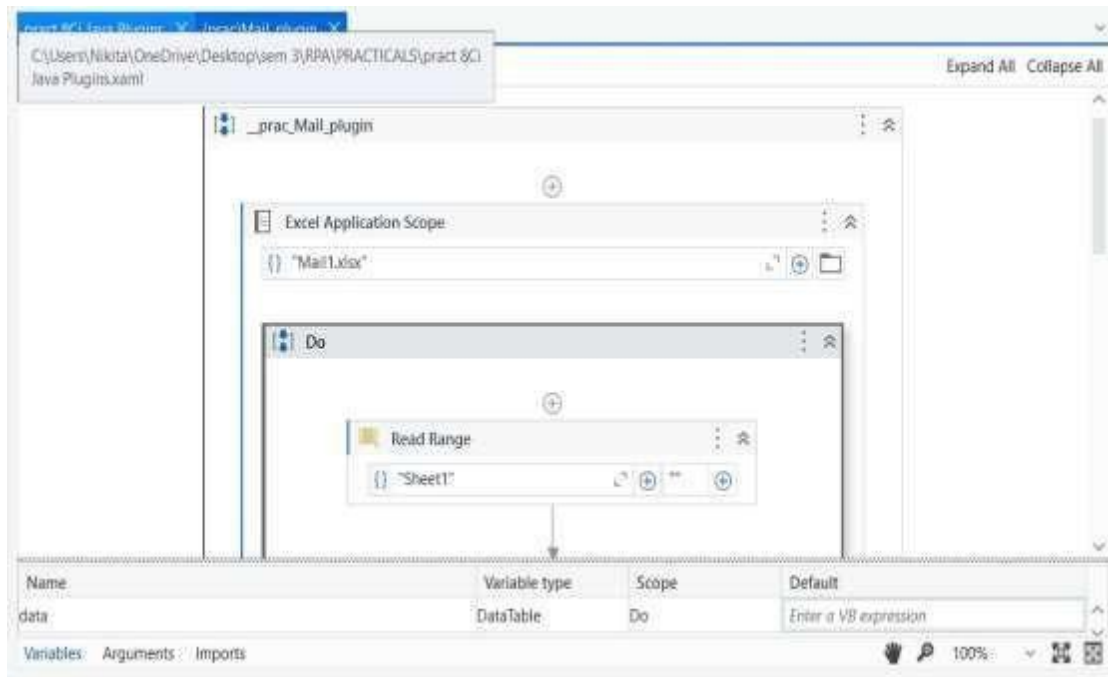
Step 1: Install UiPath.Mail.Activities package.

Step 2: Create a excel sheet as

	A	B	C
1	Srno	Name	Email
2	1	Sakshi Barbade	sakshirb2203@ascn.sies.edu.in
3	2	Sanket Deshmukh	sanketdeshmukhsd2206@ascn.sies.edu.in
4	3	Shruti Gadarla	shrutisg2207@ascn.sies.edu.in
5	4	Sakshi Singh	shrutisg2207@ascn.sies.edu.in
6			
7			
8			

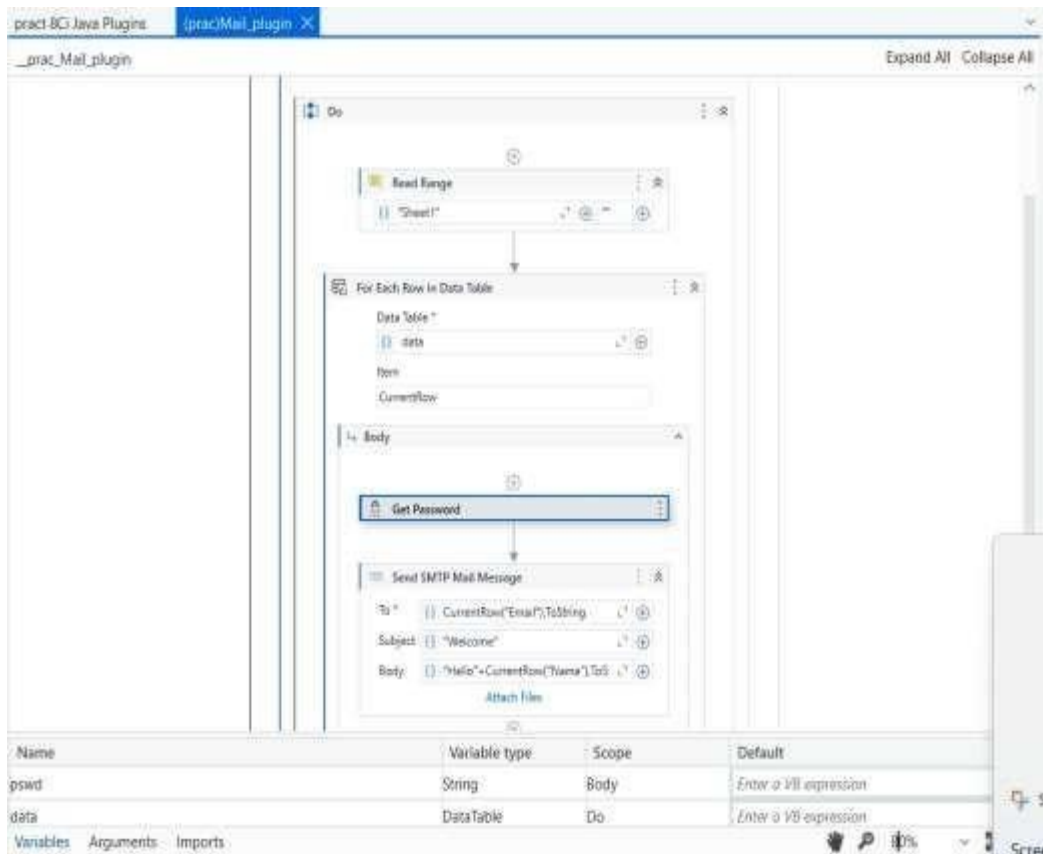
Step 3: Create a new Sequence.

Step 4: Drag and drop Excel Application scope.



Load the excel sheet created above.

Step 5: In do, drop Read Range activity to read the excel sheet, and create a variable to store the data with DataTable variable type.



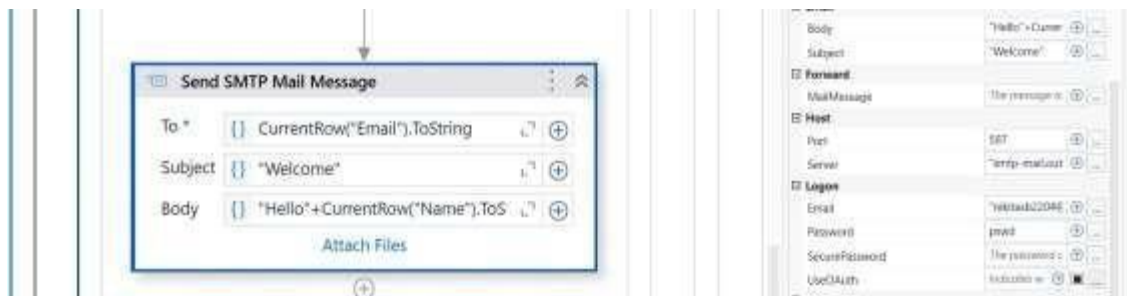
Step 6: To send automated mail to everyone in excel sheet, so drop an For Each Row in Data Table activity.

Give it data as input where data table is stored.

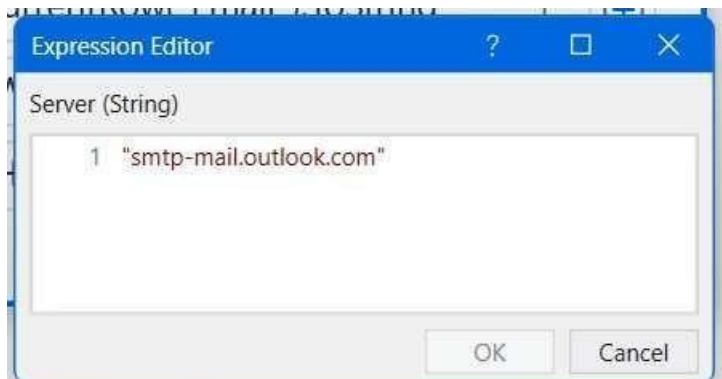
Step 7: In Body, drop an Get Password activity,



Enter password in password field and create pswd variable to store password. Drop a Send SMTP Mail Message activity.



Step 8: In properties Ribbon, Give port as 587, Server as

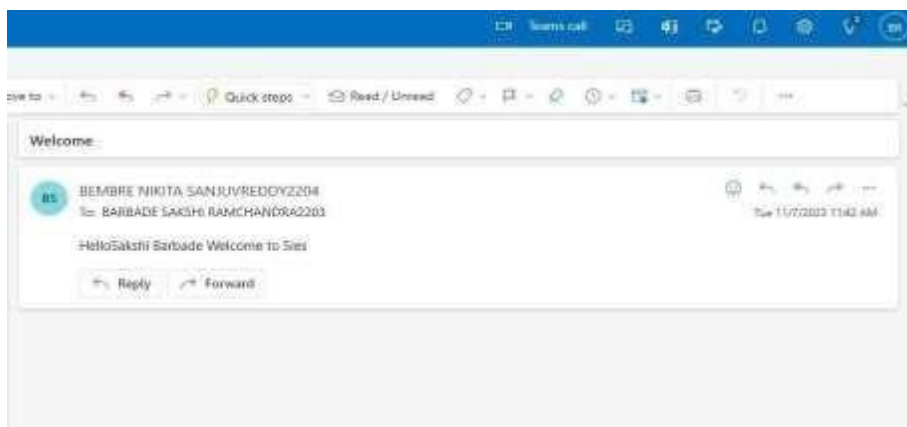


In email enter your email id, in password give a pswd as input where password is stored. In body,



Step 9: Run

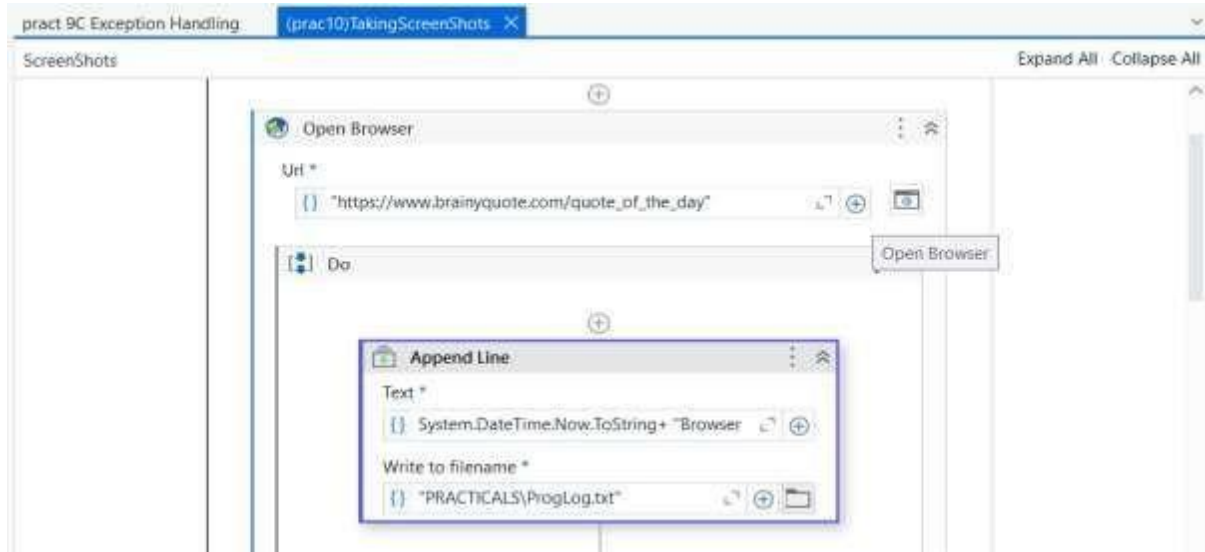
Output:-



9. b. Automate the process of logging and taking screenshots in UiPath.

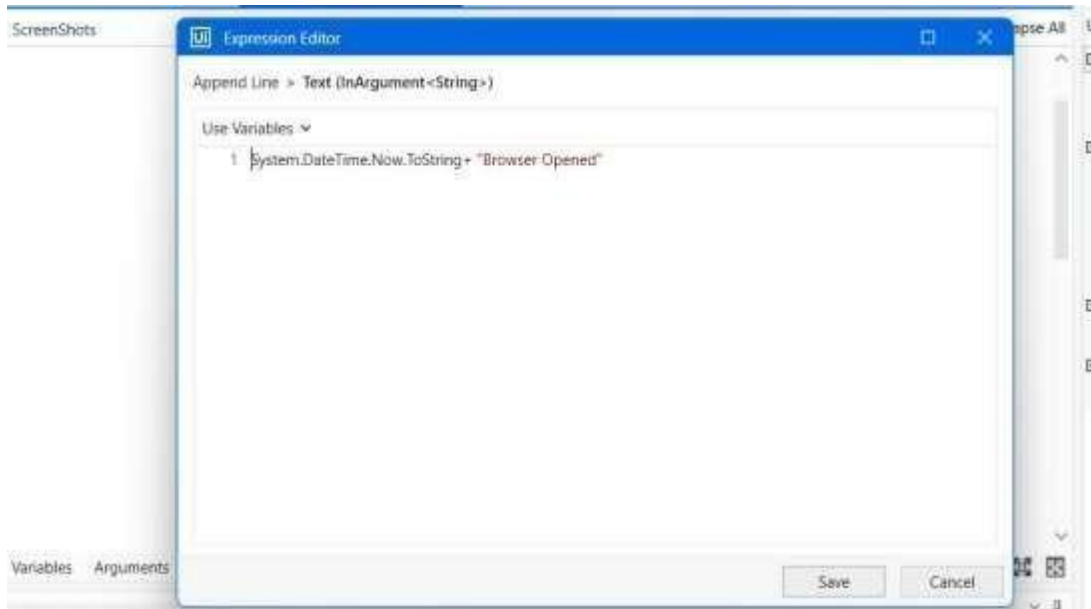
Step 1: Create a new Sequence.

Step 2: Drag and drop Open Browser activity, Give https://www.brainyquote.com/quote_of_the_day as url.



Step 3: To save time in prolog.txt

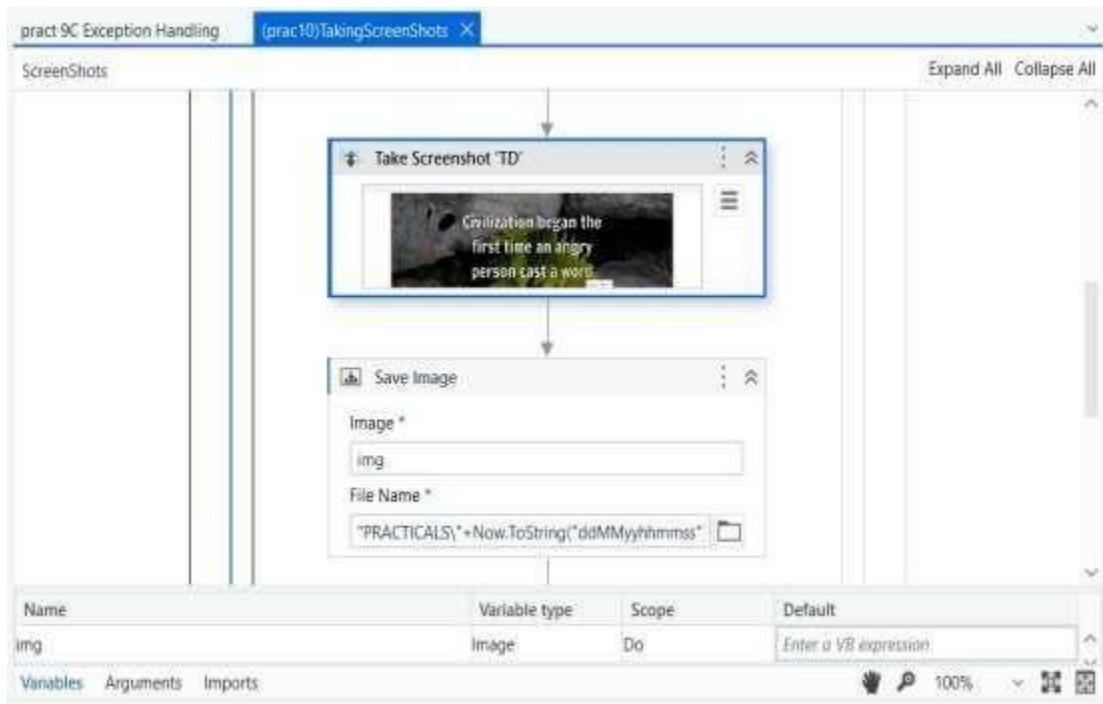
In write to filename, give a text file path to save the time.



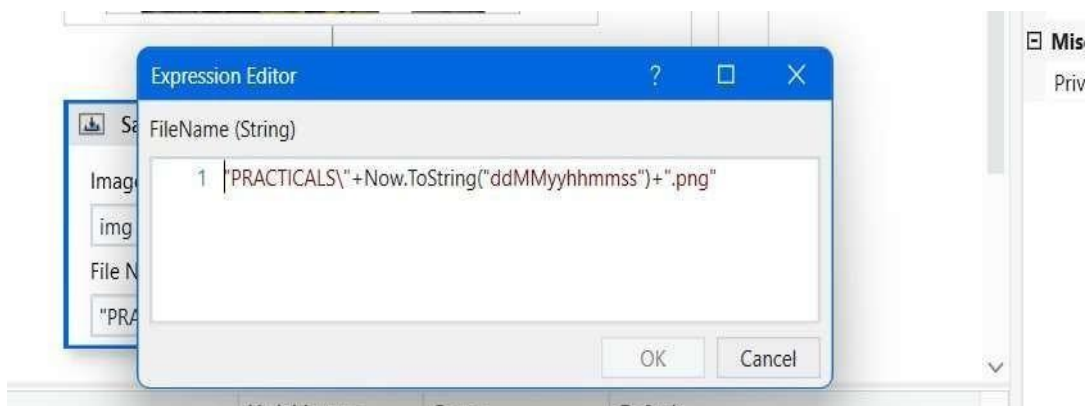
System.DateTime.Now is an inbuilt function in uith to get current time and .ToString is to convert datatype to String.

Step 4: Drop an Take Screenshot activity, indicate it on image which is to be saved. Create variable img to save the img.

Step 5: Drop Save Image activity, give img as input.



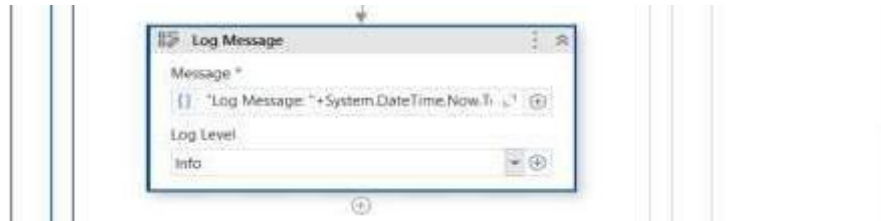
In File Name field, give “path”+filename+“.png” extension to save in image format.



Step 6: Drop Log Message activity, In Message field write

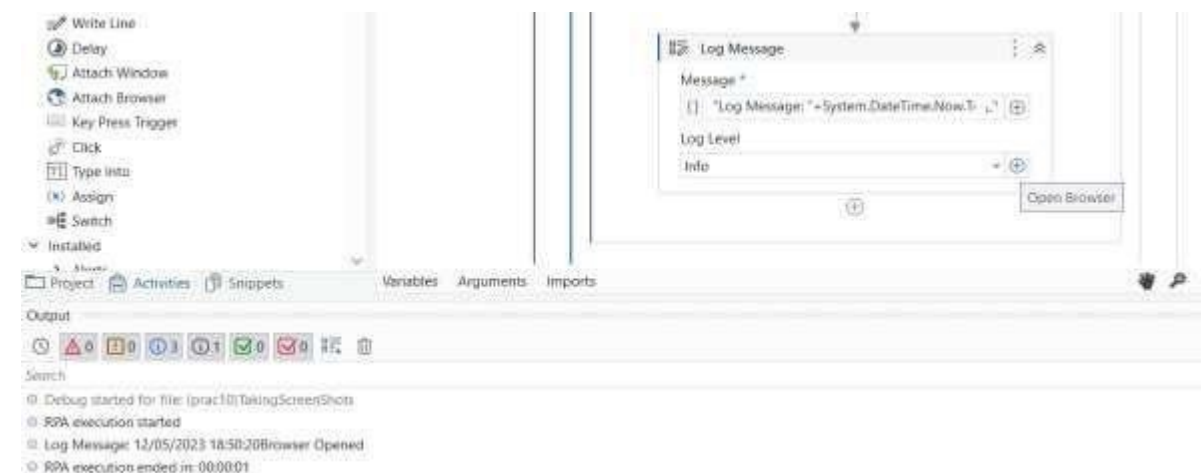
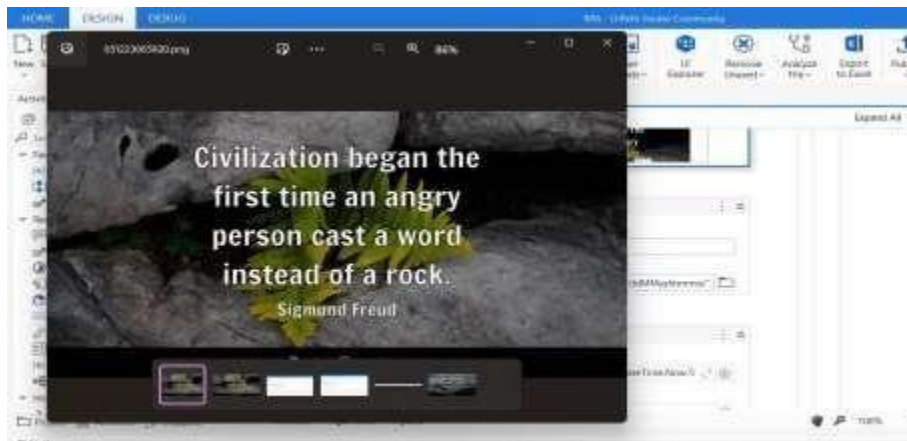
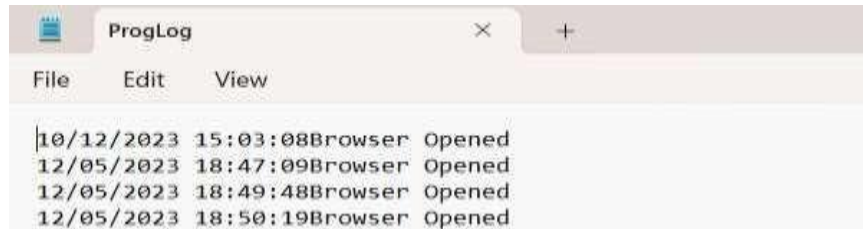
"Log Message:"+System.DateTime.Now.ToString+"Browser Opened".

Give Log Level as info



Step 7: Run

Output:-

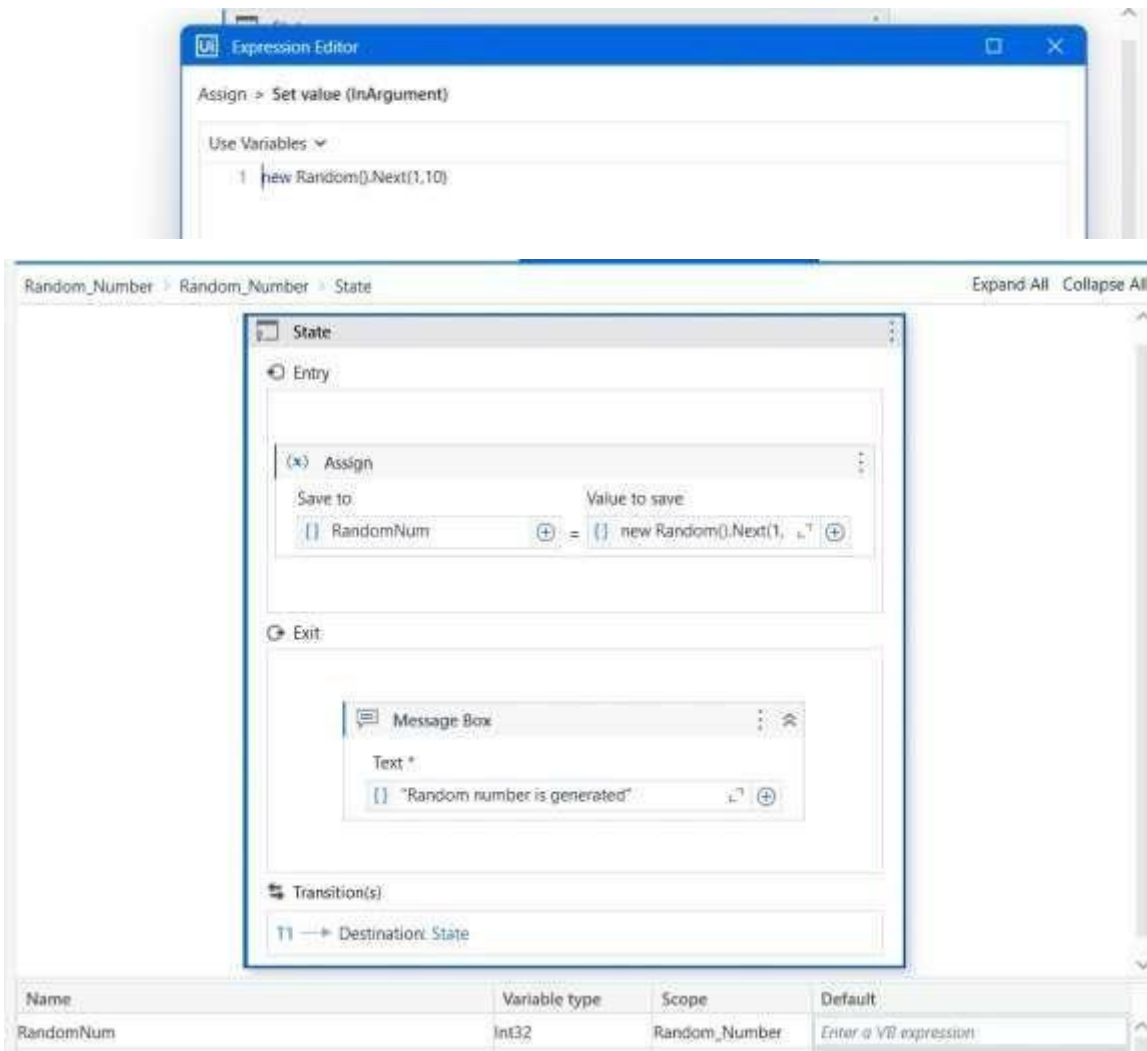


9. c. Automate any process using State Machine in UiPath.

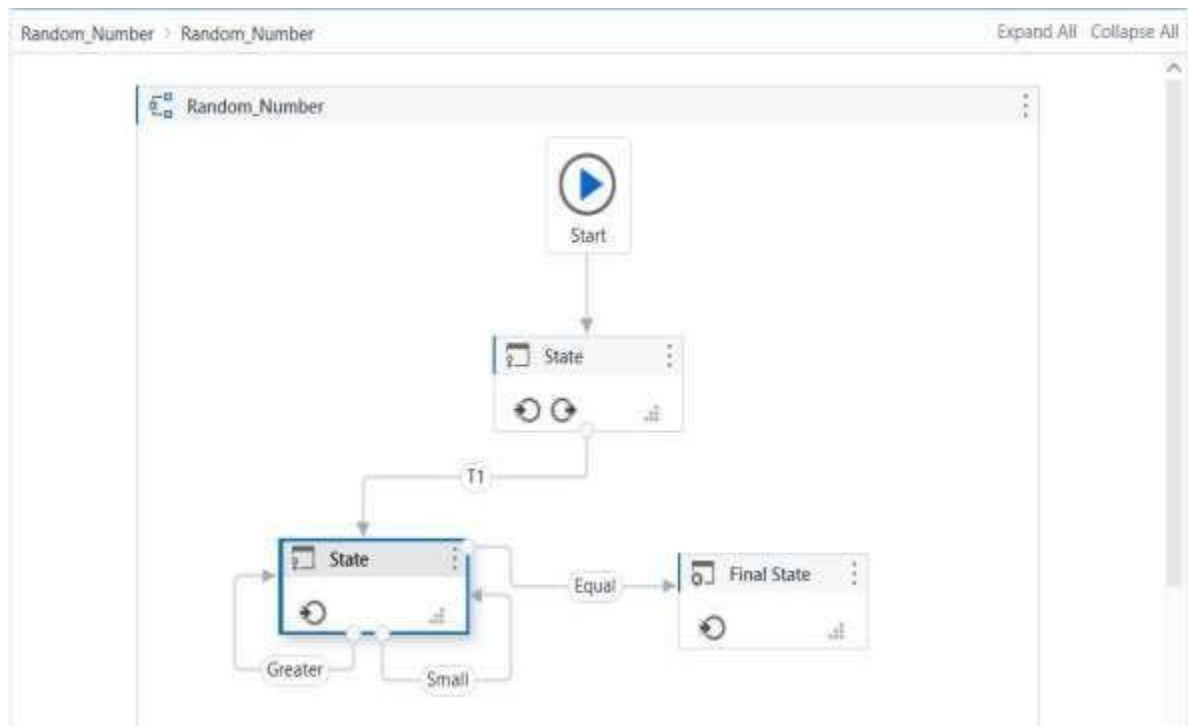
Step 1: Create new State Machine.

Step 2: Drag and drop 2 State and 1 Final State.

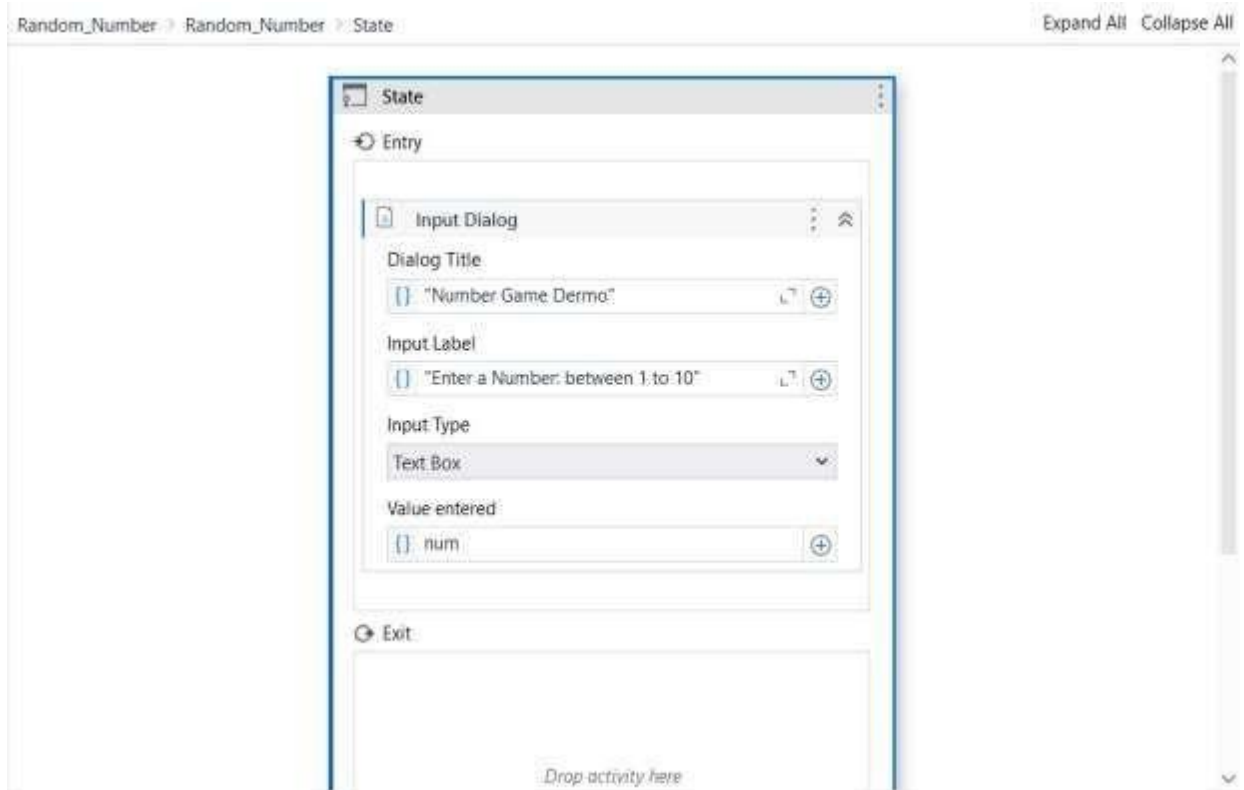
Step 3: In 1st state, drop Assign activity. create RandomNum variable to save random number. Value to save in RandomNum is generated as



In Exit, drop Message bos as “Random number is generated”. 1st state should be connected to 2nd state as



Step 4: In 2nd state, Drop an Input Dialog box. Create a variable num to store entered value.

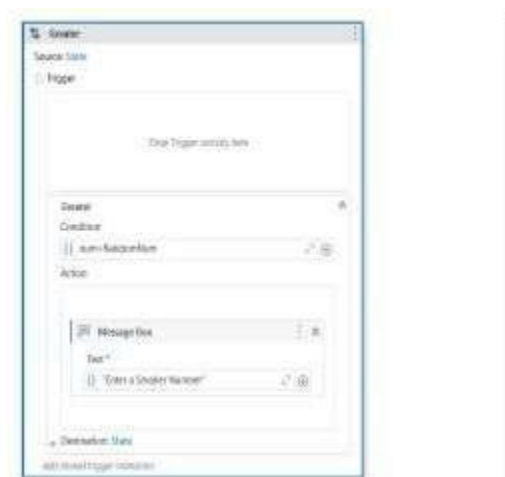


2nd will have 3 transition as

Transition(s)			
Greater	→	Destination: State	
Equal	→	Destination: Final State	
Small	→	Destination: State	

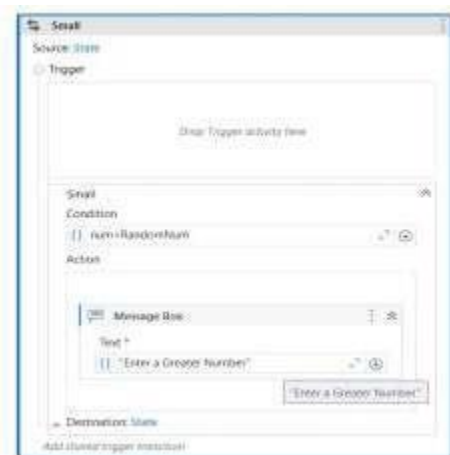
Name	Variable type	Scope	Default
RandomNum	Int32	Random_Number	Enter a VB expression
num	Int32	Random_Number	Enter a VB expression

Step 5: In 1st transition Greater,



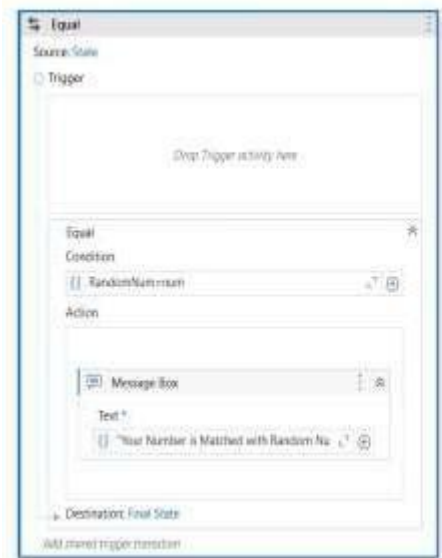
Give Condition as $num > RandomNumber$ Message box will display as “Enter a smaller Number”.

Step 6: In 2nd transition Small



Give Condition as $num < RandomNum$ Message Box will display as “Enter Greater Number”.

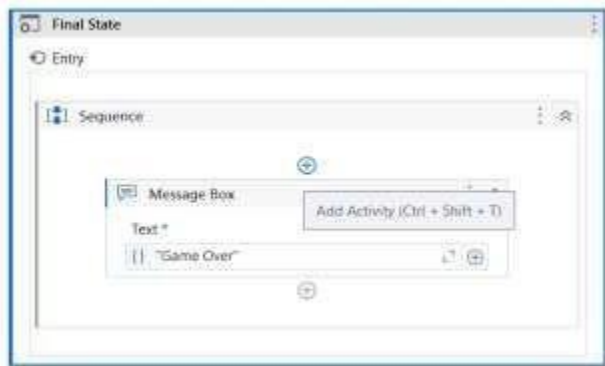
Step 7: In 3rd transition Equal Final State,



Give Condition RandomNum=num Message Box will display “Your number is matched with Random Number”.

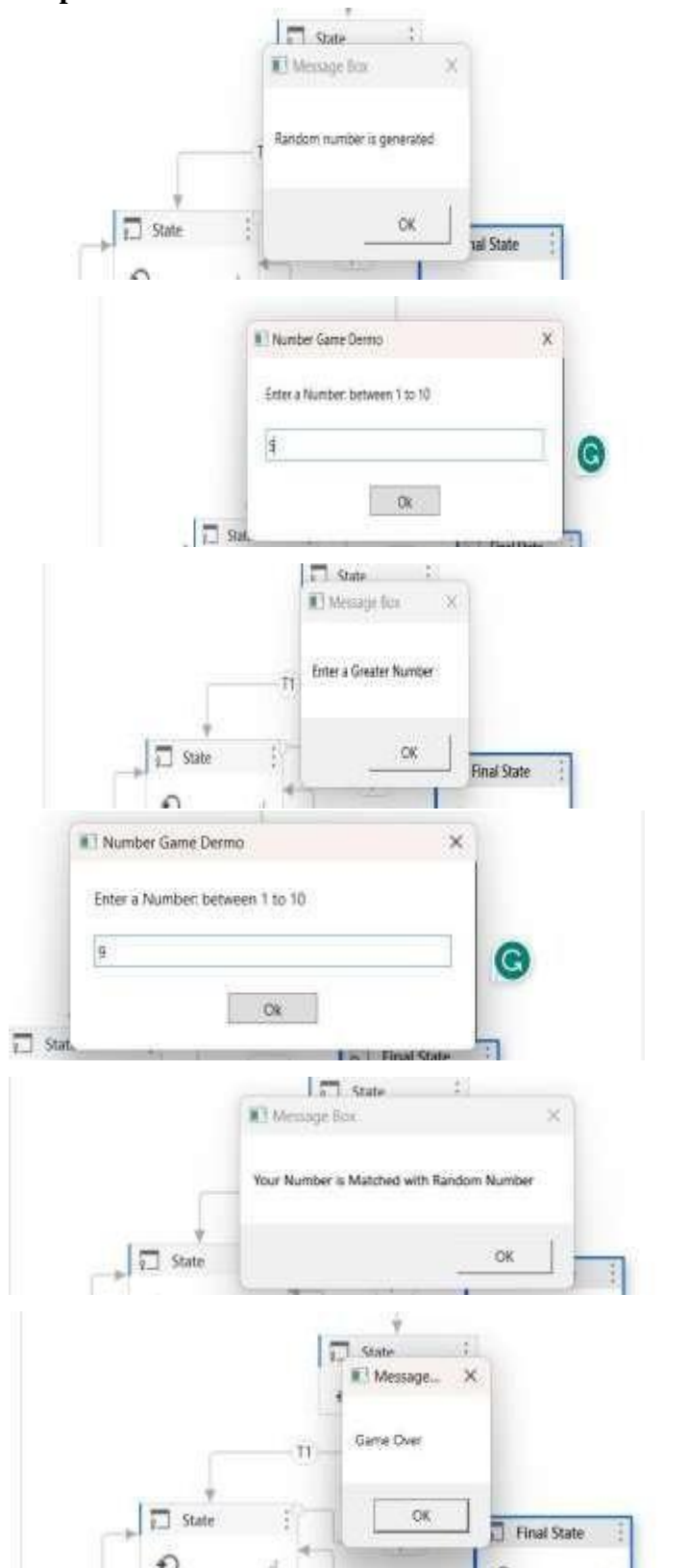
Step 8: Lastly, connect 3rd transition to Final state.

In Final State, Drop sequence, Message box will display “Game Over”.



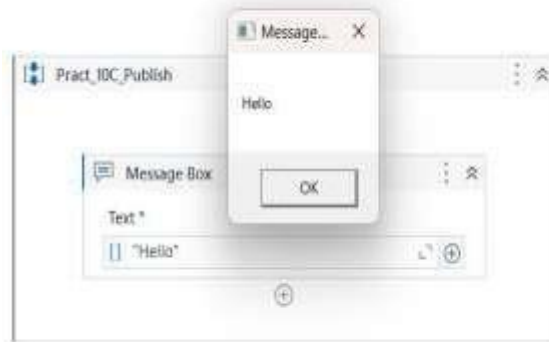
Step 9: Run.

Output: -



9. d. Demonstrate the use of publish utility.

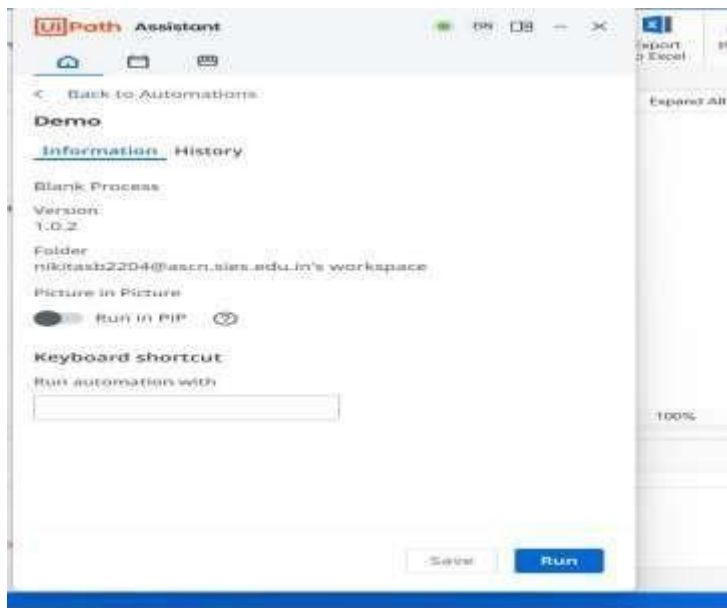
Step 1: Create a simple project and Run



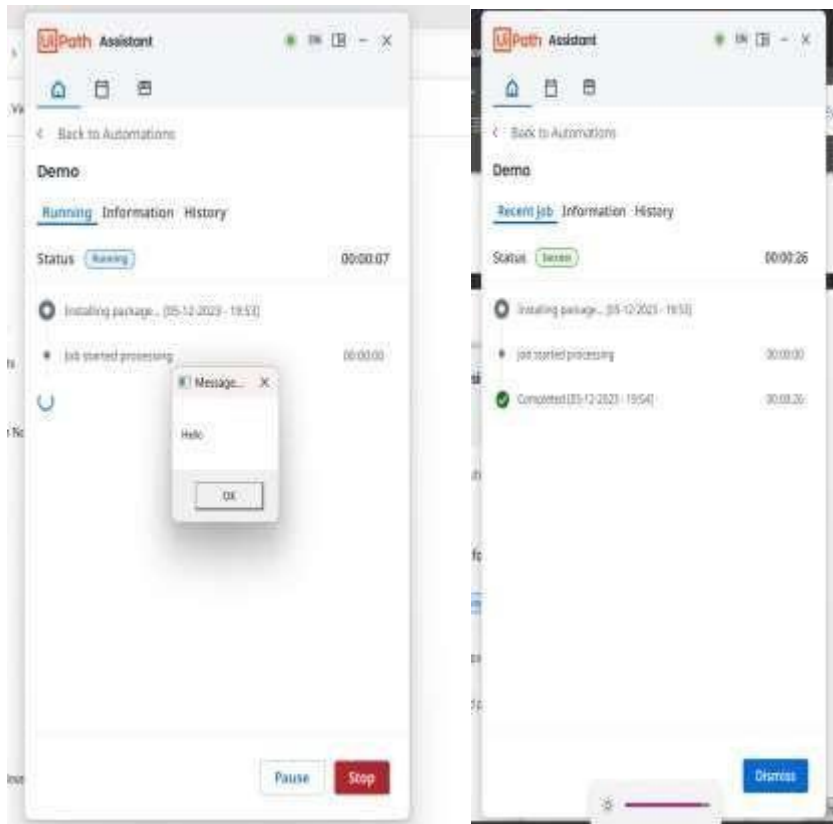
Step 2: Click on Publish in Design Ribbon.

Once it is Published.

Step 3: Open UiPath Assistant. Refresh it. And install the published project.



Step 4: Click on Run.

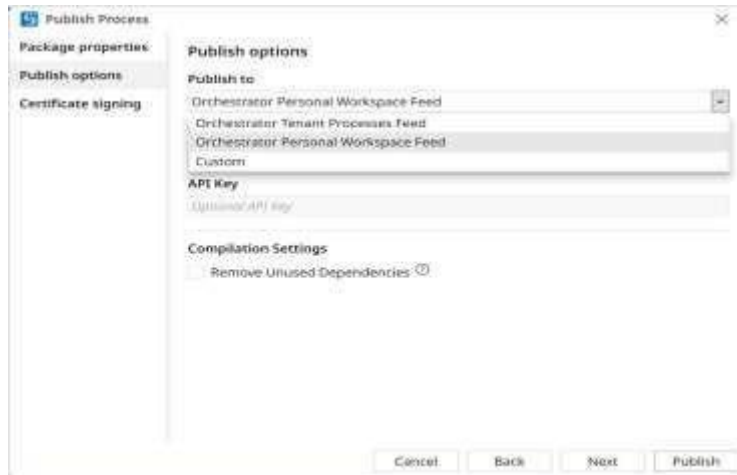


Practical No. 10

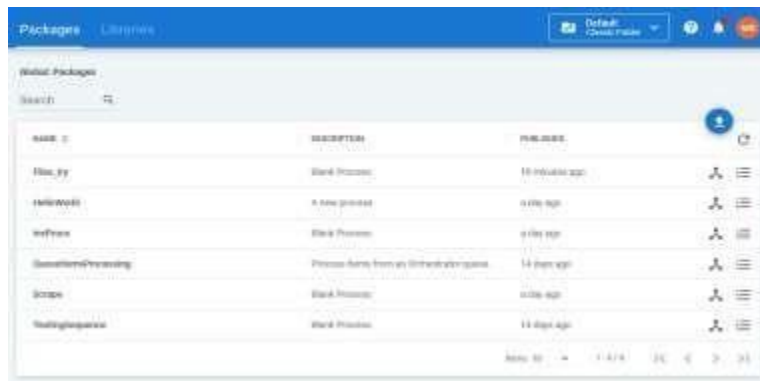
Orchestrator management and mini project:

10. a. Deploy bots to Orchestrator

Step 1: Publish the UiPath Project. Open the completed automation project in UiPath Studio and click the Publish button. Select Orchestrator Tenant Processes Feed and publish the package.

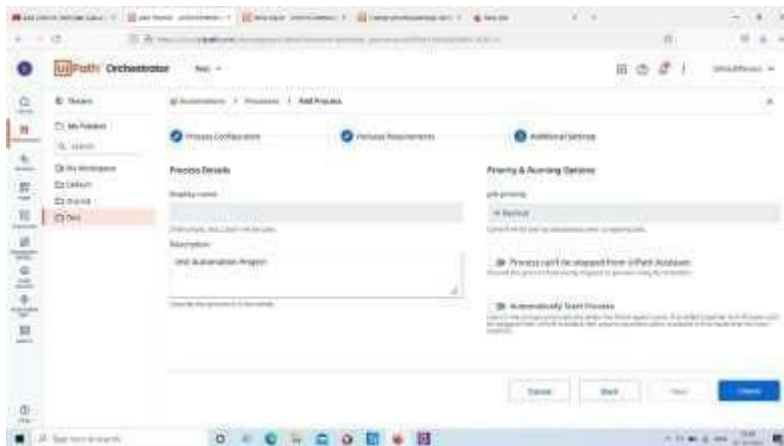


Step 2: Verify the Package. Log in to UiPath Orchestrator and navigate to the Packages section to verify that the published package is available.



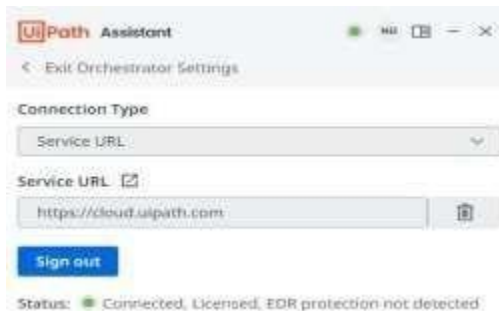
Step 3: Create a Process. Open the Processes section, click Add Process, select the published package, and save the process.

Step 4: Connect the Robot. Launch UiPath Assistant and ensure that the robot is connected to Orchestrator.



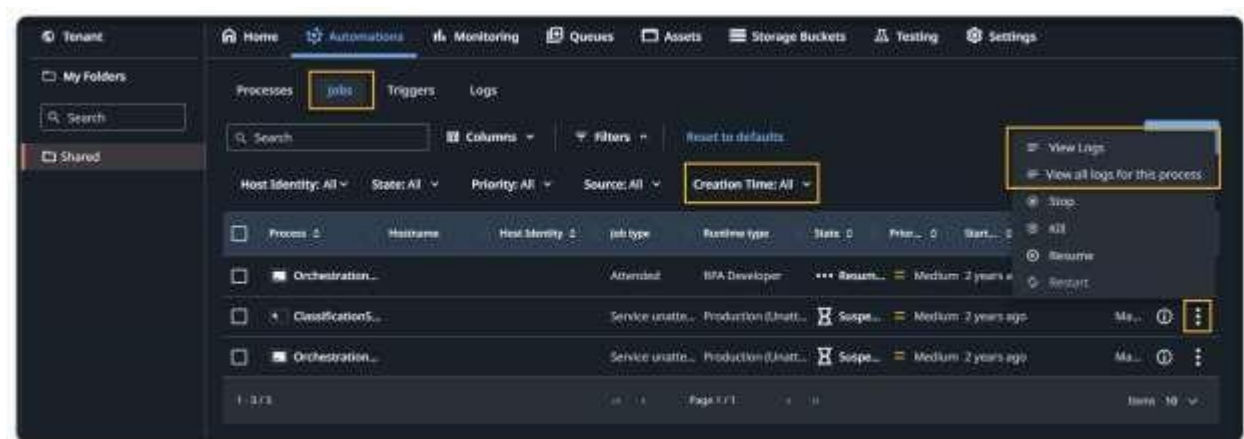
Step 5: Start the Job. Navigate to the Jobs section in Orchestrator, click Start Job, select the created process, and run it.

Output: The job executes successfully, and its status changes to Successful

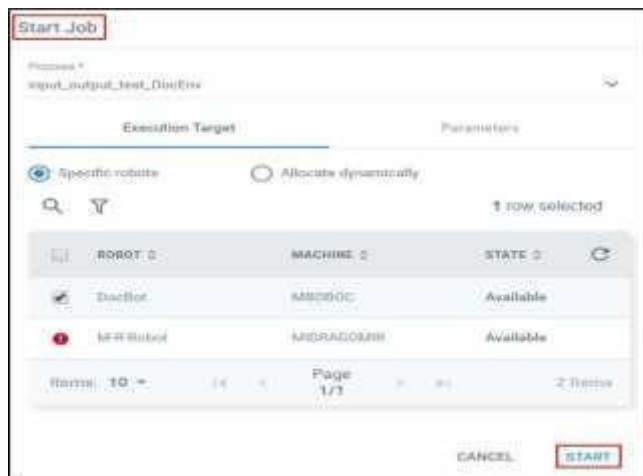


10. b. Run jobs from Orchestrator

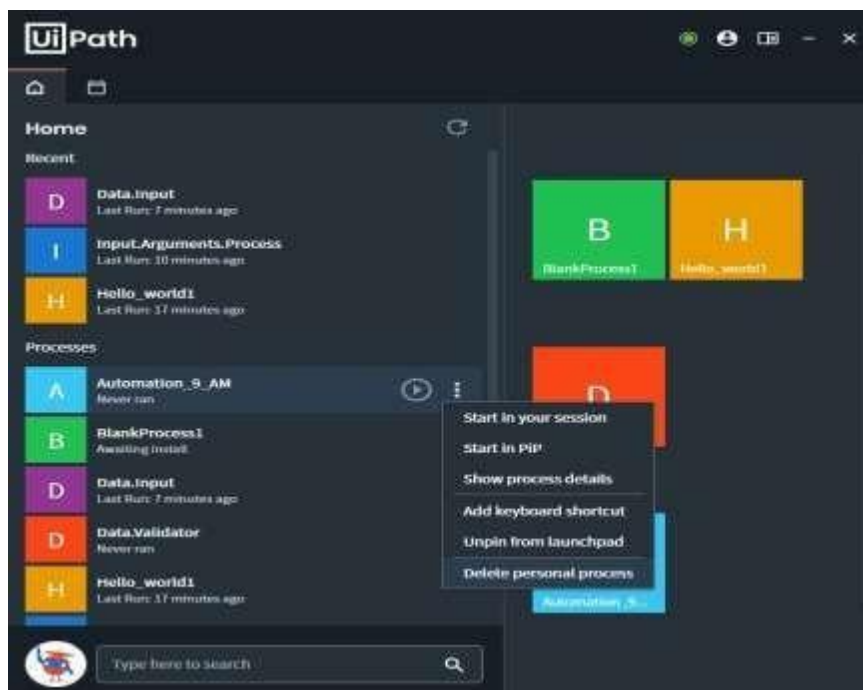
Step 1: Open the Jobs Page. Log in to UiPath Orchestrator and select the required folder. Navigate to the Jobs section.



Step 2: Start a New Job. Click the Start Job button, select the required process and robot account, and then click Start.



Step 3: Monitor Job Execution. Observe the job status on the Jobs page while the robot executes the process.

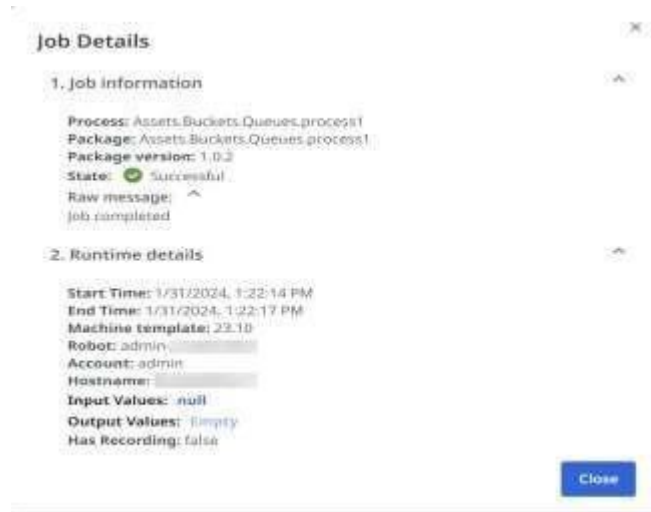


Step 4: View Execution in UiPath Assistant. Open UiPath Assistant on the robot machine to view the running process.



Step 5: Verify Successful Completion. After execution is completed, refresh the Jobs page in Orchestrator.

Output: The job status changes to Successful.

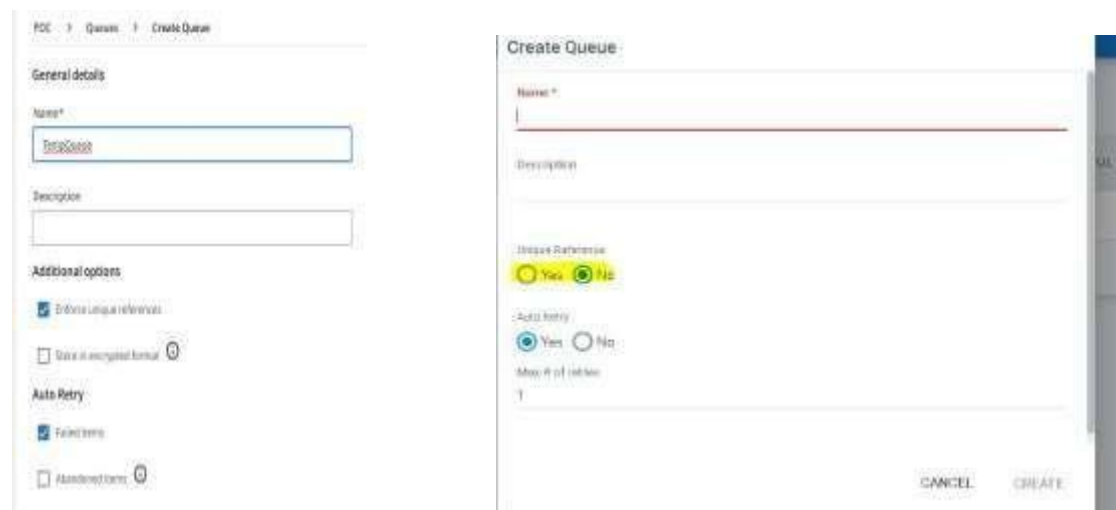


10. c. Queue Introduction:

i. Add items to Queue.

Step 1: Create a Queue in Orchestrator

Open Orchestrator → Queues → Add Queue → Enter queue name (e.g., StudentQueue) → Click **Create**.



Queue Item

General details

Name:

Description:

Additional options

☒ Define unique references

☐ Store in encrypted format

☒ Auto retry

Max # of retries:

Tags

Labels:

Run typing to get a list of possible results

Properties (key-value pairs)

The object contains the properties

Schema Definitions

Specific Data (JSON) Schema:

Output Data (JSON) Schema:

Analytics Data (JSON) Schema:

SLA Predictions

SLA for newly added queue items that don't have a SLA item

☒ Enable SLA for this queue

Retention policy

Retention policies control when and how old queue items are removed. Read the documentation

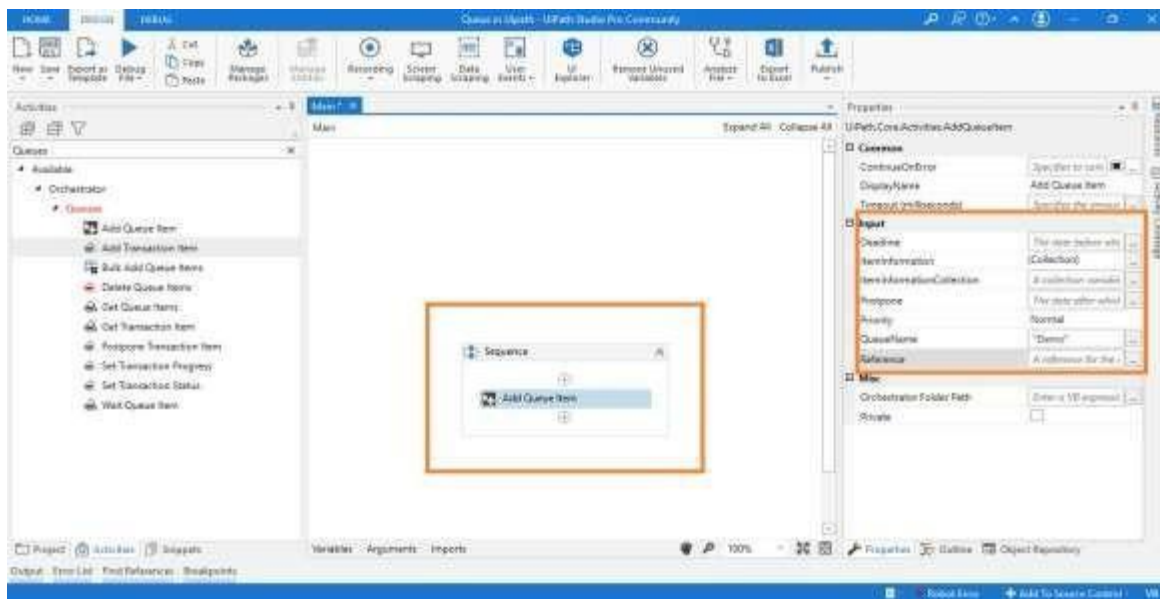
Action: Retention duration: days

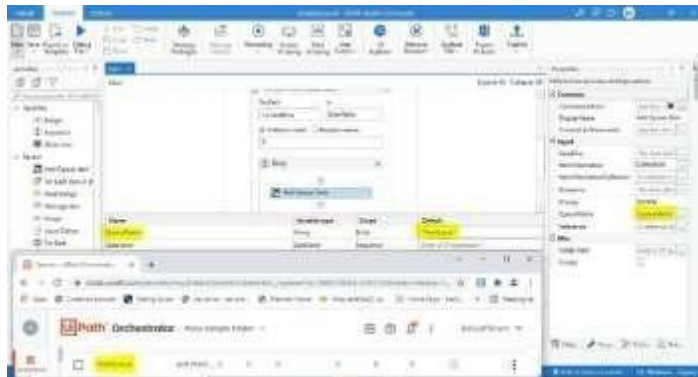
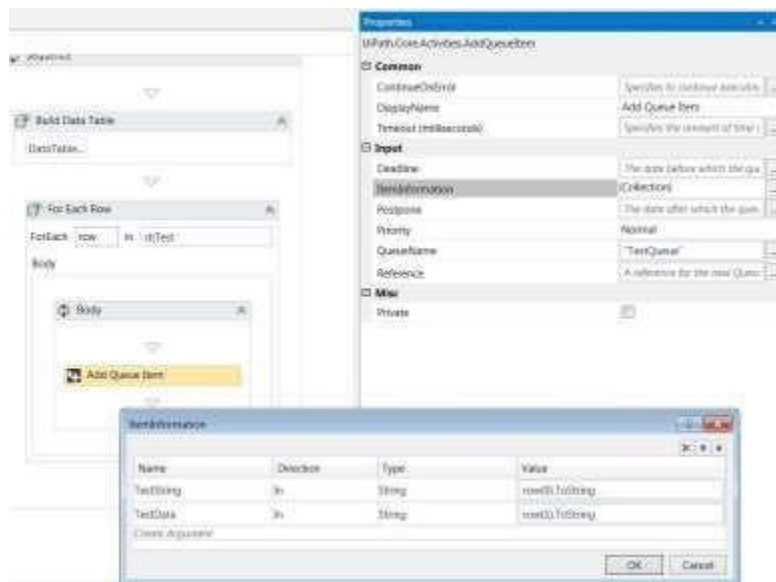
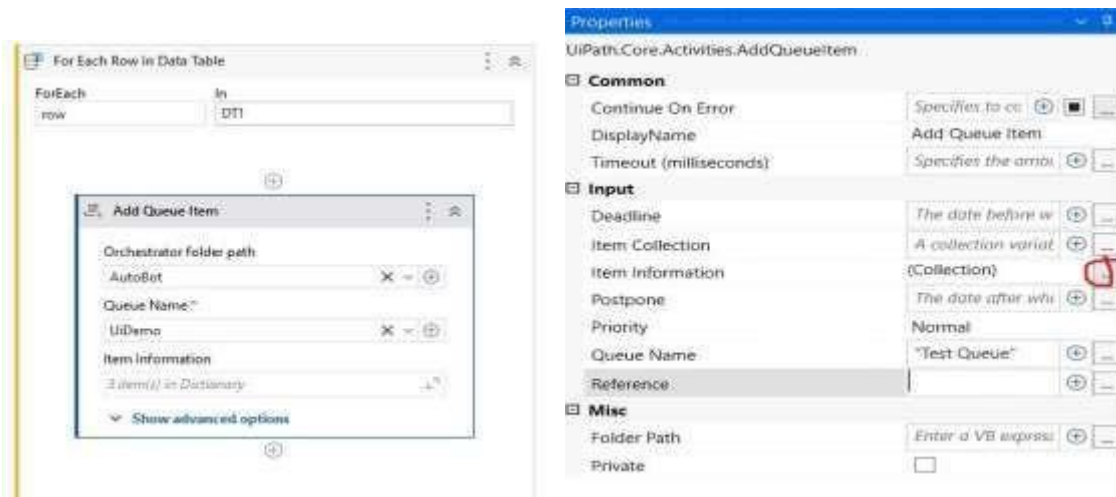
Delete items in a terminal state ☒ that haven't been updated for more than 30 days

Buttons: Cancel, Add

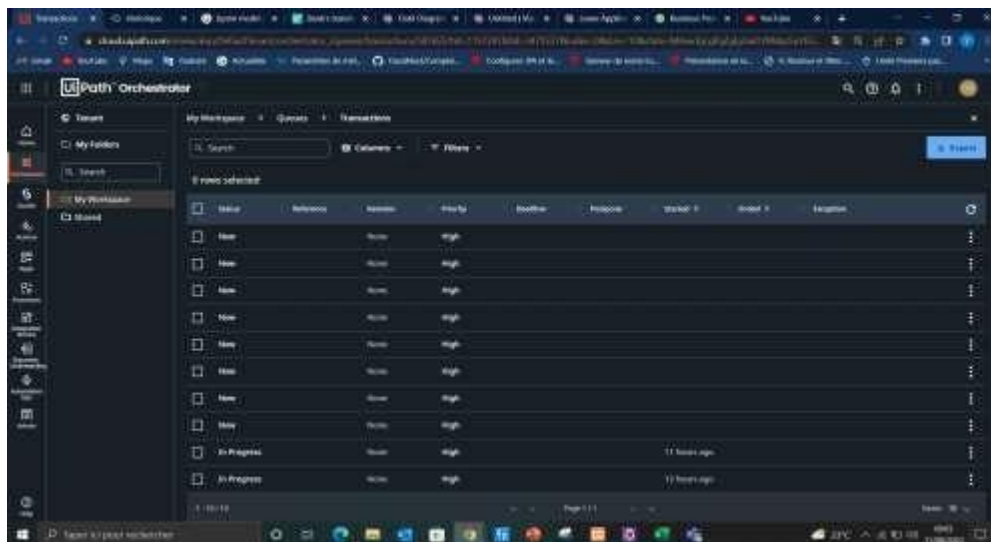
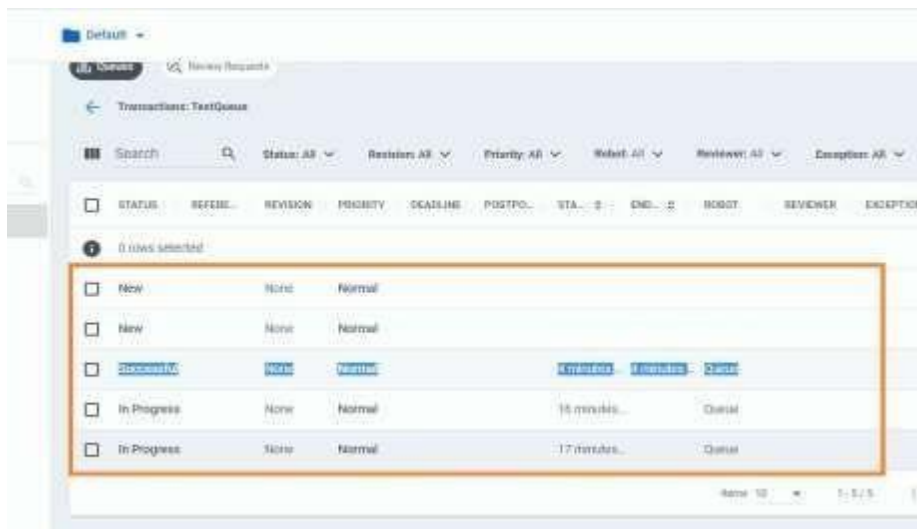
Step 2: Add Queue Item Activity

In UiPath Studio, drag the Add Queue Item activity, select StudentQueue, and enter transaction details.





Step 3: Verify Queue Transactions. Open Orchestrator → Queues → Transactions.



Queues > TestQueue

Search: Status: All #Visible: All Priority: All Robot: All Reviewer: All Exception: All

☐	STATUS	REFERENCE	REVISION	PRIORITY	DEADLINE	POSTPONE	START	ENDED	ROOT	REVIEWER	EXCEPTION
☐	0 rows selected										
☐	New	1	None	High	09/21/2021						
☐	New	2	None	Normal	09/22/2021						
☐	New	3	None	Normal	09/23/2021						
☐	New	4	None	Low	09/24/2021						
☐	New	5	None	High	09/25/2021						

Items: 10 1 / 5 / 5 |< < > >|

ii. Get Queue item from Orchestrator

Step 1: Use Get Transaction Item

Drag the **Get Transaction Item** activity and select StudentQueue.

Main > Sequence

Expand All Collapse All

Sequence

Get Transaction Item

Message Box

TransactionItem.Status

Name	Variable type	Scope	Default
TransactionItem	QueueItem	Sequence	Enter a VBA expression
Create Variable			

Properties

UiPath.Core.Activities.GetQueueItem

Common

ContinueOnError ☒ Specifies to continue

DisplayName Get Transaction Item

Timeout (milliseconds) Specifies the amount of time

Input

QueueName Files

Misc

FolderPath Enter a VBA expression

Private ☐

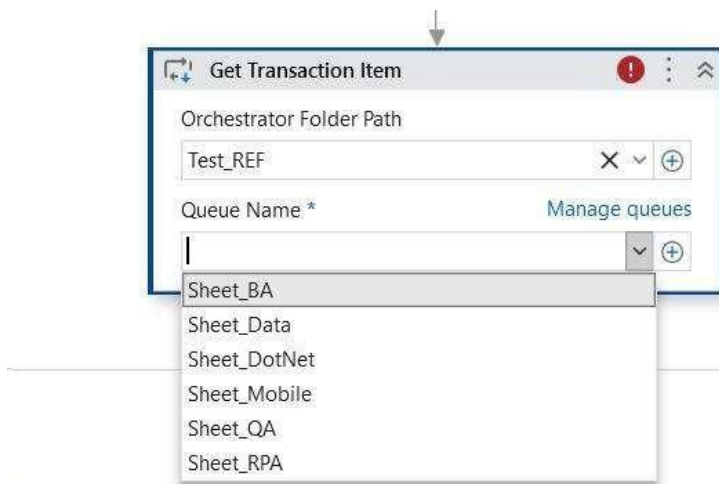
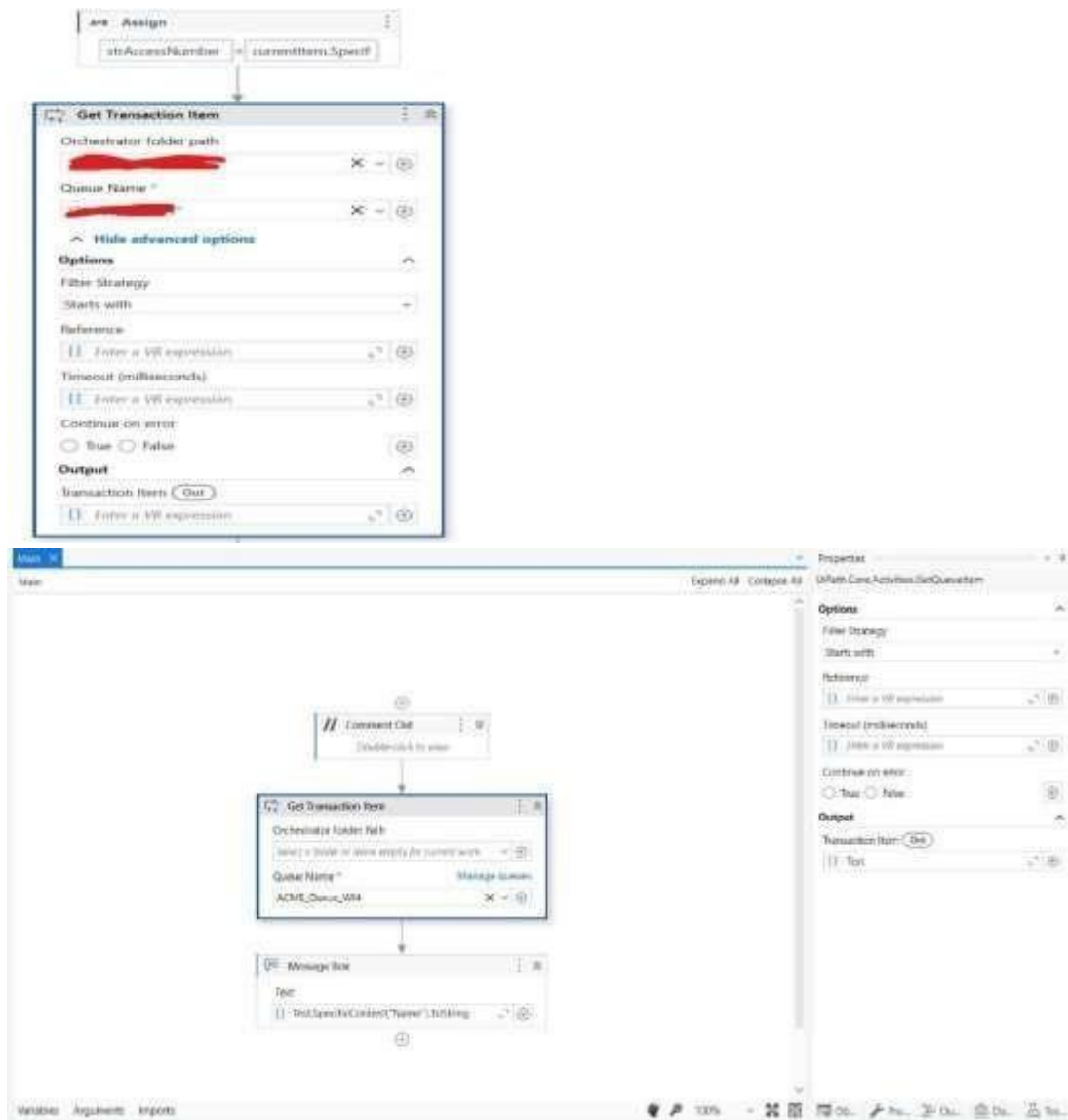
Output

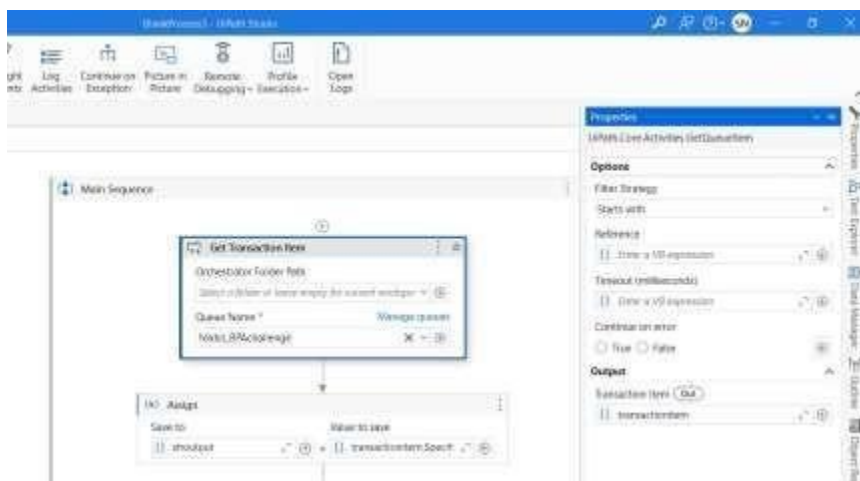
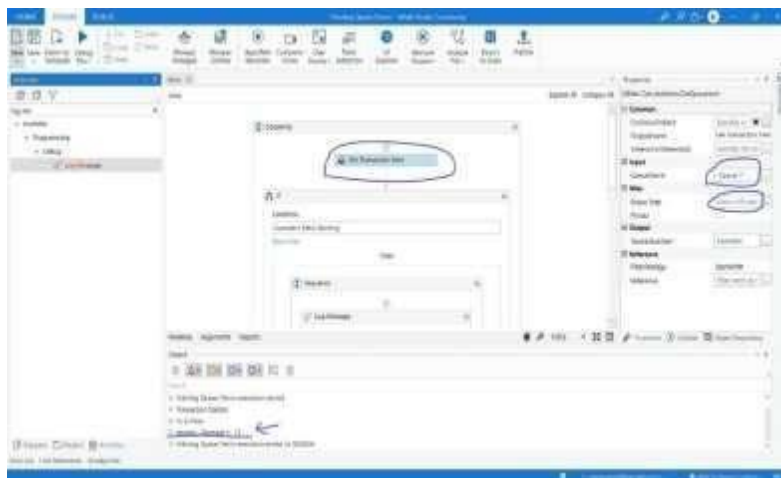
TransactionItem TransactionItem

Reference

FilterStrategy StartsWith

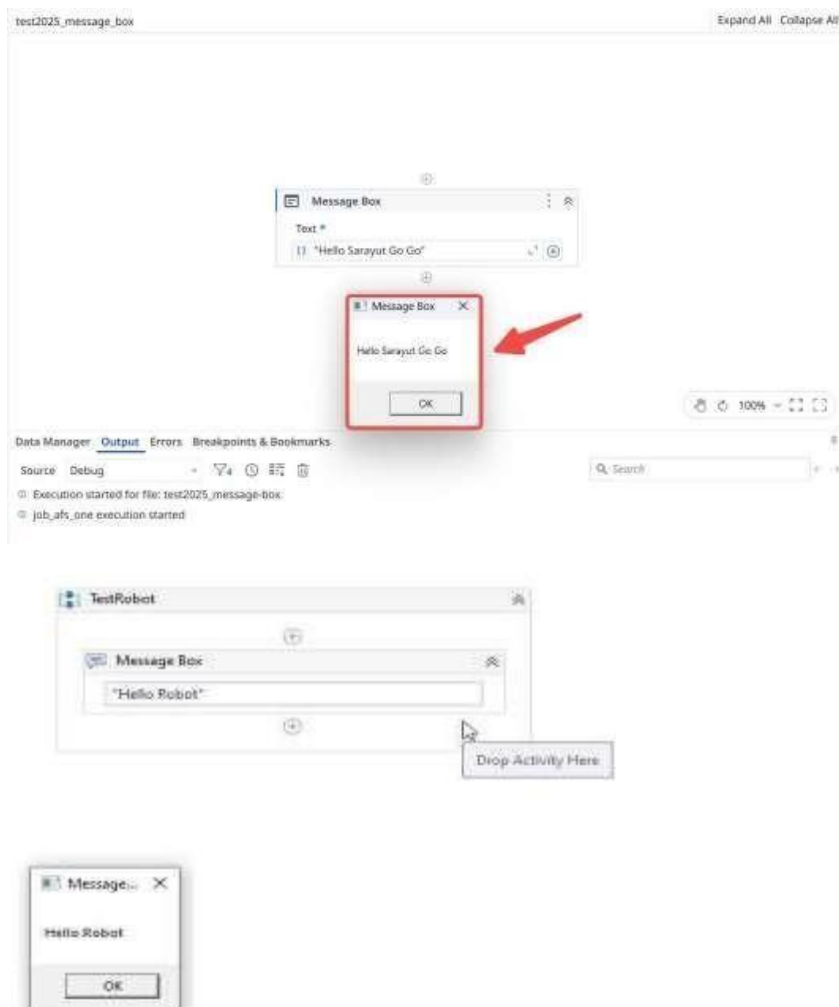
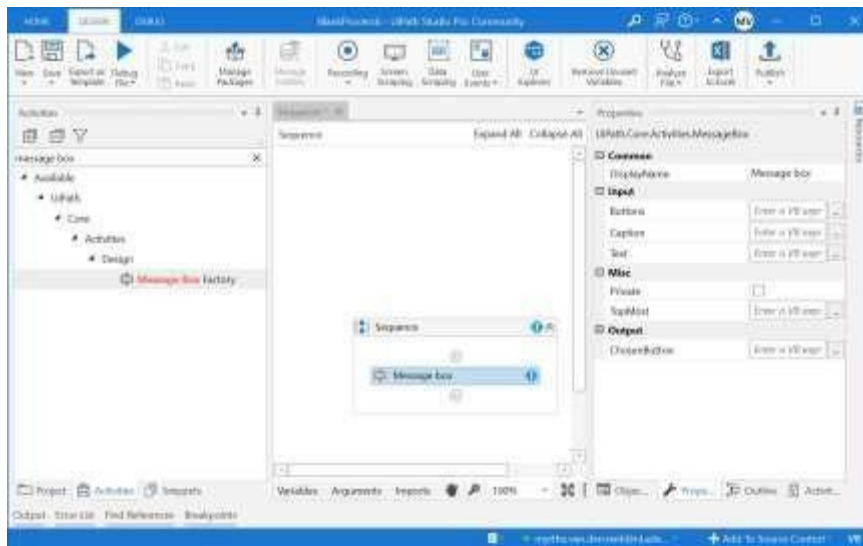
Reference Filter items by reference





Step 2: Display Queue Data. Show the retrieved transaction details using a Message Box.





Step 3: Check Transaction Status. Refresh the Transactions page in Orchestrator.

10. d. Build UiPath Chatbot using Google dialog flow

Step 1: Create a Dialogflow Agent

Open the Google Dialogflow Console and create a new agent that will handle the chatbot conversations. Enter the agent name as "UiPathChatbot," select English as the default language, choose the appropriate time zone, and save the configuration. This agent serves as the central component responsible for understanding user queries and generating responses.

Step 2: Create Intents in Dialogflow

Navigate to the Intents section and create intents to define how the chatbot should respond to different user messages. For example, create a "Greeting" intent and add training phrases such as "Hi," "Hello," and "Good Morning." Then provide suitable responses like "Hello! How can I help you?" and save the intent. These intents enable Dialogflow to identify user intentions accurately.

Step 3: Generate Dialogflow API Credentials

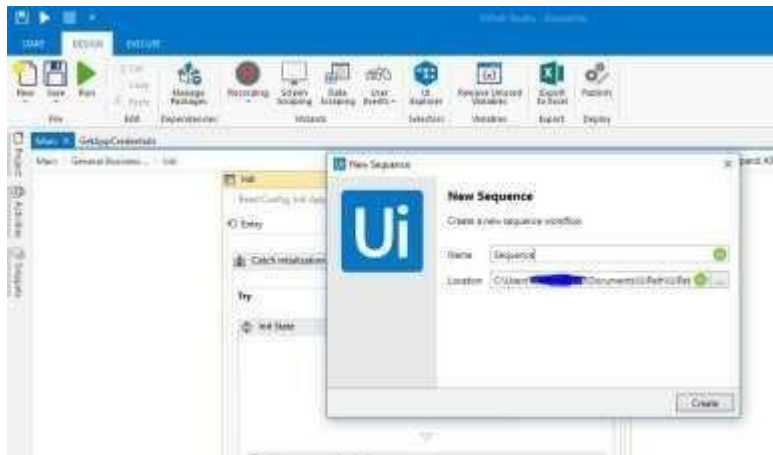
Access the agent settings and open the linked Google Cloud project. In the Google Cloud Console, navigate to APIs and Services, create a Service Account, and generate a JSON key file. Download this file, as it contains the credentials required for secure communication between UiPath and Dialogflow.

Step 4: Create a New UiPath Project

Launch UiPath Studio and select the option to create a new Process project. Enter the project name as "DialogflowChatbot" and click Create. A new automation workspace will open where the chatbot workflow can be developed.

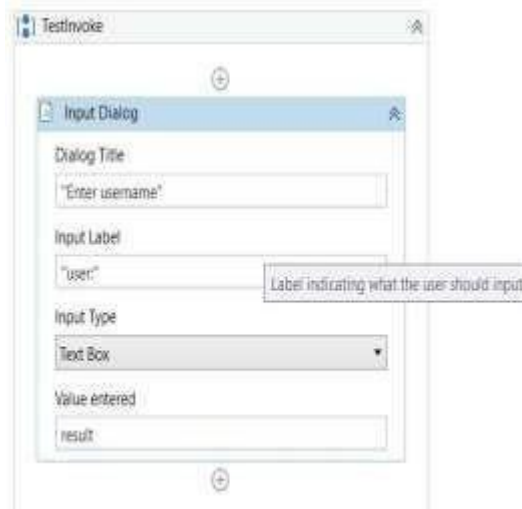
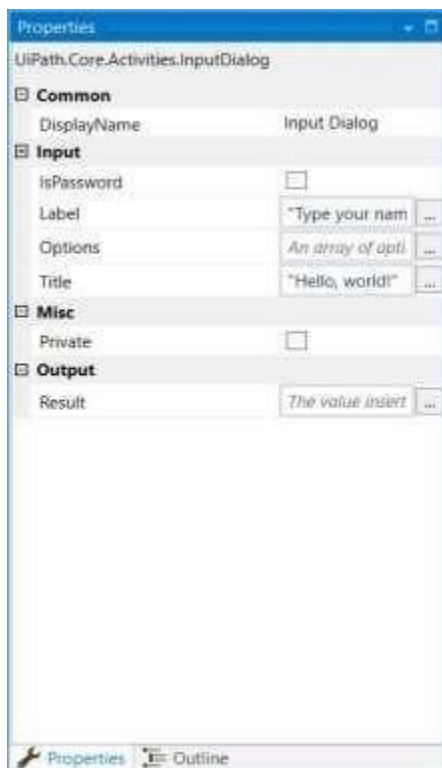
Step 5: Design the Workflow

Inside the Main sequence, add the required activities including Input Dialog, HTTP Request, Deserialize JSON, Assign, and Message Box. Arrange these activities logically so that the user's message is collected, processed through Dialogflow, and the resulting response is displayed back to the user.



Step 6: Configure the Input Dialog

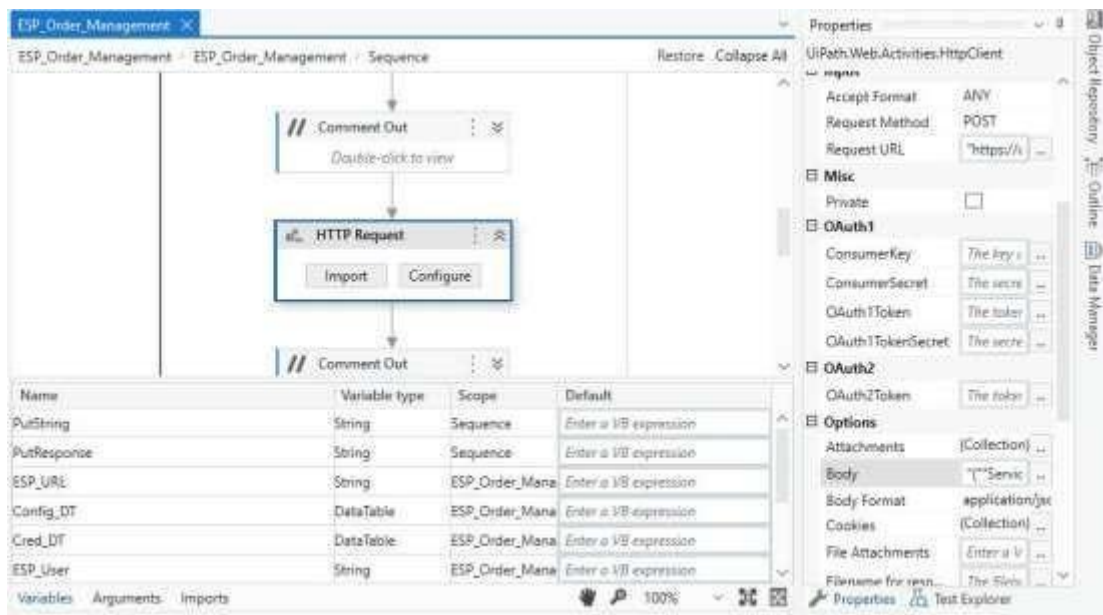
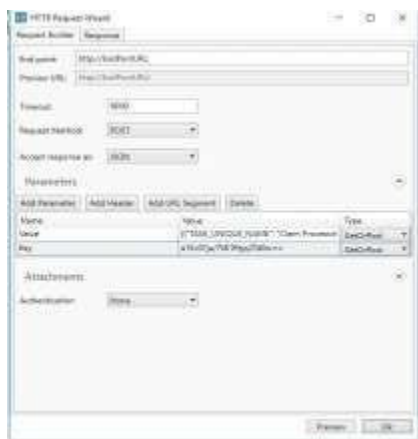
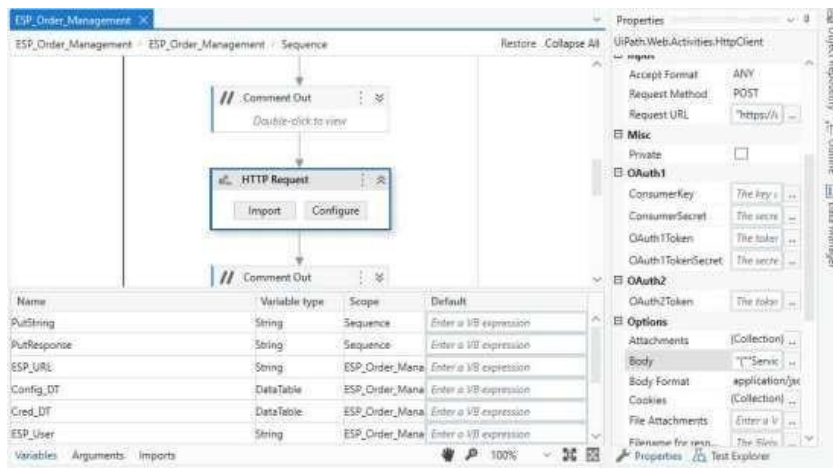
Use the Input Dialog activity to obtain input from the user during execution. Set the title as "Chatbot" and provide a label such as "Enter your message." Store the entered text in a variable named userMessage, which will later be sent to Dialogflow for processing.

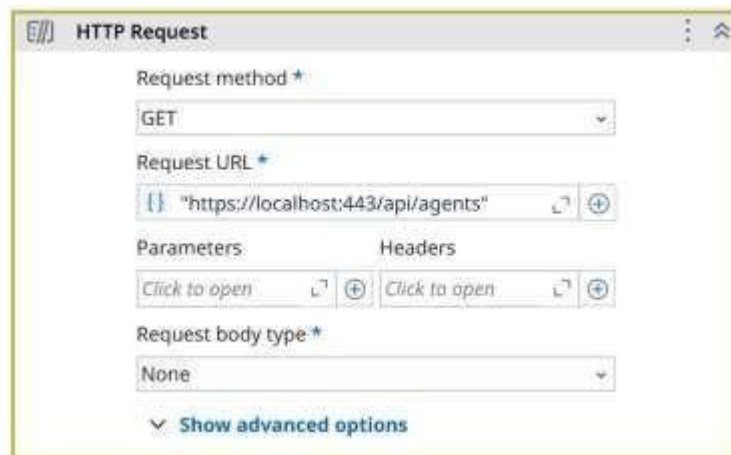


Step 7: Configure the HTTP Request Activity

Install the UiPath.WebAPI.Activities package through Manage Packages and drag the HTTP Request activity into the workflow. Configure it to send a POST request to the Dialogflow Detect Intent API endpoint. Add the authorization and content-type headers, and include the

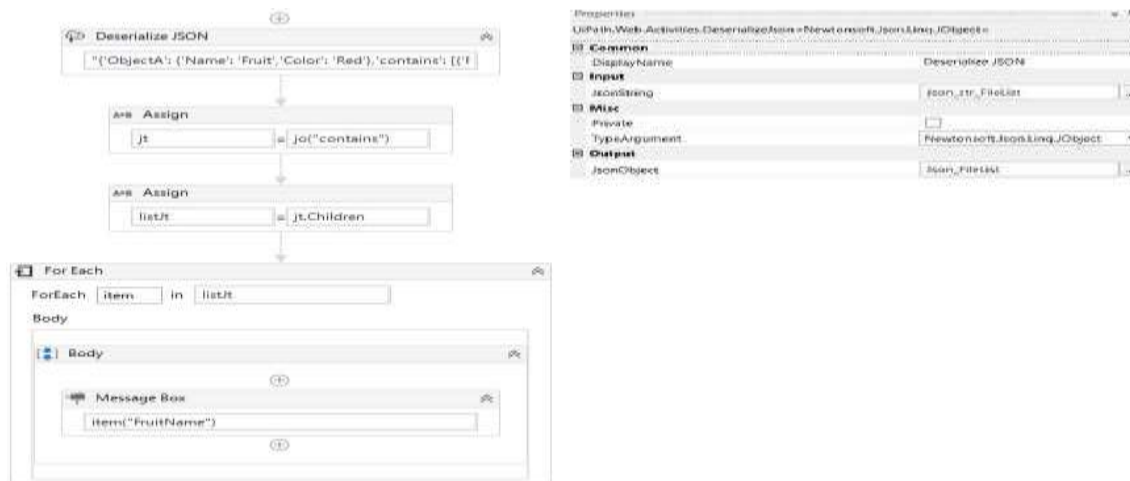
user's message in JSON format within the request body. This activity enables UiPath to communicate with Dialogflow and submit user queries.

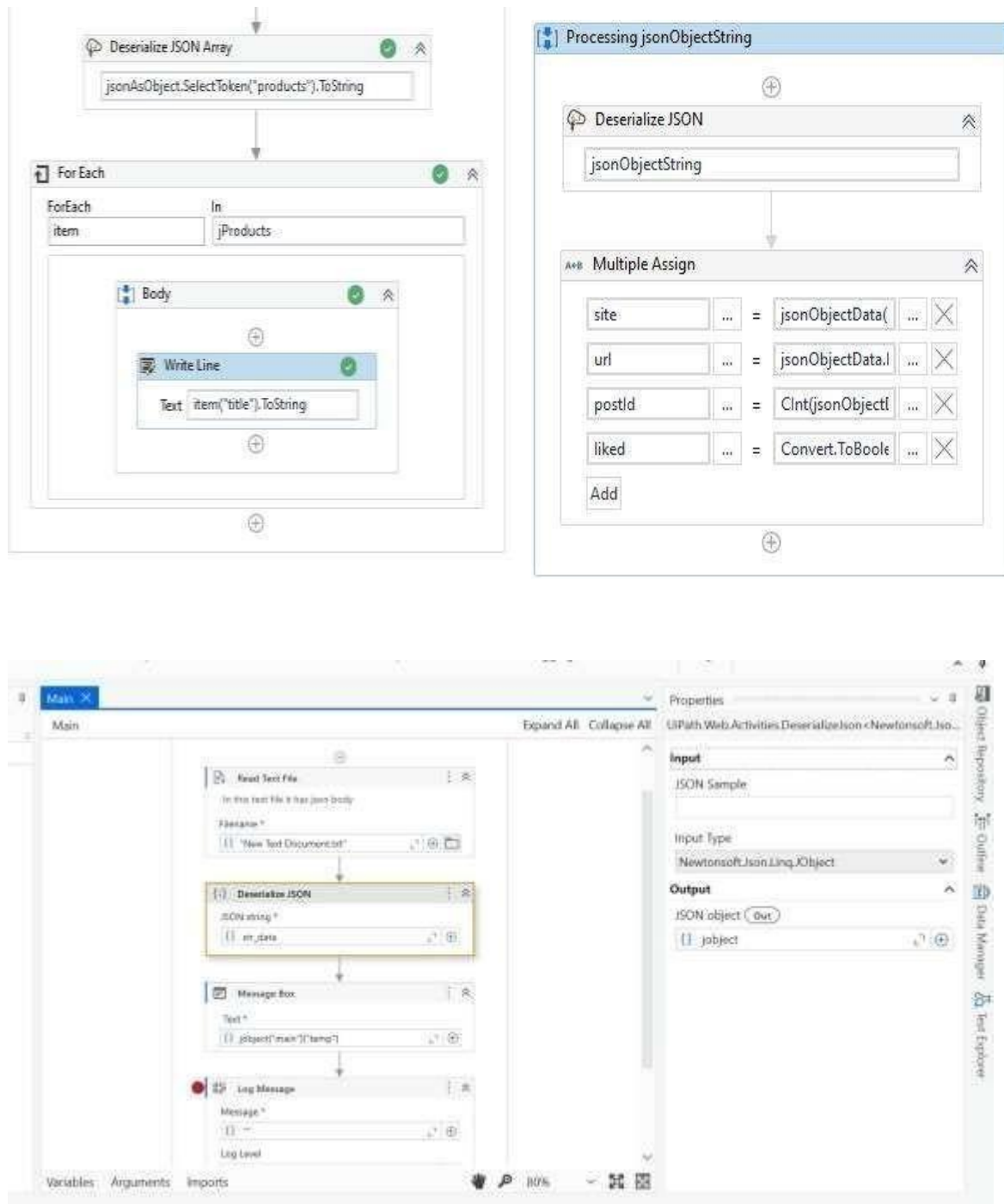




Step 8: Deserialize the JSON Response

After receiving the response from Dialogflow, use the Deserialize JSON activity to convert the JSON string into an accessible JSON object. Store the output in a variable named jsonResponse so that the chatbot's response can be extracted easily.





Step 9: Extract the Chatbot Reply

Use an Assign activity to retrieve the fulfillmentText value from the deserialized JSON object and store it in a variable named reply. This value represents the response generated by Dialogflow based on the detected intent.

Step 10: Display the Output

Finally, add a Message Box activity and display the value stored in the reply variable. When the workflow executes, the chatbot response appears on the screen, allowing the user to interact with the chatbot in a simple and user-friendly manner.

OUTPUT

